

Spectral Arithmetic Frontiers of the Fabius–Rvachev System

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Provenance

This volume consolidates the following drafts verbatim (labels, citation keys, and asset paths mechanically prefixed per part; no mathematical content altered). The absorbed drafts are deleted from the working tree; git history is the archive.

- Fabius_Half_Integer_Spectral_Frontier_Report/Fabius_Half_Integer_Spectral_Frontier_Report.tex
SHA-256 167aea6c6c2ddc55e510c496364e0a5dcca635c39d2fe04392b1f8b6d13df89d
- Fabius_Arithmetic_Rays_Frontier_Report/Fabius_Arithmetic_Rays_Frontier_Report.tex
SHA-256 428bc7020f42b5d3cd9029fcda7ce350460911a1add65d6c123ebdefad541a6a
- Spectral_Arithmetic_Pascal_Rvachev_Hierarchy/Spectral_Arithmetic_Pascal_Rvachev_Hierarchy.tex
SHA-256 34f789a6c4febdd5b6978905f7f45a0a69f84586cc061e91742387d4d8833255
- Fabius_Derivative_Norm_Spectrum_bundle/Fabius_Derivative_Norm_Spectrum.tex
SHA-256 345e0097d0476260f0de9a86ce354a01ebd203e252e2e83f7870e3b06ded7c7d

Part I

Half-Integer Spectral Arithmetic for the Fabius–Rvachev System

Abstract

Let

$$\Phi(z) := \widehat{\text{up}}(z) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{z}{2^j}\right), \quad \text{sinc } u := \frac{\sin(\pi u)}{\pi u},$$

be the Fourier transform of Rvachev’s compactly supported up-function, and put $a_k = \Phi(k + \frac{1}{2})$. The repository corpus already contains the Fourier product, exact dyadic values of the Fabius function, their Gaussian-binomial and Thue–Morse formulae, the Lambert– W endpoint asymptotic, the complete real-axis Fourier-decay theory assembled from several waves, and a recent spectral account of the integer zero divisor. This report does not relabel those results as new. It develops instead the half-integer spectral sequence (a_k) , identified in the previous spectral audit as a genuine unresolved bridge.

The first main theorem is an infinite, positive-weight Prouhet system of all orders. With $o_k = 2k + 1$, $\varepsilon_k = (-1)^{s_2(k)}$, and $w_k = |a_k|$, one has

$$\sum_{k \geq 0} a_k = \frac{1}{2}, \quad \sum_{k \geq 0} \varepsilon_k w_k o_k^{2m} = 0 \quad (m \geq 1).$$

Thus the Thue–Morse partition of the odd integers has exactly matching positive weighted even moments of every order. More generally, the same coefficients form an exact signed quadrature rule $\sum_k \varepsilon_k w_k P(o_k^2) = P(0)/2$ for every polynomial P .

The second main theorem is a finite dyadic spectral transform. If $N = 2^n$ and r is odd with $1 \leq r < N$, define

$$A_{n,r} = \frac{1}{N} \left[1 - 2 \sum_{j=1}^{N-1} F\left(\frac{j}{N}\right) \cos\left(\frac{\pi r j}{N}\right) \right].$$

Then the exact alias identity

$$A_{n,r} = \sum_{q \in \mathbb{Z}} \Phi\left(qN + \frac{r}{2}\right)$$

holds, the dyadic samples are recovered by an exact inverse discrete cosine transform, and

$$|A_{n,r} - \Phi(r/2)| \leq \frac{2(1 - 2^{-n})\zeta(n)}{\pi^n} 2^{-n(n-1)/2} \quad (n \geq 2).$$

Substitution of the repository’s finite $q = 1/2$ Gaussian-binomial formula for each $F(j/N)$ therefore gives superexponentially convergent, finite q -binomial–Thue–Morse–cyclotomic approximants to every fixed half-integer Fourier amplitude.

The folded coefficients form a Galois orbit in the maximal real 2^{n+1} -st cyclotomic field. Their orbit polynomial lies in $\mathbb{Q}[X]$; all power traces are explicit rational convolutions of dyadic Fabius values with a Ramanujan-sum congruence filter. These rational traces converge to nonlinear spectral moments $\sum_k a_k^m$. An additional exact factorization isolates a finite Thue–Morse polynomial value as the sign carrier of every alias class. Finally, the all-order Prouhet identities turn the endpoint function into an exact “beyond all algebraic orders” spectral remainder. Inserting the repo’s corrected Lambert– W asymptotic produces a direct, explicit bridge between the signed half-integer sinc amplitudes and the lower-Lambert saddle. All new claims below are proved for human verification; none is represented as already formalized in Lean. Novelty is asserted only relative to the reviewed repository corpus and targeted source searches, not as a universal priority claim.

Status convention. **IMPORTED BASELINE.** marks a theorem, identity, or notation already present in the repository or in a cited source. **NEW RESULT PROVED HERE.** marks a statement for which a proof is supplied here and which was not found in the audited TeX corpus or in targeted searches of the cited literature. **OPEN OR CONJECTURAL.** marks conjectures or computational observations. The distinction is essential because the report is designed as a frontier supplement, not a rewrite of the existing corpus.

1 Scope, recursive audit, and the non-duplication boundary

1.1 What was inspected

The documentation tree is a stratified research record rather than one article. The same-day recursive audit used for this report contains fifty-seven TeX artifacts: twenty-five living or imported sources and thirty-two frozen historical versions. The living stratum includes the main Fabius–Rvachev exposition, a mathematical glossary, the Lean walkthrough, the non-elementarity study, a large consolidated non-formalized frontier volume, a Thue–Morse formula atlas, a sequence of Fourier-decay reports and audits, and seven source-paper transcriptions. The historical stratum preserves earlier expositions, q-calculus studies, asymptotic investigations, repeated-integration reports, gap registers, and superseded formula atlases. A path ledger is supplied in [appendix A](#).

The current canonical descendants were rechecked separately, because the frontier directory is intentionally consolidated over time: processed drafts may be absorbed into the large canonical volume or removed while their frozen archive remains. Accordingly, the appendix records the audited snapshot; it is not a promise that every historical path remains a live branch-head file after future repository maintenance.

Mathematical extraction was concentrated on the current descendants and on the distinctive claims of each archived line of development. Byte-identical and superseded files were treated as provenance, not as independent evidence of multiple results. The audit was then compared with three frontier reports produced from the same corpus on spectral determinants, spline/quadrature structure, and arithmetic Fourier rays. This prevents accidental duplication of very recent deductions that are not yet part of the main exposition.

1.2 Results explicitly treated as baseline

The following are *not* claimed as discoveries here.

1. The relations between the bounded Fabius function F , its signed global extension $\mathcal{F}$, and Rvachev’s even compactly supported function $\mathfrak{u}$.
2. The refinement differential equation, the entire Fourier product, the cosine-product form, and rapid decay of Φ .
3. Exact rationality at dyadic points, the moment recurrences, the highest-bit recursion, and the finite $q = 1/2$ Gaussian-binomial–Thue–Morse formula for $\mathcal{F}(m/2^n)$.
4. The integer zero set of Φ , multiplicity $1 + \nu_2(m)$ at a nonzero integer m , strict logarithmic concavity between zeros, and the sign law $\text{sgn } \Phi(x) = (-1)^{s_2(M)}$ on $M < x < M + 1$.
5. The pointwise, metric, probabilistic, extremal, annular, and transfer-operator Fourier-decay results developed in the successive decay reports.
6. The corrected endpoint asymptotic in the lower-Lambert coordinate, including the nonconstant Gamma–zeta periodic fluctuation.
7. The spectral determinant model, spectral zeta and heat trace, all-order integer-zero jets, dyadic q-Euler determinants, and symmetric q-deformation developed in the prior spectral frontier report.
8. Exact rational-ray decay laws, cyclotomic orbit norms, resultants, Ramanujan traces, and the Mersenne half-base bridge developed in the prior arithmetic-ray report.

1.3 The gap selected for this report

The previous spectral report ends with the focused problem of classifying the half-integer amplitudes

$$\Phi\left(\frac{2k+1}{2}\right),$$

which govern the leading Taylor coefficients at every integer zero. The main repository exposition contains the exact cosine series in which the same amplitudes reconstruct up, while its dyadic arithmetic chapters contain finite q -binomial formulae for the values being reconstructed. Yet the reviewed corpus does not develop the finite transform between those two arrays, does not state the all-order weighted Prouhet identities below, and does not connect those cancellations to the corrected Lambert endpoint scale.

This is therefore a natural unfilled intersection of five mature themes already present separately in the repository:

$$\begin{aligned} \text{Thue--Morse signs} &\longleftrightarrow \text{half-integer sinc amplitudes} \\ &\longleftrightarrow \text{dyadic } q\text{-binomial values} \longleftrightarrow \text{cyclotomic traces} \longleftrightarrow \text{Lambert--}W. \end{aligned}$$

1.4 Contribution map

[table 1](#) summarizes the boundary between imported material, new deductions proved here, and conjectural extensions.

Table 1: Boundary between imported material and the new theorem package.

Result	Content	Status
Half-integer cosine spectrum	Isolates the sequence $a_k = \Phi(k + 1/2)$ as the nonzero Fourier coefficients of the two-periodic up-function.	IMPORTED BASELINE. identity, new organization
Weighted Prouhet theorem	Proves equality of every positive weighted even moment in the Thue–Morse partition of the odd integers.	NEW RESULT PROVED HERE.
Spectral quadrature	Shows $\sum_k \varepsilon_k w_k P((2k+1)^2) = P(0)/2$ for every polynomial P .	NEW RESULT PROVED HERE.
Dyadic moment filtration	Converts every derivative jet at a reduced dyadic point into a half-integer spectral moment and identifies its exact terminating depth.	NEW RESULT PROVED HERE. formulation
Beyond-all-orders remainder	Rewrites $F(x)$ after subtraction of an arbitrary cosine Taylor polynomial, with all algebraic terms cancelling exactly.	NEW RESULT PROVED HERE.
Exact dyadic alias transform	Gives a finite DCT of dyadic Fabius values equal to a complete folded half-integer alias class.	NEW RESULT PROVED HERE. Abstract cyclic engine formal; specialization remains a separate Lean obligation.
Certified convergence	Gives a uniform $2^{-n^2/2+\mathcal{O}(n)}$ error bound from the folded coefficient to $\Phi(r/2)$.	NEW RESULT PROVED HERE.
q -binomial spectral approximants	Substitutes the finite Gaussian-binomial formula into the DCT, producing exact cyclotomic approximants.	NEW RESULT PROVED HERE.
Cyclotomic orbit traces	Constructs rational orbit polynomials and explicit Ramanujan-filtered power traces.	NEW RESULT PROVED HERE.

Continued on the next page

Contribution map (continued)

Result	Content	Status
Spectral power-sum limits	Shows the rational Galois traces converge to $\sum_k a_k^m$ superexponentially up to a rapidly decreasing tail.	NEW RESULT PROVED HERE.
Thue–Morse sign carrier	Factors each alias class into a finite root-of-unity Thue–Morse product and an absolutely convergent shifted spectral sum.	NEW RESULT PROVED HERE.
Lambert bridge	Inserts the corrected lower-Lambert asymptotic into the exact Prouhet remainder.	new consequence of imported asymptotic
Alias-sign positivity	Conjectures positivity of the residual shifted spectral sum, equivalent to an exact Thue–Morse sign law for every folded coefficient.	OPEN OR CONJECTURAL.

2 Baseline notation and identities

2.1 The three functions

Let $F : [0, 1] \rightarrow [0, 1]$ be the bounded Fabius function. Let

$$\mathcal{F}(x) := \sum_{n=0}^{\infty} \varepsilon_n \operatorname{up}(x - 2n - 1), \quad \varepsilon_n := (-1)^{s_2(n)}, \quad (1)$$

be its signed global extension; the sum is locally finite. On $[-1, 1]$,

$$\operatorname{up}(t) = \mathcal{F}(t + 1) = \begin{cases} F(t + 1), & -1 \leq t \leq 0, \\ F(1 - t) = 1 - F(t), & 0 \leq t \leq 1. \end{cases} \quad (2)$$

The function up is even, nonnegative, C^∞ , supported on $[-1, 1]$, normalized by $\int_{\mathbb{R}} \operatorname{up} = 1$, and satisfies $\operatorname{up}(0) = 1$. Every positive derivative of up vanishes at 0, because on the right of zero it is $1 - F(t)$ and F is flat at the origin.

The global derivative identities are

$$\mathcal{F}^{(m)}(x) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m x), \quad \operatorname{up}^{(m)}(t) = 2^{\binom{m+1}{2}} \mathcal{F}(2^m(t + 1)). \quad (3)$$

At nonnegative integers,

$$\mathcal{F}(2b) = 0, \quad \mathcal{F}(2b + 1) = \varepsilon_b. \quad (4)$$

These are repository baseline facts, ultimately descending from Arias de Reyna’s formulation [6, 7].

2.2 The Fourier product

We use the Fourier convention

$$\widehat{f}(z) = \int_{\mathbb{R}} f(t) e^{-2\pi i t z} dt.$$

Put

$$\Phi(z) := \widehat{\operatorname{up}}(z) = \prod_{j=0}^{\infty} \operatorname{sinc}\left(\frac{z}{2^j}\right), \quad \operatorname{sinc} u = \frac{\sin(\pi u)}{\pi u}. \quad (5)$$

The product is entire and even. For every $J \geq 1$ and real $x \neq 0$, selecting the first J factors gives the elementary rapid-decay estimate

$$|\Phi(x)| \leq \frac{2^{J(J-1)/2}}{\pi^J |x|^J}. \quad (6)$$

The nonzero real zeros are exactly the integers; at $m \neq 0$ the multiplicity is $1 + \nu_2(|m|)$. If $M < x < M + 1$, then

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(M)}. \quad (7)$$

2.3 The exact dyadic q -binomial formula

For later substitution, we record the current corpus formula. With $(q; q)_n = \prod_{j=1}^n (1 - q^j)$ and the Gaussian binomial coefficient $\begin{bmatrix} n \\ k \end{bmatrix}_q$, one has for $m, n \in \mathbb{N}$ and every rational translation c

$$\begin{aligned} \mathcal{F}\left(\frac{m}{2^n}\right) &= \frac{1}{2^{n^2}(1/2; 1/2)_n} \sum_{k=0}^n \frac{\begin{bmatrix} n \\ k \end{bmatrix}_{1/2}}{2^{k(k-1)}(n+k)!} \\ &\quad \times \sum_{s=0}^{m2^k-1} \varepsilon_s \left(s - m2^k + c\right)^{n+k}. \end{aligned} \quad (8)$$

The right side is independent of c . When $0 \leq m \leq 2^n$, it equals the bounded value $F(m/2^n)$. This is **IMPORTED BASELINE**. from the main exposition; the new use below is to insert it into a cyclotomic discrete cosine transform.

2.4 The lower-Lambert endpoint theorem

Write $L = \log 2$ and, for sufficiently small $x > 0$,

$$W = W_{-1}(-Lx), \quad \lambda(x) = -\frac{W}{L}, \quad r(x) = 2^{\lambda(x)} = \frac{\lambda(x)}{x}. \quad (9)$$

Let Ψ be the smooth one-periodic correction whose nonzero Fourier coefficients are

$$\widehat{\Psi}(k) = -\frac{\Gamma(-\chi_k)\zeta(1-\chi_k)}{L}, \quad \chi_k = \frac{2\pi i k}{L}. \quad (10)$$

Put

$$\mathfrak{g} = \gamma_1 + \frac{\gamma^2}{2} - \frac{\pi^2}{12}. \quad (11)$$

The corrected endpoint theorem in the corpus states

$$F(x) = \frac{e^{\Psi(-W/L)}}{2^{7/12}\sqrt{\pi x}} \exp\left(\frac{\mathfrak{g} - W - \frac{1}{2}W^2}{L}\right) (1 + \mathcal{O}(\lambda(x)^{-1})). \quad (12)$$

The periodic factor is nonconstant and cannot be deleted from a sharp asymptotic. We shall not reprove (12); instead we derive an exact half-integer spectral identity whose left side is $F(x)$ and then insert this imported theorem.

3 The half-integer spectral sequence

3.1 Definition and first values

Define

$$a_k := \Phi\left(k + \frac{1}{2}\right), \quad o_k := 2k + 1, \quad w_k := |a_k|. \quad (13)$$

The sign law (7) gives immediately

$$a_k = \varepsilon_k w_k. \quad (14)$$

The first amplitudes are shown in [table 2](#). Their rapid shrinkage is deceptive: the exact identities below require cancellations over all indices, and no finite initial segment has the moment-vanishing property.

k	ε_k	$a_k = \Phi(k + 1/2)$
0	+1	0.553771275888810095344217416345
1	-1	-0.046202299134306712992574862210
2	-1	-0.008656867928610010895881610116
3	+1	0.001044808241827280615683472766
4	-1	-0.000356152861629792122112229855
5	+1	0.000298368336838126419196617073
6	+1	0.000091404980598947955260616141
7	-1	-0.000006555455730992891195331226

Table 2: Initial half-integer amplitudes. The signs are the Thue–Morse word.

3.2 Exact two-periodic cosine expansion

Let P denote the two-periodization of up ,

$$P(t) := \sum_{\ell \in \mathbb{Z}} \text{up}(t - 2\ell). \quad (15)$$

Because up is supported on $[-1, 1]$, $P(t) = \text{up}(t)$ for $-1 \leq t \leq 1$.

Proposition 3.1 (Half-integer cosine spectrum). *The Fourier series of P is absolutely differentiable to every order and is*

$$P(t) = \frac{1}{2} + \sum_{k=0}^{\infty} a_k \cos(\pi o_k t). \quad (16)$$

In particular, on $[-1, 1]$ the series equals $\text{up}(t)$.

Proof. The m th complex Fourier coefficient for period two is

$$c_m = \frac{1}{2} \int_{-1}^1 \text{up}(t) e^{-\pi i m t} dt = \frac{1}{2} \Phi(m/2).$$

For nonzero even m , $m/2$ is a nonzero integer and $c_m = 0$. Also $c_0 = 1/2$. For $m = \pm(2k + 1)$, evenness and reality give $c_m = a_k/2$. Thus complex Fourier inversion reduces to (16). Estimate (6) with arbitrarily large J shows that a_k is a Schwartz sequence, so the series and every differentiated series converge absolutely and uniformly. $\square$

Lean status. The displayed support-interval identity is already formalized as `Fabius.rvachev_cosine_series` in `PoissonSummation.lean`. That theorem supplies the exact odd half-integer cosine series. The all-orders differentiated-series packaging and the weighted Prouhet, specialized dyadic alias-transform, cyclotomic-trace, and Lambert consequences developed below remain separate formalization obligations.

Remark 3.2. The half-integer sequence is therefore not an auxiliary sampling choice. It is *exactly* the nonconstant Fourier spectrum of the natural two-periodic extension of up . Integer Fourier samples disappear because the entire sinc product vanishes there.

4 An all-order weighted Prouhet system

4.1 Moment cancellation

The classical Prouhet–Thue–Morse construction balances finitely many *unweighted* power sums on a finite dyadic block. Here the half-integer Fourier amplitudes supply positive, rapidly decreasing weights that balance *every* even power on the infinite set of odd integers.

Theorem 4.1 (Half-integer weighted Prouhet theorem). *The amplitudes (13) satisfy*

$$\sum_{k=0}^{\infty} a_k = \frac{1}{2}, \quad (17)$$

$$\sum_{k=0}^{\infty} a_k o_k^{2m} = 0 \quad (m \geq 1). \quad (18)$$

Equivalently, for every $m \geq 1$,

$$\boxed{\sum_{\substack{k \geq 0 \\ s_2(k) \text{ even}}} w_k (2k+1)^{2m} = \sum_{\substack{k \geq 0 \\ s_2(k) \text{ odd}}} w_k (2k+1)^{2m}.} \quad (19)$$

All displayed series converge absolutely.

Proof. Set $t = 0$ in (16). Since $P(0) = \text{up}(0) = 1$,

$$1 = \frac{1}{2} + \sum_{k \geq 0} a_k,$$

which proves (17). For $m \geq 1$, termwise differentiation gives

$$P^{(2m)}(0) = (-1)^m \pi^{2m} \sum_{k \geq 0} a_k o_k^{2m}.$$

Every positive derivative of $P = \text{up}$ vanishes at zero, hence the left side is zero. This proves (18). Absolute convergence follows by applying (6) with $J > 2m + 1$. Finally substitute $a_k = \varepsilon_k w_k$ and move the negative-sign terms to the opposite side. $\square$

Corollary 4.2 (Exact spectral quadrature at the origin). *For every polynomial $Q \in \mathbb{C}[X]$,*

$$\boxed{\sum_{k=0}^{\infty} \varepsilon_k w_k Q((2k+1)^2) = \frac{Q(0)}{2}.} \quad (20)$$

Thus the signed positive weights $\varepsilon_k w_k$ evaluate every polynomial in the squared odd variable exactly at the off-grid point 0.

Proof. Expand $Q(X) = \sum_{m=0}^d q_m X^m$ and use (17)–(18). $\square$

Remark 4.3 (Why this is not the ordinary finite Prouhet theorem). For a block $0 \leq k < 2^n$, the unweighted Thue–Morse signs annihilate powers only below degree n . In (19), the support is infinite, the nodes are the odd integers, the weights are the absolute half-integer sinc amplitudes, and every even degree is annihilated. The recent Thue–Morse transform literature continues to develop finite Prouhet–Tarry–Escott families [11]; it does not supply the spectral weights appearing here.

4.2 A nontrivial signed measure with every positive moment zero

Corollary 4.4 (Spectral moment-null measure). *The finite signed measure*

$$\nu := \frac{1}{2} \sum_{k=0}^{\infty} a_k (\delta_{o_k} + \delta_{-o_k}) \quad (21)$$

has total mass $1/2$ and satisfies

$$\int_{\mathbb{R}} x^m d\nu(x) = 0 \quad (m \geq 1). \quad (22)$$

Consequently $\nu - \frac{1}{2}\delta_0$ is a nonzero finite signed measure whose moments of every nonnegative integer order vanish.

Proof. The total variation and every polynomial moment are finite because (a_k) is Schwartz. Odd moments vanish by symmetry; positive even moments vanish by [theorem 4.1](#); and the mass is [\(17\)](#). $\square$

Remark 4.5. There is no contradiction with uniqueness results for positive moment problems: [\(21\)](#) is signed and has unbounded support. It gives a concrete moment-null object built from the exact spectrum of a positive compactly supported density.

4.3 Certified finite truncations

The identities are infinite, but the elementary product bound gives explicit certificates for truncation experiments.

Proposition 4.6 (Prouhet truncation bound). *Let $m \geq 0$, $K \geq 1$, and $J > 2m + 1$ be integers. Put $p = J - 2m$. Then*

$$\left| \sum_{k=K}^{\infty} a_k o_k^{2m} \right| \leq \frac{2^{J(J+1)/2}}{\pi^J} \sum_{k=K}^{\infty} (2k+1)^{-p} \quad (23)$$

$$\leq \frac{2^{J(J+1)/2}}{\pi^J} \left((2K+1)^{-p} + \frac{(2K+1)^{1-p}}{2(p-1)} \right). \quad (24)$$

For $m \geq 1$, the same bound is an a posteriori certificate for the residual of the first K terms, because the complete sum is zero.

Proof. Apply [\(6\)](#) with $x = o_k/2$. Since

$$|a_k| \leq \frac{2^{J(J-1)/2}}{\pi^J (k+1/2)^J} = \frac{2^{J(J+1)/2}}{\pi^J o_k^J},$$

multiplication by o_k^{2m} and summation give [\(23\)](#). For the second line use the monotonicity of $(2x+1)^{-p}$ and the integral estimate

$$\sum_{k=K}^{\infty} (2k+1)^{-p} \leq (2K+1)^{-p} + \int_K^{\infty} (2x+1)^{-p} dx.$$

$\square$

Remark 4.7. The coefficient in [\(23\)](#) records every factor of two from the half-integer substitution $x = o_k/2$. The sharper alias estimate in [section 8](#) works directly in the variable x and therefore uses the constant C_K of [\(41\)](#) without this substitution.

5 Dyadic spectral moment filtration

5.1 Every derivative as a half-integer spectral moment

For $m \geq 0$ define

$$S_m(t) := \sum_{k=0}^{\infty} a_k o_k^m \cos\left(\pi o_k t + \frac{m\pi}{2}\right). \quad (25)$$

Theorem 5.1 (Spectral derivative identity). *For $-1 \leq t \leq 1$ and every $m \geq 0$,*

$$S_m(t) = \frac{\text{up}^{(m)}(t)}{\pi^m} = \frac{2^{\binom{m+1}{2}}}{\pi^m} \mathcal{F}(2^m(t+1)). \quad (26)$$

The series is absolutely and uniformly convergent on $[-1, 1]$.

Proof. Differentiate [\(16\)](#) m times. The derivative of $\cos(\pi o_k t)$ is $(\pi o_k)^m \cos(\pi o_k t + m\pi/2)$. The second equality is the imported global derivative identity [\(3\)](#). $\square$

5.2 Exact termination at reduced dyadic points

Theorem 5.2 (Dyadic spectral filtration). *Let $n \geq 1$ and let a be odd with $0 < a < 2^n$. Put $t = a/2^n$. Then*

$$S_m(t) = 0 \quad (m > n), \quad (27)$$

$$S_n(t) = -\varepsilon_{(a-1)/2} \frac{2^{\binom{n+1}{2}}}{\pi^n}, \quad (28)$$

$$S_m(t) = \frac{2^{\binom{m+1}{2}}}{\pi^m} \mathcal{F}\left(\frac{2^n + a}{2^{n-m}}\right) \quad (0 \leq m < n). \quad (29)$$

Every value in (29) is rational up to the displayed power of π , and is given explicitly by (8). In particular, the reduced dyadic depth n is exactly the largest positive order for which the spectral moment $S_m(t)$ can be nonzero.

Proof. By theorem 5.1, the relevant argument of $\mathcal{F}$ is

$$2^m(t+1) = \frac{2^n + a}{2^{n-m}}.$$

If $m > n$, this is an even integer, so (4) gives zero. If $m = n$, the argument is the odd integer $2^n + a = 2b + 1$, with

$$b = 2^{n-1} + \frac{a-1}{2}.$$

The binary supports of the two summands are disjoint; hence $\varepsilon_b = -\varepsilon_{(a-1)/2}$, proving (28). The remaining case is simply the unreduced dyadic argument displayed in (29), to which the global q -binomial formula applies. The critical value is nonzero, while all larger orders vanish. $\square$

Corollary 5.3 (Spectral detection of the denominator). *For an interior reduced dyadic point $t = a/2^n$ with odd a ,*

$$n = \max\{m \geq 1 : S_m(t) \neq 0\}. \quad (30)$$

Thus the denominator depth is encoded intrinsically in the half-integer Fourier spectrum, without inspecting the binary expansion of t .

Remark 5.4. The underlying finite derivative jet at a dyadic point is baseline material in the repository. The new content is its exact re-expression as a hierarchy of weighted half-integer spectral moments and the resulting denominator detector.

6 An exact beyond-all-orders spectral remainder

6.1 Cancellation of every cosine Taylor coefficient

For $M \geq 0$ define

$$\rho_M(y) := 1 - \cos y - \sum_{m=1}^M (-1)^{m+1} \frac{y^{2m}}{(2m)!}, \quad (31)$$

with the sum empty when $M = 0$.

Theorem 6.1 (Exact all-order spectral remainder). *For every $x \in [0, 1]$ and every integer $M \geq 0$,*

$$F(x) = \sum_{k=0}^{\infty} a_k (1 - \cos(\pi o_k x)), \quad (32)$$

and, more strongly,

$$F(x) = \sum_{k=0}^{\infty} \varepsilon_k w_k \rho_M(\pi o_k x). \quad (33)$$

The series in (33) converges absolutely for every fixed M and uniformly for $x \in [0, 1]$.

Proof. For $0 \leq x \leq 1$, (2) gives $F(x) = 1 - \text{up}(x)$. Subtract (16) from 1 and use $\sum_k a_k = 1/2$ to obtain (32). Subtracting the first M nonconstant Taylor terms of $1 - \cos y$ changes the right side by

$$\sum_{m=1}^M \frac{(-1)^{m+1} (\pi x)^{2m}}{(2m)!} \sum_{k=0}^{\infty} a_k o_k^{2m},$$

which is zero by [theorem 4.1](#). Rapid decay of a_k dominates the polynomial growth of $\rho_M(\pi o_k x)$ uniformly on $[0, 1]$. $\square$

Remark 6.2 (A spectral explanation of flatness). For a fixed individual frequency, $1 - \cos(\pi o_k x)$ has an ordinary even Taylor series at zero. Formula (33) says that after the frequencies are summed with their exact Thue–Morse sinc weights, *every* algebraic Taylor coefficient disappears. The endpoint value survives only as a collective remainder beyond all finite algebraic orders. This is stronger than checking $F^{(m)}(0) = 0$ one derivative at a time: the same exact identity holds after subtraction at any prescribed order M .

6.2 The direct Lambert– W bridge

Corollary 6.3 (Thue–Morse–sinc–Lambert bridge). *Let $M \geq 0$ be fixed, $L = \log 2$, and $W = W_{-1}(-Lx)$. As $x \downarrow 0$,*

$$\sum_{k=0}^{\infty} \varepsilon_k w_k \rho_M(\pi o_k x) = \frac{e^{\Psi(-W/L)}}{2^{7/12} \sqrt{\pi x}} \exp\left(\frac{\mathfrak{g} - W - \frac{1}{2}W^2}{L}\right) \times (1 + \mathcal{O}(\lambda(x)^{-1})). \quad (34)$$

The left side is independent of M , although its individual summands are not.

Proof. The left side equals $F(x)$ exactly by [theorem 6.1](#). Apply the imported endpoint theorem (12). $\square$

Remark 6.4 (What has and has not been proved). Equation (34) is a rigorous bridge, but it does not yet identify a sharply localized set of indices k that carries the asymptotic. The natural lower-Lambert scale $r(x) = \lambda(x)/x$ strongly suggests such a localization problem. Establishing it requires estimates that preserve the massive cancellations in (33); termwise absolute values would destroy the phenomenon. We formulate this as an open problem in [section 15](#).

7 The exact dyadic alias transform

7.1 The abstract cyclic aliasing foundation

The reusable summation mechanism underlying this section is now formal. In its most general convergence form, if a summable family $a : B \rightarrow E$ is multiplied by scalars that factor through a finite label type, the twisted family remains summable. This is valid in every complete uniform additive group with a continuous distributive scalar action; no norm is required.

For a summable bilateral family $a : \mathbb{Z} \rightarrow E$, define

$$\mathcal{R}_s(a) := \sum_{\substack{\ell \in \mathbb{Z} \\ \bar{\ell} = s}} a(\ell).$$

If E is also Hausdorff, then for every natural modulus, including zero, the residue family $s \mapsto \mathcal{R}_s(a)$ has sum $\sum_{\ell \in \mathbb{Z}} a(\ell)$. For a positive modulus M and a complete Hausdorff uniform complex module, put

$$\mathcal{A}_k(a) := \sum_{\ell \in \mathbb{Z}} \chi_M(-\bar{\ell}k) a(\ell).$$

Every twisted series is summable, and the unnormalized DFT satisfies

$$\mathcal{D}_M(\mathcal{R}_\bullet(a))(k) = \mathcal{A}_k(a), \quad \mathcal{R}_s(a) = \frac{1}{M} \sum_{k \in \mathbb{Z}/M\mathbb{Z}} \chi_M(ks) \mathcal{A}_k(a).$$

Lean status. The finite-range engine is `Fabius.summable_finite_range_smul`. The residue and Fourier sums are `Fabius.intResidueTsum` and `Fabius.intFourierTsum`; the remaining statements are `Fabius.intFourierSummand_summable`, `Fabius.hasSum_intResidueTsum`, `Fabius.sum_intResidueTsum`, `Fabius.dft_intResidueTsum`, and `Fabius.intResidueTsum_eq_dft_average`, all in `SummableCyclicAlias.lean`. This foundation does not by itself prove (36): that specialization still requires inserting the absolutely convergent Fourier series of P and identifying its finite DFT with the sampled values $A_{n,r}$.

7.2 Folded coefficients

Fix $n \geq 1$ and put $N = 2^n$. For every integer r , define the length- $2N$ folded Fourier coefficient

$$A_{n,r} := \frac{1}{N} \sum_{j=0}^{2N-1} P\left(\frac{j}{N}\right) e^{-\pi i r j / N}. \quad (35)$$

It depends only on $r \bmod 2N$ and is real because P is real and even. The nontrivial coefficients are represented by odd r with $1 \leq r < N$.

Theorem 7.1 (Exact half-integer alias identity). *For every admissible n, r ,*

$$A_{n,r} = \sum_{q \in \mathbb{Z}} \Phi\left(qN + \frac{r}{2}\right). \quad (36)$$

The series converges absolutely. Moreover,

$$A_{n,r} = \frac{1}{N} \left[1 + 2 \sum_{j=1}^{N-1} \text{up}\left(\frac{j}{N}\right) \cos\left(\frac{\pi r j}{N}\right) \right] \quad (37)$$

$$= \frac{1}{N} \left[1 - 2 \sum_{j=1}^{N-1} F\left(\frac{j}{N}\right) \cos\left(\frac{\pi r j}{N}\right) \right]. \quad (38)$$

Thus a finite discrete cosine transform of exact dyadic Fabius values equals a complete folded alias class of the infinite sinc product.

Proof. Write the complex Fourier expansion of P as

$$P(t) = \sum_{m \in \mathbb{Z}} c_m e^{\pi i m t}, \quad c_m = \frac{1}{2} \Phi(m/2).$$

Substitute it in (35). Absolute convergence permits interchange, and the finite character sum is $2N$ when $m \equiv r \pmod{2N}$ and zero otherwise. Therefore

$$A_{n,r} = 2 \sum_{m \equiv r \pmod{2N}} c_m = \sum_{q \in \mathbb{Z}} \Phi\left(\frac{r + 2Nq}{2}\right),$$

which is (36). Absolute convergence follows from rapid decay.

Pair the terms j and $2N - j$ in (35). The endpoint $j = N$ contributes zero because $P(1) = \text{up}(1) = 0$, while $P(0) = 1$; this gives (37). On $0 \leq j \leq N$, $\text{up}(j/N) = 1 - F(j/N)$. Since r is odd,

$$\sum_{j=1}^{N-1} \cos(\pi r j / N) = 0,$$

which yields (38). □

7.3 Exact inverse transform

Theorem 7.2 (Dyadic spectral reconstruction). *For every integer $0 \leq j \leq 2N - 1$,*

$$P\left(\frac{j}{N}\right) = \frac{1}{2} + \sum_{\substack{1 \leq r < N \\ r \text{ odd}}} A_{n,r} \cos\left(\frac{\pi r j}{N}\right). \quad (39)$$

In particular, all dyadic values of up at denominator N are reconstructed from the $N/2$ folded half-integer classes.

Proof. The normalization in (35) is twice the standard length- $2N$ DFT normalization. Its inverse is therefore

$$P(j/N) = \frac{1}{2} \sum_{r=0}^{2N-1} A_{n,r} e^{\pi i r j / N}.$$

For $r = 0$, the alias sum contains $\Phi(0) = 1$ and only zero-valued nonzero integer samples, so $A_{n,0} = 1$. For nonzero even r , every alias argument is a nonzero integer and $A_{n,r} = 0$. Finally $A_{n,2N-r} = A_{n,r}$. Pairing the odd frequencies gives (39). □

Corollary 7.3 (Exact discrete energy identity). *One has*

$$\sum_{\substack{1 \leq r < N \\ r \text{ odd}}} A_{n,r}^2 = \frac{1}{N} \left[1 + 2 \sum_{j=1}^{N-1} \text{up}\left(\frac{j}{N}\right)^2 \right] - \frac{1}{2}. \quad (40)$$

The right side is rational.

Proof. Apply finite Parseval to (39). The constant mode contributes $1/4$, and each nonconstant cosine has mean square $1/2$ on the length- $2N$ grid. Use $P(0) = 1$, $P(1) = 0$, and symmetry of the remaining samples. □

Remark 7.4 (Continuous limit). As $n \rightarrow \infty$, the right side of (40) tends

$$2 \int_0^1 \text{up}(t)^2 dt - \frac{1}{2} = \sum_{k=0}^{\infty} a_k^2,$$

which is also ordinary Fourier Parseval for (16). The finite identity is stronger: before taking a limit it expresses the complete Galois-square trace of the folded spectrum using rational dyadic samples.

8 Certified superexponential recovery

8.1 A uniform product estimate

Lemma 8.1 (Selected-factor bound). *For every integer $K \geq 1$ and real $x \neq 0$,*

$$|\Phi(x)| \leq C_K |x|^{-K}, \quad C_K := \frac{2^{K(K-1)/2}}{\pi^K}. \quad (41)$$

Proof. For $0 \leq j < K$,

$$|\operatorname{sinc}(x/2^j)| \leq \frac{2^j}{\pi|x|}.$$

Multiply these K inequalities and bound every remaining sinc factor by one.

This estimate is now machine-checked in the stronger total form

$$(\pi|x|)^K |\Phi(x)| = 2^{\binom{K}{2}} \left(\prod_{j < K} |\sin(\pi x/2^j)| \right) |\Phi(x/2^K)| \leq 2^{\binom{K}{2}},$$

which includes both $x = 0$ and $K = 0$ before division. The identity, total bound, and divided corollary are `norm_rvachevFourierProduct_prefix_cross`, `norm_rvachevFourierProduct_prefix_cross_le`, and `norm_rvachevFourierProduct_le_prefix_div` in `SincPrefixBound.lean`. $\square$

Theorem 8.2 (Uniform aliasing error). *Let $K > 1$ be an integer. For $N = 2^n$ and odd $1 \leq r < N$,*

$$|A_{n,r} - \Phi(r/2)| \leq 2C_K N^{-K} (2^K - 1) \zeta(K). \quad (42)$$

Choosing $K = n \geq 2$ gives the explicit uniform estimate

$$|A_{n,r} - \Phi(r/2)| \leq E_n := \frac{2(1 - 2^{-n})\zeta(n)}{\pi^n} 2^{-n(n-1)/2}. \quad (43)$$

In particular $E_n = 2^{-n^2/2 + \mathcal{O}(n \log n)}$, uniformly over all odd $r < N$.

Proof. Remove the $q = 0$ term from (36). For $q \geq 1$ and $0 < r < N$,

$$|qN \pm r/2| \geq (q - 1/2)N.$$

Hence lemma 8.1 gives

$$\begin{aligned} |A_{n,r} - \Phi(r/2)| &\leq 2C_K N^{-K} \sum_{q=1}^{\infty} (q - 1/2)^{-K} \\ &= 2C_K N^{-K} (2^K - 1) \zeta(K), \end{aligned}$$

using $\sum_{q \geq 1} (q - 1/2)^{-K} = (2^K - 1) \zeta(K)$. Substitute $N = 2^n$ and $K = n$; collecting powers of two gives (43). $\square$

Remark 8.3 (Why the rate is so fast). At level n , the unwanted aliases begin a distance comparable with 2^n from the target. The same sinc product permits an n -factor power bound, so the alias tail behaves like $(2^n)^{-n}$ multiplied by the factor $2^{n(n-1)/2}$ accumulated from dyadic scaling. The net exponent is $-n(n-1)/2$.

8.2 Numerical convergence for the leading amplitude

The distinguished coefficient $A_{n,1}$ converges to

$$h := \Phi(1/2) = 0.553771275888810095344217416344548881284886642554 \dots \quad (44)$$

The finite values in [table 3](#) were computed from exact rational $F(j/2^n)$ values, with only the final cyclotomic evaluation performed at high precision.

n	$A_{n,1}$	actual error	rigorous E_n
2	0.5544487530108746285614747	$6.7748 \cdot 10^{-4}$	$1.2500 \cdot 10^{-1}$
3	0.5537613501982070710846451	$9.9257 \cdot 10^{-6}$	$8.4805 \cdot 10^{-3}$
4	0.5537712795003417162450708	$3.6115 \cdot 10^{-9}$	$3.2552 \cdot 10^{-4}$
5	0.5537712758798224271695171	$8.9877 \cdot 10^{-12}$	$6.4112 \cdot 10^{-6}$
6	0.5537712758888101658815125	$7.0537 \cdot 10^{-17}$	$6.3578 \cdot 10^{-8}$
7	0.5537712758888100953109457	$3.3272 \cdot 10^{-20}$	$3.1590 \cdot 10^{-10}$
8	0.5537712758888100953442174	$5.0949 \cdot 10^{-27}$	$7.8534 \cdot 10^{-13}$
9	0.5537712758888100953442174	$4.8386 \cdot 10^{-31}$	$9.7639 \cdot 10^{-16}$
10	0.5537712758888100953442174	$1.3837 \cdot 10^{-39}$	$6.0700 \cdot 10^{-19}$

Table 3: Superexponential recovery of the first half-integer amplitude. The bound is deliberately elementary and uniform in r , so it is not expected to track the much smaller observed error closely.

9 Finite q -binomial–cyclotomic approximants

9.1 Explicit substitution

The alias transform is finite, while every dyadic value in it has the finite Gaussian-binomial form (8). Their composition is the promised bridge between the two coefficient webs.

Theorem 9.1 (q -binomial half-integer approximant). *Let $N = 2^n$ and let r be odd with $1 \leq r < N$. Then*

$$A_{n,r} = \frac{1}{N} - \frac{2}{N 2^{n^2} (1/2; 1/2)_n} \sum_{j=1}^{N-1} \cos\left(\frac{\pi r j}{N}\right) \times \sum_{k=0}^n \frac{[n]_{1/2}}{2^{k(k-1)} (n+k)!} \sum_{s=0}^{j2^k-1} \varepsilon_s (s - j2^k)^{n+k}. \quad (45)$$

For every fixed odd r , the right side converges to $\Phi(r/2)$ as $n \rightarrow \infty$ through levels with $r < 2^n$, and the error is at most E_n .

Proof. Insert (8) with $m = j$ and $c = 0$ into (38). The convergence statement is [theorem 8.2](#). $\square$

Remark 9.2 (Three structures in one finite expression). Formula (45) simultaneously contains

1. Gaussian binomial coefficients at $q = 1/2$;
2. finite Thue–Morse power sums;
3. real 2^{n+1} -st cyclotomic numbers $\cos(\pi r j / 2^n)$.

No infinite product or numerical quadrature is needed to construct the approximant. The infinite sinc product enters only as the certified limit.

9.2 A finite algebraic tower

Let

$$\zeta_{2N} = e^{\pi i/N}, \quad K_n := \mathbb{Q}(\zeta_{2N} + \zeta_{2N}^{-1}). \quad (46)$$

Since every $F(j/N)$ is rational,

$$A_{n,r} \in K_n. \quad (47)$$

The fields form the nested real dyadic cyclotomic tower $K_1 \subset K_2 \subset \dots$. The approximants therefore approach the transcendence-unknown numbers $\Phi(r/2)$ through an explicit tower of exact algebraic numbers with a superexponential analytic certificate.

Remark 9.3. The phrase “transcendence-unknown” is intentional. No irrationality or transcendence assertion for $h = \Phi(1/2)$ is proved here or in the reviewed corpus. The algebraic approximants are useful precisely because they separate exact finite arithmetic from the unresolved nature of the limiting constants.

10 Cyclotomic orbit polynomials and rational traces

10.1 The Galois orbit

The real cyclotomic Galois group is

$$G_n \simeq (\mathbb{Z}/2N\mathbb{Z})^\times / \{\pm 1\}. \quad (48)$$

Its elements may be represented by odd integers u modulo $2N$, with u and $-u$ identified. The representatives $1 \leq r < N$, r odd, form a complete set.

Proposition 10.1 (Galois action). *If $\sigma_u(\zeta_{2N}) = \zeta_{2N}^u$, then*

$$\sigma_u(A_{n,r}) = A_{n,r'}, \quad (49)$$

where r' is the unique odd representative in $[1, N)$ congruent to $\pm ur \pmod{2N}$.

Proof. Apply σ_u to (38). It fixes the rational Fabius values and sends $\cos(\pi r j/N)$ to $\cos(\pi u r j/N)$. Reduction modulo $2N$ and evenness of cosine give the stated representative. $\square$

Definition 10.2 (Orbit polynomial and power traces). Define

$$\mathcal{P}_n(X) := \prod_{\substack{1 \leq r < N \\ r \text{ odd}}} (X - A_{n,r}) \quad (50)$$

and, for $m \geq 1$,

$$p_m(n) := \sum_{\substack{1 \leq r < N \\ r \text{ odd}}} A_{n,r}^m. \quad (51)$$

The polynomial is called an orbit polynomial rather than a minimal polynomial: if the stabilizer of $A_{n,1}$ is nontrivial, repeated conjugates may occur.

Theorem 10.3 (Rational orbit polynomial). *For every $n \geq 1$,*

$$\mathcal{P}_n(X) \in \mathbb{Q}[X], \quad p_m(n) \in \mathbb{Q} \quad (m \geq 1). \quad (52)$$

The coefficients of $\mathcal{P}_n$ are recovered from the $p_m(n)$ by Newton’s identities.

Proof. The multiset of roots is invariant under the full Galois group by [proposition 10.1](#); hence every elementary symmetric function is fixed by G_n and lies in $\mathbb{Q}$. The same argument applies to power sums. Newton’s identities are purely rational algebra. $\square$

10.2 Ramanujan-filtered trace formula

Put

$$P_j := P(j/N), \quad 0 \leq j < 2N. \quad (53)$$

Every P_j is rational, since it is a dyadic value of ω .

Theorem 10.4 (Exact rational power traces). *For every $m \geq 1$,*

$$p_m(n) = \frac{1}{2} N^{1-m} \sum_{\substack{0 \leq j_1, \dots, j_m < 2N \\ N \mid j_1 + \dots + j_m}} (-1)^{(j_1 + \dots + j_m)/N} \prod_{\ell=1}^m P_{j_\ell}. \quad (54)$$

In particular,

$$p_1(n) = \frac{1}{2} \quad (n \geq 1). \quad (55)$$

For $m = 2$, (54) reduces to the discrete energy identity (40).

Proof. Expand $A_{n,r}^m$ from (35) and sum over the $N/2$ odd representatives. Since $A_{n,r} = A_{n,2N-r}$, the representative sum is one half of the sum over all odd residues modulo $2N$. Thus

$$p_m(n) = \frac{1}{2N^m} \sum_{0 \leq j_1, \dots, j_m < 2N} \left(\prod_{\ell=1}^m P_{j_\ell} \right) c_{2N}(j_1 + \dots + j_m),$$

where

$$c_{2N}(S) = \sum_{\substack{r \bmod 2N \\ r \text{ odd}}} e^{-\pi i r S / N}$$

is the Ramanujan sum for the power-of-two modulus $2N$. Its elementary value is

$$c_{2N}(S) = \begin{cases} N(-1)^{S/N}, & N \mid S, \\ 0, & N \nmid S. \end{cases} \quad (56)$$

Substitution gives (54). For $m = 1$, the only indices with $N \mid j$ are $j = 0, N$, so $p_1(n) = \frac{1}{2}(P_0 - P_N) = 1/2$. $\square$

Remark 10.5 (A trace that does not disperse). The orbit degree is $N/2 \rightarrow \infty$, yet the unnormalized first trace remains exactly $1/2$. Most conjugates become tiny; a sparse collection of low-frequency conjugates carries the persistent trace. This is quantified below.

10.3 Orbit polynomials at the first levels

At $n = 1$ there is one coefficient, $A_{1,1} = 1/2$, so

$$\mathcal{P}_1(X) = X - \frac{1}{2}. \quad (57)$$

At $n = 2$,

$$A_{2,1} = \frac{1}{4} + \frac{31\sqrt{2}}{144}, \quad A_{2,3} = \frac{1}{4} - \frac{31\sqrt{2}}{144}, \quad (58)$$

and

$$\mathcal{P}_2(X) = X^2 - \frac{1}{2}X - \frac{313}{10368}. \quad (59)$$

At $n = 3$, direct use of (54) and Newton identities gives

$$\mathcal{P}_3(X) = X^4 - \frac{1}{2}X^3 - \frac{1627}{55296}X^2 - \frac{4603}{23887872}X + \frac{32257}{220150628352}. \quad (60)$$

The corresponding rational power traces are

$$p_1(3) = \frac{1}{2}, \quad p_2(3) = \frac{8539}{27648}, \quad (61)$$

$$p_3(3) = \frac{1351363}{7962624}, \quad p_4(3) = \frac{5175727235}{55037657088}. \quad (62)$$

The four conjugates are approximately

$$\begin{aligned} &0.553761350198207071, & -0.046098883415321902, \\ &-0.008349869595552727, & 0.000687402812667557. \end{aligned} \quad (63)$$

They already resemble the first four half-integer amplitudes, but each includes all aliases in its congruence class.

At $n = 4$ the eight conjugates are approximately

$$\begin{aligned} &0.553771279500341716, & -0.046202534866953151, \\ &-0.008657957107189789, & 0.001046027211381176, \\ &-0.000358624398713618, & 0.000308087511637062, \\ &0.000103651451631249, & -0.000009929302134645. \end{aligned}$$

Their sum is exactly $1/2$.

10.4 Rational traces converging to nonlinear sinc moments

Define the nonlinear half-integer spectral sums

$$\mathfrak{s}_m := \sum_{k=0}^{\infty} a_k^m \quad (m \geq 1). \quad (64)$$

They converge absolutely. The case $m = 1$ is $\mathfrak{s}_1 = 1/2$; for $m = 2$,

$$\mathfrak{s}_2 = 2 \int_0^1 \text{up}(t)^2 dt - \frac{1}{2}. \quad (65)$$

Theorem 10.6 (Cyclotomic trace limits). *For every fixed integer $m \geq 1$,*

$$\boxed{\lim_{n \rightarrow \infty} p_m(n) = \mathfrak{s}_m}. \quad (66)$$

For $m = 1$ the equality is exact at every level. For fixed $m \geq 2$, the error splits into

$$|p_m(n) - \mathfrak{s}_m| \leq \frac{Nm}{2}(1 + E_n)^{m-1}E_n + \sum_{k \geq N/2} |a_k|^m, \quad (67)$$

where both terms decay faster than every exponential in n after applying an n -factor sinc bound to the tail.

Proof. Index the admissible residues by $r = 2k + 1$, $0 \leq k < N/2$. Then [theorem 8.2](#) gives $|A_{n,2k+1} - a_k| \leq E_n$. Since $|a_k| \leq 1$ and $E_n \leq 1/8$ for $n \geq 2$, the mean-value inequality for $x \mapsto x^m$ gives

$$|A_{n,2k+1}^m - a_k^m| \leq m(1 + E_n)^{m-1} E_n.$$

Summing over $k < N/2$ and adding the omitted tail proves (67). The first term is superexponential because $NE_n = 2^{-n^2/2 + \mathcal{O}(n)}$. For the tail, use [lemma 8.1](#) with $K = n$:

$$|a_k| \leq C_n(k + 1/2)^{-n}.$$

The resulting integral bound is

$$\mathcal{O}(C_n^m (N/2)^{1-nm}) = 2^{-mn^2/2 + \mathcal{O}(n)},$$

where the implied constant may depend on the fixed exponent m . □

Corollary 10.7 (Algebraic approximation to spectral norms). *Every $\mathfrak{s}_m$ is a limit of explicitly computable rational numbers $p_m(n)$ whose defining algebraic conjugates are given by the finite q -binomial formula (45). In particular, $2 \int_0^1 \text{up}^2 - 1/2$ admits a superexponentially convergent rational sequence coming from cyclotomic Galois square traces.*

10.5 Collapse of the normalized Galois orbit

Let

$$\mu_n := \frac{2}{N} \sum_{\substack{1 \leq r < N \\ r \text{ odd}}} \delta_{A_{n,r}}. \quad (68)$$

This is a probability measure on $\mathbb{R}$.

Theorem 10.8 (Galois-orbit collapse). *The measures μ_n converge weakly to the point mass δ_0 . More quantitatively,*

$$\int x^2 d\mu_n(x) = \frac{2}{N} p_2(n) \leq \frac{3}{N}. \quad (69)$$

Nevertheless the unnormalized trace remains $p_1(n) = 1/2$.

Proof. From (40) and $0 \leq \text{up} \leq 1$,

$$p_2(n) \leq \frac{1}{N}(1 + 2(N-1)) - \frac{1}{2} < \frac{3}{2}.$$

This proves (69). Chebyshev's inequality shows that the proportion of conjugates outside any fixed neighborhood of zero tends to zero, hence $\mu_n \Rightarrow \delta_0$. □

Remark 10.9. This is an unusual arithmetic picture: the degree of the orbit grows, the normalized orbit collapses to zero, but its unnormalized first trace remains fixed. The low-index half-integer amplitudes survive as a vanishing proportion of the conjugates and carry the entire trace in the limit.

11 A finite Thue–Morse sign carrier for every alias class

11.1 Exact factorization of a folded class

Let $N = 2^n$, let r be odd with $1 \leq r < N$, and put

$$\alpha := \frac{r}{2N} \in (0, 1/2). \quad (70)$$

Define the finite prefactor

$$\Gamma_{n,r} := \frac{\prod_{h=1}^n \sin(\pi r/2^h)}{\pi^n 2^{n(n+1)/2}}. \quad (71)$$

Theorem 11.1 (Alias-class factorization). *For every $q \in \mathbb{Z}$,*

$$\Phi\left(qN + \frac{r}{2}\right) = \Gamma_{n,r} \frac{\Phi(q + \alpha)}{(q + \alpha)^n}. \quad (72)$$

Consequently

$$A_{n,r} = \Gamma_{n,r} \sum_{q \in \mathbb{Z}} \frac{\Phi(q + \alpha)}{(q + \alpha)^n}. \quad (73)$$

The shifted sum converges absolutely.

Proof. Split the product (5) after the first n factors. Since $qN + r/2 = N(q + \alpha)$, the tail is

$$\prod_{j=n}^{\infty} \operatorname{sinc}\left(\frac{qN + r/2}{2^j}\right) = \Phi(q + \alpha).$$

For $0 \leq j < n$,

$$\sin\left(\pi \frac{qN + r/2}{2^j}\right) = \sin\left(\pi q 2^{n-j} + \frac{\pi r}{2^{j+1}}\right) = \sin\left(\frac{\pi r}{2^{j+1}}\right),$$

because $q 2^{n-j}$ is even. The denominators of the first n sinc factors contribute

$$\prod_{j=0}^{n-1} \frac{2^j}{\pi N(q + \alpha)} = \frac{1}{\pi^n 2^{n(n+1)/2} (q + \alpha)^n}.$$

Multiplication proves (72); summing over q and using (36) gives (73). $\square$

11.2 The finite Thue–Morse polynomial

Let

$$T_n(z) := \prod_{\ell=0}^{n-1} (1 - z^{2^\ell}) = \sum_{m=0}^{2^n-1} \varepsilon_m z^m. \quad (74)$$

Proposition 11.2 (Root-of-unity magnitude and sign). *For odd $1 \leq r < N$,*

$$\left| T_n\left(e^{2\pi i r/N}\right) \right| = 2^n \left| \prod_{h=1}^n \sin(\pi r/2^h) \right|, \quad (75)$$

$$\operatorname{sgn} \Gamma_{n,r} = (-1)^{\sum_{h=1}^n \lfloor r/2^h \rfloor} = \varepsilon_{(r-1)/2}. \quad (76)$$

Thus the finite prefactor in (73) carries exactly the Thue–Morse sign attached to the target amplitude $\Phi(r/2)$.

Proof. For the magnitude, use $|1 - e^{2\pi i x}| = 2|\sin \pi x|$ in each factor of (74) and reverse the order of scales. For the sign, $\sin(\pi x)$ has sign $(-1)^{\lfloor x \rfloor}$ away from integers. Since $r < N$, all floors beyond $h = n$ vanish, and

$$\sum_{h=1}^n \left\lfloor \frac{r}{2^h} \right\rfloor = r - s_2(r).$$

Write $r = 2c + 1$. Then $s_2(r) = s_2(c) + 1$, so $r - s_2(r) \equiv s_2(c) \pmod{2}$. Hence the sign is ε_c with $c = (r - 1)/2$. $\square$

Remark 11.3. The earlier arithmetic-ray report uses root-of-unity values of finite Thue–Morse polynomials to control products along rational doubling orbits. The mechanism here is different: a finite root-of-unity product is extracted from a *Poisson alias class* of the sinc product. The same automatic sign word therefore governs both real-axis intervals and finite dyadic spectral folding.

11.3 A positivity conjecture

Computations through $n = 7$ show

$$\operatorname{sgn} A_{n,r} = \varepsilon_{(r-1)/2} \quad (77)$$

for every admissible r . By (76), this is equivalent to positivity of the residual shifted sum.

Conjecture 11.4 (Shifted spectral positivity). *For every $n \geq 1$ and odd $1 \leq r < 2^n$, with $\alpha = r/2^{n+1}$,*

$$\sum_{q \in \mathbb{Z}} \frac{\Phi(q + \alpha)}{(q + \alpha)^n} > 0. \quad (78)$$

Equivalently, every folded coefficient has the exact Thue–Morse sign (77).

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Remark 11.5 (Why the conjecture is nontrivial). The summands in (78) have mixed signs. Neither the positivity of up nor the elementary alias error bound proves the claim, because for residues near N the target amplitude can be much smaller than the uniform alias bound. A proof may require a quasi-periodic Green-kernel representation or a total-positivity property of the dyadic sample sequence.

12 Certified arithmetic of integer-zero jets

12.1 The imported local formula

Let $m \neq 0$ be an integer and write

$$m = 2^v q, \quad q \text{ odd}, \quad d = v + 1. \quad (79)$$

The spectral determinant report proves the exact leading-jet identity

$$\Phi^{(d)}(m) = -\frac{d!}{m^d} \Phi(q/2). \quad (80)$$

It also gives every higher coefficient, but the leading term already shows why the half-integer amplitudes are arithmetically central: every nonzero integer zero reduces to one odd half-integer value.

12.2 Finite exact approximants with a rigorous error

Theorem 12.1 (q -binomial approximation of every leading zero jet). *Let $m = 2^v q \neq 0$ with q odd and $d = v + 1$. Choose $n \geq 2$ with $2^n > |q|$, and put $r = |q|$. Define*

$$J_{m,n} := -\frac{d!}{m^d} A_{n,r}. \quad (81)$$

Then $J_{m,n}$ is an explicitly computable element of the real dyadic cyclotomic field K_n , given by the finite q -binomial expression (45), and

$$|\Phi^{(d)}(m) - J_{m,n}| \leq \frac{d!}{|m|^d} E_n. \quad (82)$$

Thus every leading integer-zero jet has a superexponentially convergent sequence of exact algebraic approximants.

Proof. Evenness gives $\Phi(q/2) = \Phi(r/2)$. Combine the imported identity (80) with theorem 8.2 and multiply the error by $d!/|m|^d$. $\square$

Example 12.2 (The zero at $m = 12$). Here $12 = 2^2 \cdot 3$, so $d = 3$ and

$$\Phi^{(3)}(12) = -\frac{6}{12^3} \Phi(3/2) = \frac{-\Phi(3/2)}{288}.$$

Using the level-4 folded coefficient $A_{4,3} = -0.04620253486695315055 \dots$ gives

$$J_{12,4} = 0.000160425468288031773 \dots,$$

while the exact target is

$$\Phi^{(3)}(12) = 0.000160424649771898309 \dots$$

The rigorous error follows immediately from (82). Increasing n replaces the displayed algebraic number by a much sharper one without changing the proof architecture.

Warning 12.3. The final decimal in the target value in example 12.2 is included only as a numerical orientation. The certified content is the exact formula and the bound (82); the report's reproducibility code computes as many digits as desired.

13 Computational verification

13.1 Independent checks performed

The numerical work was designed to check identities at several logically independent interfaces rather than merely reproduce one formula in two notations. The following were evaluated using exact rational arithmetic whenever possible and 80–100 decimal digits for the final transcendental operations.

1. The moment recurrence for c_k and the closed moment–Thue–Morse formula for $F(a/2^n)$, including $F(1/4) = 5/72$, $F(1/8) = 1/288$, $F(3/8) = 73/288$, and $F(5/16) = 305857/2073600$.
2. Direct sinc-product values of a_k against the cosine reconstruction of up.
3. The weighted moment residuals $\sum_{k < K} a_k o_k^{2m}$ for increasing K , together with rapid-decay tail bounds.
4. The exact DCT formula (38) against direct alias summation (36).
5. The convergence table table 3 and the rigorous bound (43).
6. The Galois conjugates at levels $n = 2, 3, 4$, their rational power traces, and the orbit polynomials (59)–(60).
7. The single-alias factorization (72) for positive and negative alias indices.
8. The observed sign law (77) for every odd residue at levels $1 \leq n \leq 7$.
9. The leading zero-jet approximation (82) for integers of several different dyadic valuations.

13.2 Exact arithmetic workflow

The most stable exact path uses the even moments of up. Set $c_0 = 1$ and compute

$$(2m+1)(2^{2m}-1)c_m = \sum_{j=0}^{m-1} \binom{2m+1}{2j} c_j. \quad (83)$$

Then the repository's closed formula is

$$F\left(\frac{a}{2^n}\right) = \frac{2^{-\binom{n+1}{2}}}{n!} \sum_{h=0}^{a-1} \varepsilon_h \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k} (2a-2h-1)^{n-2k} c_k. \quad (84)$$

Every quantity in (84) is rational. It is then inserted into (38); exact algebraic simplification is needed only for the cosine values.

13.3 A compact reproducibility listing

Listing 1: Core exact computation used for the tables.

```
from fractions import Fraction
from math import comb, factorial

def tm_sign(k: int) -> int:
    return -1 if k.bit_count() % 2 else 1

def up_even_moments(max_k: int) -> list[Fraction]:
    c = [Fraction(1)]
    for m in range(1, max_k + 1):
        numerator = sum(
            Fraction(comb(2*m + 1, 2*j)) * c[j]
            for j in range(m)
        )
        denominator = (2*m + 1) * (2**(2*m) - 1)
        c.append(numerator / denominator)
    return c

def fabius_dyadic(a: int, n: int, c: list[Fraction]) -> Fraction:
    total = Fraction(0)
    for h in range(a):
        inner = sum(
            Fraction(comb(n, 2*k) * (2*a - 2*h - 1)**(n - 2*k))
            * c[k]
            for k in range(n//2 + 1)
        )
        total += tm_sign(h) * inner
    return total / Fraction(2**(n*(n + 1)//2) * factorial(n))

# A_{n,r} = 1/N - (2/N) sum_j F(j/N) cos(pi*r*j/N).
# Use exact Fraction values for F and a high-precision cyclotomic backend
# (or symbolic radicals for small n) for the cosine evaluation.
```

Remark 13.1 (Reproducibility versus proof). The computations are checks, not substitutes for the proofs. Every central identity in the report is established symbolically before numerical evidence is invoked. The code is included to make constants, orbit polynomials, and error tables independently reproducible.

14 Lean formalization roadmap

The new theorem package separates naturally into finite algebra, elementary analysis, and imported high-level asymptotics. A low-risk formalization order is as follows.

14.1 Layer 1: finite Fourier algebra

Formalize first the length- $2N$ DFT identities on a finite cyclic group:

1. orthogonality of characters modulo $2N$;
2. vanishing of all nonzero even folded modes;
3. the inverse cosine transform (39);
4. finite Parseval and (40);
5. the power-of-two Ramanujan sum (56);
6. the rational trace formula (54).

This layer uses only finite sums, rational dyadic values already available in the project, roots of unity, and elementary complex algebra.

14.2 Layer 2: Fourier-series extraction

Use the established rapid decay of $\widehat{u_P}$ to formalize:

1. the periodization P and its complex Fourier coefficients;
2. the reduction to odd modes and the cosine series (16);
3. termwise differentiation to every order;
4. the weighted Prouhet identities and polynomial quadrature;
5. the derivative filtration (27)–(29);
6. the exact spectral remainder (33).

The main analytic input is a reusable theorem that a Schwartz coefficient sequence permits uniform termwise differentiation of a Fourier series.

14.3 Layer 3: alias limits and estimates

The complete-uniform-module residue decomposition and its forward and inverse cyclic DFT are formalized in `SummableCyclicAlias.lean`. What remains is to instantiate that engine with the absolutely convergent Fourier coefficients of P , identify the resulting finite transform with $A_{n,r}$, prove the selected-factor bound, and sum the shifted zeta tail. The exact remaining target statements are (36), (42), and (43). No contour integration is needed.

14.4 Layer 4: cyclotomic structure

Two implementation choices are possible.

1. Work concretely in $\mathbb{C}$ with a primitive root ζ_{2N} and prove invariance of symmetric polynomials under the explicit automorphisms.
2. Work in an abstract cyclotomic extension, use the maximal real subfield only at the interface, and obtain rationality from fixed-field lemmas.

The first route is likely shorter for the DCT formula; the second gives cleaner Galois traces. The orbit polynomial should be introduced without asserting it is minimal.

14.5 Layer 5: imported endpoint asymptotics

Once (33) is formalized, the Lambert bridge is a one-line rewrite using the project’s existing endpoint theorem. This is much easier than formalizing a new saddle analysis. A later project can address the genuinely new localization problem posed in [conjecture 15.5](#).

14.6 Indicative theorem interfaces

Listing 2: Schematic Lean theorem signatures.

```
-- Half-integer weighted Prouhet cancellation.
theorem halfIntegerSpectrum_evenMoment_zero (m : Nat) (hm : 0 < m) :
  HasSum (fun k => phi ((2*k+1 : Real) / 2) * (2*k+1)^(2*m)) 0

-- Exact folded DFT / alias identity.
theorem dyadicDCT_eq_aliasSum
  (n r : Nat) (hn : 0 < n) (hr : Odd r) (hrN : r < 2^n) :
  dyadicDCT n r = tsum (fun q : Int => phi (q * 2^n + r / 2))

-- Uniform certified recovery.
theorem dyadicDCT_sub_halfInteger_le
  (n r : Nat) (hn : 2 <= n) (hr : Odd r) (hrN : r < 2^n) :
  abs (dyadicDCT n r - phi (r / 2)) <= aliasError n

-- Rational Galois power trace.
theorem foldedPowerTrace_mem_rat
  (n m : Nat) : IsRat (sumOddResidues n (fun r => (dyadicDCT n r)^m))
```

15 Further frontier problems

15.1 Exact signs of the folded coefficients

[conjecture 11.4](#) is the most immediate problem. A proof would turn the finite dyadic DCT into a second exact occurrence of the Thue–Morse sign word, not merely an approximation to it. Useful routes include:

1. a quasi-periodic Green-kernel representation of $\sum_q \Phi(q + \alpha)/(q + \alpha)^n$;
2. variation-diminishing or total-positivity properties of dyadic samples of up;
3. a recursion in n for the quotient $A_{n,r}/\Gamma_{n,r}$;
4. an integral representation with an explicitly positive kernel.

15.2 Minimal polynomials and stabilizers

Problem 15.1. *Determine when $\mathcal{P}_n$ is the minimal polynomial of $A_{n,1}$ and when it is a proper power. More generally, classify all coincidences $A_{n,r} = A_{n,s}$ and determine the discriminant and local factorization of the orbit polynomial.*

The examples $n = 2, 3$ are irreducible over $\mathbb{Q}$. The exact Galois action makes this a finite arithmetic problem at each level, but no general primitivity criterion is presently known.

15.3 Arithmetic nature of the limiting amplitudes

Problem 15.2. *Prove irrationality, transcendence, or nontrivial algebraic independence results for*

$$\Phi(1/2), \Phi(3/2), \Phi(5/2), \dots$$

The cyclotomic approximants (45) come with unusually strong error bounds, but their algebraic degrees and heights also grow. Determine whether the approximation quality can beat the relevant Liouville-type height thresholds after sharp height estimates are included.

15.4 Spectral moments and nonlinear functionals of $\mathfrak{u}_p$

The trace limits (66) define constants $\mathfrak{s}_m = \sum_k a_k^m$. The cases $m \geq 3$ do not have an immediate ordinary Parseval interpretation.

Problem 15.3. *Find direct integral, probabilistic, or operator-theoretic representations of $\mathfrak{s}_m$ for $m \geq 3$. Determine whether the rational trace sequences $p_m(n)$ satisfy recurrences or congruences inherited from the dyadic sample array.*

15.5 A genuine transform between the two Gaussian-binomial webs

Formula (45) composes the exact dyadic-value web with a cyclotomic DCT. The spectral determinant report constructs a different family of Gaussian-binomial coefficients from finite q -Euler determinants.

Problem 15.4. *Construct an intrinsic matrix or incidence-algebra transform that takes the Gaussian-binomial coefficients governing exact Fabius dyadic values directly to the Gaussian-binomial coefficients governing the spectral q -Euler determinants. The DCT developed here supplies an explicit intermediate layer, but not yet a closed transform between the two coefficient arrays.*

15.6 Lambert spectral localization

Conjecture 15.5 (Lower-Lambert frequency localization). *Let $x \downarrow 0$ and put $r(x) = \lambda(x)/x$ as in (9). There should exist windows $\mathcal{W}_x$ centered on odd frequencies of order $r(x)$, with relative width tending to zero on the logarithmic scale, such that the exact signed sum (33) is asymptotically unchanged when restricted to $\mathcal{W}_x$, after an appropriate multiscale grouping that preserves Thue–Morse cancellation.*

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A naive termwise truncation is unlikely to work: the all-order moment identities show that cancellations across scales are essential. A successful formulation may require dyadic blocks or a Mellin transform of the signed spectral measure (21).

15.7 A two-variable transform

The corrected endpoint asymptotic is controlled by poles on the vertical lattice $2\pi i\mathbb{Z}/\log 2$. The dyadic spectral tower has the same scale lattice, and the folded DCT coefficients are finite projections of its half-integer spectrum.

Problem 15.6. *Construct a meromorphic two-variable transform whose residues simultaneously produce*

1. *the Gamma–zeta periodic endpoint function Ψ ;*
2. *the rational cyclotomic traces $p_m(n)$;*
3. *the weighted Prouhet moment annihilation;*
4. *the log-periodic heat trace of the dyadic spectral determinant.*

Such a transform would unify four presently separate manifestations of dyadic self-similarity.

16 Conclusion

The half-integer values of the Rvachev sinc product sit at a previously underdeveloped junction of the repository's main themes. They are at once the nonzero Fourier coefficients of the two-periodic up-function, the leading constants in every integer-zero jet, and a Thue–Morse-signed sequence. The present report shows that these roles are not merely parallel observations.

Flatness at the center forces the sequence to solve an infinite positive-weight Prouhet problem of every order. Dyadic sampling folds the sequence into exact finite discrete cosine transforms whose entries are rational Fabius values. The finite $q = 1/2$ Gaussian-binomial formula then turns each folded coefficient into a completely explicit q -binomial–Thue–Morse–cyclotomic number. An elementary selected-factor estimate unfolds those coefficients with a uniform superexponential error. Galois conjugacy supplies rational orbit polynomials and power traces, and those traces converge to nonlinear moments of the original half-integer spectrum. Splitting the first dyadic sinc factors reveals a finite root-of-unity Thue–Morse product as the sign carrier of every alias class. Finally, the all-order Prouhet cancellation rewrites the Fabius endpoint as an exact spectral remainder beyond every algebraic order; the repository's corrected Lambert– W theorem then evaluates that same signed remainder asymptotically.

The strongest immediate next theorem would be positivity of the residual shifted spectral sum (78). Beyond it lie arithmetic questions about the cyclotomic tower, transcendence of the limiting amplitudes, and a true lower-Lambert localization theorem. The finite identities and error bounds developed here are sufficiently elementary to support a staged Lean formalization before those deeper questions are settled.

A Corpus inventory

The same-day recursive audit snapshot counted fifty-seven TeX files: twenty-five in the living or imported documentation stratum and thirty-two frozen historical versions. The current canonical frontier source is maintained by consolidation, so capitalization and the continued presence of processed draft paths can change after the snapshot. The list below records the path relative to `Analysis/FabiusFunction/docs`. Historical items are grouped by the archive directory; filenames follow the base name shown.

A.1 Living documentation: 25 TeX files

No.	Relative path
1	<code>FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>
2	<code>Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>
3	<code>Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex</code>
4	<code>fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>
5	<code>semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex</code>
6	<code>semi-formalized-research-frontiers/drafts/Fabius_Thue_Morse_Formula_Atlas/Fabius_Thue_Morse_Formula_Atlas.tex</code>
7	<code>semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex</code>
8	<code>semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex</code>

No.	Relative path
9	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
16	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
18	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
19	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
20	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
21	papers/arXiv-1502.06738v2/Lacunary_products.tex
22	papers/arXiv-1702.05442v1/09-Function.tex
23	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24	papers/arXiv-2211.13570v1/document.tex
25	papers/ubiq15/ubiq15-corrected.tex

A.2 Frozen archive: 32 TeX files

No.	Relative path under archive/
1	early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
2	early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
3	standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
5	standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex

No.	Relative path under archive/
6	primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
7	primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
8	primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
9	primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
10	primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
11	primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
12	primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
13	primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
14	q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
15	q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
16	lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
17	lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
18	glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
19	glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
20	research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
21	research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
22	research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
23	research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
24	research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
25	research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
26	research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
27	research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
28	research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
29	research-frontiers/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex

No.	Relative path under archive/
30	research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
31	research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
32	research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Selected source map

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Part II

Arithmetic Dyadic Rays of the Rvachev Fourier Product

Abstract

The documentation corpus under `Analysis/FabiusFunction/docs` contains a mature, Lean-backed exposition of the Fabius function and Rvachev’s compactly supported up-function, a large quarantine volume of non-formalized results, a Thue–Morse formula atlas, twelve competing and synthesized studies of the Fourier decay of up, seven source-paper transcriptions, and thirty-two frozen historical TeX revisions. This report audits all fifty-seven TeX corpus items, distinguishes results already present from genuinely unfilled directions, and develops an arithmetic theory for

$$\mathcal{U}(x) := \widehat{\text{up}}(x) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^n}\right).$$

The central result is an exact decay law on every non-dyadic rational ray $x_N = 2^N(1 + a/q)$. If q is odd, $d = \text{ord}_q(2)$, and

$$A_{a,q} = \left| \prod_{j=0}^{d-1} (1 - \zeta_q^{a2^j}) \right|,$$

then

$$\log |\mathcal{U}(x_N)| = -\frac{(\log x_N)^2}{2 \log 2} - \kappa_{a,q} \log x_N + \Omega_{a,q}(N), \quad \kappa_{a,q} = \frac{1}{2} + \frac{\log(2\pi)}{\log 2} - \frac{\log A_{a,q}}{d \log 2},$$

where $\Omega_{a,q}$ is explicitly periodic in N . The multiplier is a relative cyclotomic norm of a Thue–Morse polynomial value. We construct its integral orbit polynomial by a resultant, express its power traces through Thue–Morse signs and Ramanujan sums, and prove an exact mean-exponent law over Galois orbits. A short periodic-orbit argument gives the sharp bound $A_{a,q}^{1/d} \leq \sqrt{3}$, with equality only for the orbit $\{1/3, 2/3\}$.

For the denominators $2^d - 1$ and $2^d + 1$ we obtain a refined two-sided expansion. Both exponents grow as $d/2 + 1 - (\log h)/(d \log 2)$, where $h = \prod_{k \geq 1} \text{sinc}(\pi/2^k) = -\mathcal{U}'(1)$, while the first exponentially small corrections have opposite signs. Combining this with $\mathfrak{M}_d = \prod_{j=1}^d (2^j - 1) = 2^{d(d+1)/2} (1/2; 1/2)_d$ yields a direct limit identity joining the half-base q -Pochhammer normalization, a cyclotomic orbit norm, and the Rvachev sinc tail. All statements marked “new” are proved below. Novelty is claimed relative to the reviewed repository corpus and the targeted literature search; the root-of-unity recurrence itself is explicitly credited to Sobolewski’s broader automatic-polynomial theory.

Proof-status convention. **IMPORTED RESULT.** means that a result is already in the repository or cited literature. **NEW RESULT PROVED HERE.** means that a complete proof is supplied here and the statement was not found in the reviewed corpus or targeted literature search. **OPEN OR CONJECTURAL.** labels questions and numerical conjectures. This convention prevents known root-of-unity identities from being relabeled as discoveries merely because they acquire a new application to the Rvachev product.

1 Executive summary and frontier boundary

1.1 The contribution map

Result	Content	Status	
Rational-ray factorization	Splits $\mathcal{U}(2^N y)$ into a universal log-Gaussian factor, a smooth mantissa tail, and a finite Thue–Morse product.	NEW	RESULT PROVED HERE.
Exact periodic ray law	Gives $\kappa_{a,q}$ and an explicit period- d correction $\Omega_{a,q}((N + 1) \bmod d)$, with no error term.	NEW	RESULT PROVED HERE.
Cyclotomic orbit polynomial	Constructs $\mathcal{P}_{q,2}(X) \in \mathbb{Z}[X]$ and proves a resultant identity whose right side is a perfect d th power.	NEW	RESULT PROVED HERE.
Ramanujan trace bridge	Expresses all power traces of the orbit multipliers as finite Thue–Morse–Ramanujan sums.	NEW	RESULT PROVED HERE.
Mean exponent law	Computes the Galois-orbit mean of $\kappa_{a,q}$ from the single integer $\Phi_q(1)$.	NEW	RESULT PROVED HERE.
Sharp endpoint	Proves $A_{a,q}^{1/d} \leq \sqrt{3}$, uniquely extremized by $q = 3$, and obtains the exact minimum rational-ray exponent.	NEW	RESULT PROVED HERE.
Mersenne–Fermat twin expansion	Gives matching main terms and opposite first corrections for denominators $2^d \mp 1$.	NEW	RESULT PROVED HERE.
q -binomial bridge constant	Shows that two Mersenne-normalized cyclotomic norms converge to $(1/2; 1/2)_{\infty} h$.	NEW	RESULT PROVED HERE.
Root-of-unity multiplier	The recurrence $T_{n+d}(\zeta) = \Gamma T_n(\zeta)$ and fixed-field interpretation.	IMPORTED	RESULT. [7]
Metric cocycle theory	Exact variance, central limit theorem, law of the iterated logarithm, transfer operator, and multifractal discussion.	IMPORTED	RESULT. from the Fourier-decay drafts

Table 6: The result boundary used throughout the report.

1.2 Why arithmetic rays are the genuine gap

The Fourier-decay subcorpus already goes far beyond the elementary statement that $\widehat{\text{up}}$ decreases faster than every power. Across its twelve drafts it develops the logarithmic cocycle for the doubling map, almost-everywhere behavior, exact variance, central-limit and iterated-logarithm fluctuations, extremal envelopes, transfer operators, and a moment spectrum. Re-deriving that material would add polish rather than frontier mathematics. The Thue–Morse atlas likewise contains

$$T_n(z) = \sum_{m=0}^{2^n-1} (-1)^{s_2(m)} z^m = \prod_{j=0}^{n-1} (1 - z^{2^j}),$$

its Mahler recursion, rarefied sums, roots-of-unity methods, cyclotomic factors, and extensive arithmetic subsequence material.

The missing composition was

Rvachev sinc product $\longleftrightarrow$ doubling-orbit sine product $\longleftrightarrow$ cyclotomic Thue–Morse norm.

For irrational mantissas the middle term is a nontrivial ergodic cocycle. For a rational mantissa it is periodic, and periodicity promotes an asymptotic heuristic to an exact finite-state law. That promotion is the organizing idea of this report.

1.3 Corpus and novelty audit

The review treated every TeX path listed in Appendix A as a corpus item. The live primary exposition, frontier volume, formula atlas, Fourier-decay studies, and source-paper transcriptions were reviewed

at source level. The thirty-two frozen archive items were compared as revision chains, so historical duplicates were not counted as independent mathematical claims. The archive's provenance notes were used to identify material unique to older revisions.

The external search concentrated on Fabius/Rvachev Fourier products, lacunary trigonometric products, Thue–Morse polynomials at roots of unity, cyclotomic properties of automatic polynomials, and exact dyadic arithmetic. Sobolewski's root-of-unity recurrence is the closest precursor: it supplies the multiplier and fixed-field structure, but the located work does not state the Rvachev rational-ray law, the orbit resultant in the form used below, the exponent mean law, the sharp $\sqrt{3}$ endpoint, or the Mersenne–Fermat/ q -Pochhammer bridge. The novelty claim is therefore deliberately limited: these deductions are new relative to the reviewed corpus and were not found explicitly in the located literature.

2 Established platform from the corpus

2.1 Fabius, Rvachev, and the Fourier product

The primary exposition carefully separates the bounded Fabius distribution function F , the even compactly supported Rvachev function up, and the signed global continuation governed by Thue–Morse signs [1, 2, 3]. For this report only up and its Fourier transform are needed. In the repository normalization,

$$\mathcal{U}(x) := \widehat{\text{up}}(x) = \prod_{n=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^n}\right), \quad \text{sinc } u = \frac{\sin u}{u}, \quad \text{sinc } 0 = 1. \quad (85)$$

The product is entire and obeys

$$\mathcal{U}(x) = \text{sinc}(\pi x) \mathcal{U}(x/2). \quad (86)$$

Repeated use of the double-angle identity also yields

$$\mathcal{U}(x) = \prod_{m=1}^{\infty} \left(\cos \frac{\pi x}{2^m} \right)^m. \quad (87)$$

2.2 The finite Thue–Morse polynomial

Write $\varepsilon_m = (-1)^{s_2(m)}$, where $s_2(m)$ counts the binary ones in m . The block polynomial is

$$T_n(z) := \sum_{m=0}^{2^n-1} \varepsilon_m z^m = \prod_{j=0}^{n-1} (1 - z^{2^j}). \quad (88)$$

The factorization follows from independent binary digit choices and encodes Prouhet cancellation: $z = 1$ is a zero of order n , hence

$$\sum_{m < 2^n} \varepsilon_m m^r = 0 \quad (0 \leq r < n).$$

The bridge to the Fourier product is

$$|T_n(e^{2\pi i t})| = \prod_{j=0}^{n-1} |1 - e^{2\pi i 2^j t}| = \prod_{j=0}^{n-1} |2 \sin(\pi 2^j t)|. \quad (89)$$

2.3 The half-base q -calculus layer

The exact dyadic-value theory uses the base $q = 1/2$ Pochhammer symbol and Gaussian binomial coefficients. In the repository normalization,

$$(z; 1/2)_n = \sum_{k=0}^n (-1)^k 2^{-\binom{k}{2}} \left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} z^k. \quad (90)$$

With the Mersenne factorial

$$\mathfrak{M}_n := \prod_{j=1}^n (2^j - 1), \quad (91)$$

one has

$$\left[\begin{matrix} n \\ k \end{matrix} \right]_{1/2} = \frac{\mathfrak{M}_n}{\mathfrak{M}_k \mathfrak{M}_{n-k} 2^{k(n-k)}}, \quad (1/2; 1/2)_n = 2^{-n(n+1)/2} \mathfrak{M}_n. \quad (92)$$

The first q here is an analytic base fixed at $1/2$; later q denotes an odd cyclotomic denominator. Section 8 gives a limit in which these two roles meet naturally rather than notationally.

2.4 What is imported and what is not

The twelve Fourier-decay manuscripts repeatedly derive a decomposition of $\log |\mathcal{U}(2^N y)|$ into a universal quadratic term and a Birkhoff sum for $x \mapsto 2x \pmod{1}$. Their final syntheses include the typical linear exponent, exact covariance calculations, a central limit theorem, a law of the iterated logarithm, transfer-operator formulas, and a multifractal moment spectrum. Those results are context, not new contributions here.

For a primitive q th root ζ_q and $d = \text{ord}_q(2)$, the sequence $T_n(\zeta_q^a)$ obeys a two-term recurrence of span d . This is an elementary special case of Sobolewski's general recurrence theory for automatic-sequence polynomials at roots of unity [7]. The same work places the multiplier in the fixed field generated by $\langle 2 \rangle$ and proves integrality properties. We therefore mark the recurrence itself as imported. What follows from inserting it into the exact Rvachev factorization is the new contribution.

3 Exact fixed-mantissa factorization

Fix

$$1 < y < 2, \quad t := y - 1 \in (0, 1), \quad x_N := 2^N y,$$

and define

$$H(y) := \prod_{r=1}^{\infty} \text{sinc}\left(\frac{\pi y}{2^r}\right), \quad Q_n(t) := \prod_{j=0}^{n-1} |2 \sin(\pi 2^j t)|, \quad Q_0(t) := 1. \quad (93)$$

The tail H is positive and real analytic on $(0, 2)$. By (89),

$$Q_n(t) = |T_n(e^{2\pi i t})|. \quad (94)$$

Thus all ray-dependent arithmetic is concentrated in a finite Thue–Morse product.

Theorem 3.1 (Exact ray factorization). *For every $N \geq 0$ and $y \in (1, 2)$,*

$$\boxed{|\mathcal{U}(2^N y)| = 2^{-N(N+1)/2} (\pi y)^{-(N+1)} H(y) 2^{-(N+1)} Q_{N+1}(y-1).} \quad (95)$$

The identity remains valid when both sides vanish.

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Proof. Split the defining product at $n = N$ and put $k = N - n$ in the finite part:

$$\mathcal{U}(2^N y) = \prod_{k=0}^N \operatorname{sinc}(\pi 2^k y) \prod_{r=1}^{\infty} \operatorname{sinc}\left(\frac{\pi y}{2^r}\right).$$

The second product is $H(y)$. Since $y = 1 + t$ and 2^k is an integer, $|\sin(\pi 2^k y)| = |\sin(\pi 2^k t)|$. The denominator of the finite product is

$$\prod_{k=0}^N \pi 2^k y = (\pi y)^{N+1} 2^{N(N+1)/2},$$

whereas its numerator is $2^{-(N+1)} Q_{N+1}(t)$. Multiplication proves the claim. $\square$

Put

$$\begin{aligned} \lambda &:= \log 2, & L_N &:= \log x_N = N\lambda + \log y, \\ \kappa_0 &:= \frac{1}{2} + \frac{\log(2\pi)}{\lambda} = 3.1514961294723187980 \dots \end{aligned} \quad (96)$$

and define

$$\Psi(y) := \log H(y) - \log(2\pi) - \frac{1}{2} \log y + \frac{\log(2\pi) \log y}{\lambda} + \frac{(\log y)^2}{2\lambda}. \quad (97)$$

Corollary 3.2 (Master cocycle identity). *If $Q_{N+1}(t) \neq 0$, then*

$$\boxed{\log |\mathcal{U}(2^N y)| = -\frac{L_N^2}{2\lambda} - \kappa_0 L_N + \Psi(y) + \log Q_{N+1}(t).} \quad (98)$$

If $Q_{N+1}(t) = 0$, both sides are interpreted as $-\infty$.

NEW RESULT PROVED HERE.

Proof. Take logarithms in (95). The deterministic finite part is

$$-\frac{N(N+1)}{2} \lambda - (N+1) \log(2\pi y) + \log H(y).$$

Substituting $N = (L_N - \log y)/\lambda$ and collecting powers of L_N gives (98); the constant is exactly (97). $\square$

Remark 3.3 (Relation with the metric theory). For variable t ,

$$\log Q_n(t) = \sum_{j=0}^{n-1} \log |2 \sin(\pi 2^j t)|$$

is the Birkhoff sum studied in the Fourier-decay drafts. The present report fixes t rational. The doubling orbit then closes, turning the Birkhoff sum into a finite algebraic cycle.

4 Exact arithmetic laws on rational dyadic rays

4.1 The cycle multiplier

Let $q > 1$ be odd, $(a, q) = 1$, and write

$$d := \operatorname{ord}_q(2), \quad \zeta_q := e^{2\pi i/q}, \quad H_q := \langle 2 \rangle \subseteq (\mathbb{Z}/q\mathbb{Z})^\times. \quad (99)$$

Define

$$\Gamma_{a,q} := \prod_{j=0}^{d-1} (1 - \zeta_q^{a2^j}) = T_d(\zeta_q^a), \quad A_{a,q} := |\Gamma_{a,q}| > 0, \quad \alpha_{a,q} := \frac{1}{d} \log A_{a,q}. \quad (100)$$

The value depends only on the coset aH_q .

Proposition 4.1 (Root-of-unity block recurrence). *For every $n \geq 0$,*

$$T_{n+d}(\zeta_q^a) = \Gamma_{a,q} T_n(\zeta_q^a). \quad (101)$$

Consequently, if $n = md + r$, $0 \leq r < d$, then

$$Q_n(a/q) = A_{a,q}^m Q_r(a/q). \quad (102)$$

IMPORTED RESULT. in its general automatic-polynomial setting; the special-case proof is included.

Proof. Using (88) and $2^d \equiv 1 \pmod{q}$,

$$\begin{aligned} T_{n+d}(\zeta_q^a) &= \prod_{j=0}^{n+d-1} (1 - \zeta_q^{a2^j}) \\ &= \left(\prod_{j=0}^{d-1} (1 - \zeta_q^{a2^j}) \right) \left(\prod_{j=d}^{n+d-1} (1 - \zeta_q^{a2^j}) \right) \\ &= \Gamma_{a,q} \prod_{j=0}^{n-1} (1 - \zeta_q^{a2^j}). \end{aligned}$$

Iteration and absolute values give (102). □

4.2 The exact log-quadratic law

Set

$$y_{a,q} := 1 + \frac{a}{q} \in (1, 2), \quad x_N := 2^N y_{a,q}, \quad (103)$$

and define the arithmetic ray exponent

$$\kappa_{a,q} := \kappa_0 - \frac{\alpha_{a,q}}{\lambda} = \frac{1}{2} + \frac{\log(2\pi)}{\log 2} - \frac{\log A_{a,q}}{d \log 2}. \quad (104)$$

For $0 \leq r < d$, put

$$\Omega_{a,q}(r) := \Psi(y_{a,q}) + \left(1 - \frac{\log y_{a,q}}{\lambda}\right) \alpha_{a,q} + \log Q_r(a/q) - r \alpha_{a,q}. \quad (105)$$

Theorem 4.2 (Exact rational-ray decay law). *For every $N \geq 0$, let r_N be the residue of $N + 1$ modulo d in $\{0, \dots, d-1\}$. Then*

$$\log |\mathcal{U}(2^N y_{a,q})| = -\frac{(\log x_N)^2}{2 \log 2} - \kappa_{a,q} \log x_N + \Omega_{a,q}(r_N). \quad (106)$$

The complete remainder is periodic in N with period dividing d ; there is no asymptotic error term.

NEW RESULT PROVED HERE.

Proof. Write $n = N + 1 = md + r_N$. Proposition 4.1 gives

$$\log Q_{N+1}(a/q) = (N + 1)\alpha_{a,q} + \log Q_{r_N}(a/q) - r_N\alpha_{a,q}.$$

Since $N + 1 = (L_N - \log y_{a,q})/\lambda + 1$, substitution into (98) changes the coefficient of L_N from $-\kappa_0$ to $-\kappa_{a,q}$ and leaves (105). $\square$

Corollary 4.3 (Exact d -step recurrence along a ray). *Under the hypotheses above,*

$$|\mathcal{U}(2^{N+d}y_{a,q})| = A_{a,q} 2^{-d(d+3)/2} \pi^{-d} (2^N y_{a,q})^{-d} |\mathcal{U}(2^N y_{a,q})|. \quad (107)$$

NEW RESULT PROVED HERE.

Proof. Take the quotient of (95) at $N + d$ and N . The tail cancels and $Q_{N+d+1}/Q_{N+1} = A_{a,q}$; simplifying the remaining powers gives the stated factor. $\square$

Remark 4.4 (Discrete scale invariance). After removal of the universal log-Gaussian and linear factors, the values on a rational ray lie on a finite cycle. The scale is 2^d , and the multiplier is not a fitted phenomenological constant but the cyclotomic norm (100).

4.3 All rational mantissas

Every reduced rational $t \in (0, 1)$ has a unique representation

$$t = \frac{a}{2^e q}, \quad e \geq 0, \quad q \text{ odd}, \quad (a, 2q) = 1. \quad (108)$$

For $n > e$,

$$Q_n(t) = Q_e(t) Q_{n-e}(a/q). \quad (109)$$

Corollary 4.5 (Zero-or-periodic dichotomy). *Let t be as in (108).*

1. *If $q = 1$, then $Q_n(t) = 0$ for every $n > e$, and $\mathcal{U}(2^N(1+t)) = 0$ for every $N \geq e$.*
2. *If $q > 1$, the finite prefix $Q_e(t)$ is followed by the exact law of theorem 4.2, with period $\text{ord}_q(2)$.*

Thus a rational ray is either eventually an exact zero ray or an eventually periodic nonzero ray.

NEW RESULT PROVED HERE.

Proof. Split the product defining Q_n at $j = e$. If $q = 1$, the factor at $j = e$ is $2|\sin(\pi a)| = 0$. If $q > 1$, the subsequent reduced odd-denominator orbit never hits zero and (102) applies. $\square$

Example 4.6 (The denominator-three ray). For $q = 3$, $d = 2$ and

$$\Gamma_{1,3} = (1 - \zeta_3)(1 - \zeta_3^2) = 3, \quad A_{1,3} = 3.$$

Hence

$$\kappa_{1,3} = \frac{1}{2} + \frac{\log(2\pi/\sqrt{3})}{\log 2} = 2.3590148791117407073 \dots$$

The periodic correction has at most two values.

Example 4.7 (A neutral cycle at $q = 15$). The subgroup $\langle 2 \rangle$ has order 4. Both cosets have multiplier -1 , so $A_{a,15} = 1$ and the linear exponent is exactly κ_0 . The orbit polynomial is $(X + 1)^2$, showing that it can have repeated roots when the multiplier lies in a smaller subfield.

Example 4.8 (The first visible split at $q = 17$). Here $d = 8$ and the multiplier polynomial is

$$X^2 - 34X + 17.$$

Its roots are $17 \pm 4\sqrt{17}$, with amplitudes $0.50757749752936 \dots$ and $33.492422502471 \dots$. The corresponding ray exponents are $3.2737836274 \dots$ and $2.5182757764 \dots$

5 Cyclotomic algebra of the ray spectrum

5.1 Relative norms and orbit polynomials

Let $K_q = \mathbb{Q}(\zeta_q)$, $G_q = (\mathbb{Z}/q\mathbb{Z})^\times$, and let $K_q^{H_q}$ be the fixed field of $H_q = \langle 2 \rangle$. Then

$$\Gamma_{a,q} = N_{K_q/K_q^{H_q}}(1 - \zeta_q^a). \quad (110)$$

This imported fixed-field interpretation explains that every multiplier is an algebraic integer. The quotient G_q/H_q indexes its conjugates, with possible repetitions. Write

$$c_q^* := |G_q/H_q| = \frac{\varphi(q)}{d}. \quad (111)$$

Definition 5.1 (Cyclotomic orbit polynomial). Choose a representative a_C from every coset $C \in G_q/H_q$ and set

$$\mathcal{P}_{q,2}(X) := \prod_{C \in G_q/H_q} (X - \Gamma_{a_C,q}). \quad (112)$$

Theorem 5.2 (Integrality and resultant construction). Let $F_d(z) := T_d(z) = \prod_{j=0}^{d-1} (1 - z^{2^j})$. Then

$$\mathcal{P}_{q,2}(X) \in \mathbb{Z}[X] \quad (113)$$

and

$$\boxed{\text{Res}_z(\Phi_q(z), X - F_d(z)) = \mathcal{P}_{q,2}(X)^d.} \quad (114)$$

The polynomial on the right need not be squarefree.

NEW RESULT PROVED HERE.

Proof. The coefficients of (112) are symmetric polynomials in algebraic integers, hence algebraic integers. The Galois group permutes the cosets, so every coefficient lies in $\mathbb{Q}$ and therefore in $\mathbb{Z}$.

Since Φ_q is monic with roots ζ_q^a , $a \in G_q$,

$$\text{Res}_z(\Phi_q(z), X - F_d(z)) = \prod_{a \in G_q} (X - F_d(\zeta_q^a)).$$

The value $F_d(\zeta_q^a) = \Gamma_{a,q}$ is constant on a coset aH_q , and every coset contains d elements. Grouping factors proves (114). $\square$

5.2 Ramanujan traces

Recall the Ramanujan sum

$$c_q(m) := \sum_{a \in G_q} \zeta_q^{am} = \mu\left(\frac{q}{(q,m)}\right) \frac{\varphi(q)}{\varphi(q/(q,m))}. \quad (115)$$

Theorem 5.3 (Power traces as finite digital sums). For every integer $r \geq 1$,

$$\boxed{\sum_{C \in G_q/H_q} \Gamma_{a_C,q}^r = \frac{1}{d} \sum_{0 \leq m_1, \dots, m_r < 2^d} \varepsilon_{m_1} \cdots \varepsilon_{m_r} c_q(m_1 + \cdots + m_r).} \quad (116)$$

In particular,

$$\text{Tr}_{K_q^{H_q}/\mathbb{Q}}(\Gamma_{1,q}) = \frac{1}{d} \sum_{m=0}^{2^d-1} \varepsilon_m c_q(m), \quad (117)$$

with the natural embedding multiplicities if $\Gamma_{1,q}$ lies in a smaller subfield.

NEW RESULT PROVED HERE.

Proof. Expand

$$\Gamma_{a,q} = T_d(\zeta_q^a) = \sum_{m=0}^{2^d-1} \varepsilon_m \zeta_q^{am}.$$

The value is constant on each H_q -coset, so summing over cosets is $1/d$ times summing over all $a \in G_q$. Raise the finite sum to the r th power, interchange sums, and apply (115). $\square$

Corollary 5.4 (Newton reconstruction). *The first c_q^* power sums in (116), together with Newton's identities, reconstruct $\mathcal{P}_{q,2}(X)$ using integer arithmetic only. This is an independent check on the resultant method.*

5.3 A one-integer law for the mean exponent

Theorem 5.5 (Norm, constant term, and mean ray exponent). *With $c_q^* = \varphi(q)/d$,*

$$\prod_{C \in G_q/H_q} \Gamma_{a_C,q} = \Phi_q(1), \quad (118)$$

$$\mathcal{P}_{q,2}(0) = (-1)^{c_q^*} \Phi_q(1), \quad (119)$$

$$\prod_{C \in G_q/H_q} A_{a_C,q} = \Phi_q(1), \quad (120)$$

and

$$\frac{1}{c_q^*} \sum_{C \in G_q/H_q} \kappa_{a_C,q} = \kappa_0 - \frac{\log \Phi_q(1)}{\varphi(q) \log 2}. \quad (121)$$

For $q > 1$,

$$\Phi_q(1) = \begin{cases} p, & q = p^s \text{ is a prime power,} \\ 1, & q \text{ has at least two distinct prime divisors.} \end{cases} \quad (122)$$

Thus every non-prime-power denominator has mean exponent exactly κ_0 .

NEW RESULT PROVED HERE.

Proof. By transitivity of norms,

$$\prod_C \Gamma_{a_C,q} = N_{K_q^{H_q}/\mathbb{Q}} \left(N_{K_q/K_q^{H_q}}(1 - \zeta_q) \right) = N_{K_q/\mathbb{Q}}(1 - \zeta_q) = \Phi_q(1).$$

This proves (118); the constant term and absolute-value product follow. Taking logarithms in (120), inserting (104), and using $dc_q^* = \varphi(q)$ gives (121). Formula (122) is the classical value of a cyclotomic polynomial at one. $\square$

Remark 5.6 (Compensation between conjugate rays). For a denominator with at least two prime factors, a coset with $A_{a,q} > 1$ and hence a smaller-than- κ_0 exponent must be compensated geometrically by cosets with $A_{a,q} < 1$. The $q = 17$ and $q = 31$ tables below show this splitting explicitly.

6 The sharp rational-ray endpoint

Theorem 6.1 (Sharp periodic-orbit sine-product bound). *For every odd $q > 1$ and every $(a, q) = 1$,*

$$A_{a,q}^{1/d} \leq \sqrt{3}. \quad (123)$$

Equality holds if and only if $q = 3$ (with $a = 1$ or 2).

NEW RESULT PROVED HERE.

Proof. Let

$$t_j = \left\{ \frac{a2^j}{q} \right\}, \quad c_j = \cos(2\pi t_j), \quad 0 \leq j < d,$$

with indices cyclic. Doubling gives $c_{j+1} = 2c_j^2 - 1$. If $s = d^{-1} \sum_j c_j$, cyclic averaging and Cauchy-Schwarz give

$$s = 2 \frac{1}{d} \sum_j c_j^2 - 1 \geq 2s^2 - 1.$$

Hence $(2s + 1)(s - 1) \leq 0$ and $s \geq -1/2$. By arithmetic-geometric mean,

$$\begin{aligned} A_{a,q}^{2/d} &= \left(\prod_{j=0}^{d-1} |1 - e^{2\pi i t_j}|^2 \right)^{1/d} \\ &= \left(\prod_{j=0}^{d-1} (2 - 2c_j) \right)^{1/d} \leq \frac{1}{d} \sum_{j=0}^{d-1} (2 - 2c_j) = 2 - 2s \leq 3. \end{aligned}$$

If equality holds, then $s = -1/2$ and equality in arithmetic-geometric mean forces all $c_j = -1/2$. Every orbit point is therefore $1/3$ or $2/3$ modulo one, so the reduced orbit is exactly $\{1/3, 2/3\}$ and $q = 3$. The converse is immediate. $\square$

Corollary 6.2 (Sharp minimum of the rational-ray exponent). *Every nonzero rational ray satisfies*

$$\kappa_{a,q} \geq \kappa_{\min} := \frac{1}{2} + \frac{\log(2\pi/\sqrt{3})}{\log 2} = 2.3590148791117407073 \dots \quad (124)$$

Equality is attained only by the denominator-three doubling orbit.

NEW RESULT PROVED HERE.

Proof. Insert (123) into (104). $\square$

Remark 6.3 (A spectral interpretation). The arithmetic exponents form a spectrum of cycle averages of $\log |2 \sin(\pi t)|$. The theorem identifies the exact lower endpoint without using transfer-operator machinery. Section 7 constructs exponents tending to $+\infty$, so the spectrum has a sharp bottom and no finite top.

7 Mersenne and plus-denominator rays

The preceding theory turns a fast-decay construction into an arithmetic design problem: choose a rational doubling cycle for which $A_{a,q}$ is very small. The simplest such cycles come from the denominators immediately adjacent to a power of two.

7.1 Exact orbit products

For $d \geq 2$ put

$$q_d^- := 2^d - 1, \quad q_d^+ := 2^d + 1, \quad (125)$$

and let A_d^- and A_d^+ denote the amplitudes $A_{1,q}$ for these denominators.

Lemma 7.1 (Orders of two). *One has*

$$\text{ord}_{q_d^-}(2) = d, \quad \text{ord}_{q_d^+}(2) = 2d. \quad (126)$$

Proof. For q_d^- , if $2^r \equiv 1 \pmod{2^d - 1}$ with $0 < r < d$, then the positive integer $2^r - 1$ would be divisible by the larger integer $2^d - 1$, impossible.

For q_d^+ , $2^d \equiv -1$, so the order r is even. Inside the cyclic group generated by 2, the unique element of order two is $2^{r/2}$; hence $d = (2m + 1)r/2$ for some $m \geq 0$. If $m \geq 1$, then $r \leq 2d/3 < d$, but q_d^+ would divide $2^r - 1 < 2^d + 1$, again impossible. Thus $m = 0$ and $r = 2d$. $\square$

Proposition 7.2 (Exact product forms). *For $d \geq 2$,*

$$A_d^- = \prod_{k=1}^d 2 \sin\left(\frac{\pi 2^{-k}}{1 - 2^{-d}}\right), \quad (127)$$

$$A_d^+ = \left[\prod_{k=1}^d 2 \sin\left(\frac{\pi 2^{-k}}{1 + 2^{-d}}\right) \right]^2. \quad (128)$$

Proof. For q_d^- , the orbit is $1, 2, \dots, 2^{d-1}$ under multiplication by two. Reindexing $j = d - k$ gives

$$\frac{2^j}{2^d - 1} = \frac{2^{-k}}{1 - 2^{-d}}.$$

For q_d^+ the order is $2d$ and $2^{j+d} \equiv -2^j \pmod{q_d^+}$, so the second half of the orbit contributes the same absolute factors as the first. The same reindexing proves (128). $\square$

7.2 The universal sinc-tail constant

Define

$$B_d := \prod_{k=1}^d 2 \sin\left(\frac{\pi}{2^k}\right), \quad H_d(1) := \prod_{k=1}^d \operatorname{sinc}\left(\frac{\pi}{2^k}\right). \quad (129)$$

Then

$$B_d = (2\pi)^d 2^{-d(d+1)/2} H_d(1). \quad (130)$$

The finite sinc tail converges to

$$h := \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{\pi}{2^k}\right) = 0.5537712758888100953442174163445488813 \dots \quad (131)$$

Lemma 7.3 (Three forms of the tail constant). *The constant h satisfies*

$$h = -\mathcal{U}'(1), \quad (132)$$

$$\log h = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m(4^m - 1)}. \quad (133)$$

Moreover,

$$\log H_d(1) = \log h + \mathcal{O}(4^{-d}). \quad (134)$$

Proof. From (86),

$$\mathcal{U}(x) = \operatorname{sinc}(\pi x) \prod_{k=1}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{2^k}\right).$$

The derivative of $\operatorname{sinc}(\pi x)$ at $x = 1$ is -1 , proving (132). Euler's product for the sine gives, for $|z| < \pi$,

$$\log \operatorname{sinc} z = - \sum_{m=1}^{\infty} \frac{\zeta(2m)}{m\pi^{2m}} z^{2m}.$$

Substitute $z = \pi/2^k$, sum over $k \geq 1$, and use $\sum_{k \geq 1} 4^{-km} = 1/(4^m - 1)$. Finally, $\log \operatorname{sinc}(\pi/2^k) = \mathcal{O}(4^{-k})$, which yields (134). $\square$

The first exponentially small correction is controlled by the convergent constant

$$C_{\cot} := \sum_{k=1}^{\infty} \left(\pi 2^{-k} \cot(\pi 2^{-k}) - 1 \right) = -1.2837124730461260445945438959726042474 \dots \quad (135)$$

Convergence follows from $x \cot x = 1 - x^2/3 + \mathcal{O}(x^4)$ at the origin.

Theorem 7.4 (Refined twin product expansion). *As $d \rightarrow \infty$,*

$$\log \frac{A_d^-}{B_d} = (d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d}), \quad (136)$$

$$\log \frac{\sqrt{A_d^+}}{B_d} = -(d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d}). \quad (137)$$

Equivalently,

$$A_d^- = h(2\pi)^d 2^{-d(d+1)/2} \exp\left((d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d})\right), \quad (138)$$

$$A_d^+ = h^2(2\pi)^{2d} 2^{-d(d+1)} \exp\left(-2(d + C_{\cot}) 2^{-d} + \mathcal{O}(d 4^{-d})\right). \quad (139)$$

NEW RESULT PROVED HERE.

Proof. Put $\delta = 2^{-d}$, $x_k = \pi 2^{-k}$, and $g(x) = \log(2 \sin x)$. From [proposition 7.2](#),

$$\begin{aligned} \log(A_d^-/B_d) &= \sum_{k=1}^d \left[g\left(\frac{x_k}{1-\delta}\right) - g(x_k) \right], \\ \log(\sqrt{A_d^+}/B_d) &= \sum_{k=1}^d \left[g\left(\frac{x_k}{1+\delta}\right) - g(x_k) \right]. \end{aligned}$$

On the relevant interval, $xg'(x) = x \cot x$ and $x^2g''(x)$ are uniformly bounded. Taylor expansion under a multiplicative perturbation therefore gives, uniformly in k ,

$$g\left(\frac{x_k}{1 \mp \delta}\right) - g(x_k) = \pm \delta x_k \cot x_k + \mathcal{O}(\delta^2).$$

Summing produces an error $\mathcal{O}(d\delta^2)$. Further,

$$\sum_{k=1}^d x_k \cot x_k = d + C_{\cot} + \mathcal{O}(4^{-d}).$$

This proves (136)–(137). Equations (138)–(139) follow from (130) and (134). $\square$

7.3 Exponent asymptotics and unboundedness

Let κ_d^- and κ_d^+ be the ray exponents for q_d^- and q_d^+ , respectively, and put

$$K_d := \frac{d}{2} + 1 - \frac{\log h}{d \log 2}. \quad (140)$$

Corollary 7.5 (Opposite first corrections). *As $d \rightarrow \infty$,*

$$\kappa_d^- = K_d - \frac{2^{-d}}{\log 2} \left(1 + \frac{C_{\cot}}{d} \right) + \mathcal{O}(4^{-d}), \quad (141)$$

$$\kappa_d^+ = K_d + \frac{2^{-d}}{\log 2} \left(1 + \frac{C_{\cot}}{d} \right) + \mathcal{O}(4^{-d}). \quad (142)$$

In particular,

$$\kappa_d^\pm = \frac{d}{2} + 1 - \frac{\log h}{d \log 2} + \mathcal{O}(2^{-d}), \quad (143)$$

and the rational-ray exponent spectrum is unbounded above.

NEW RESULT PROVED HERE.

Proof. Insert (138) into (104) with order d , and (139) with order $2d$. The terms $\log(2\pi)$ cancel against κ_0 , and the remaining algebra gives (141) and (142). Since $K_d \sim d/2$, the exponents tend to infinity. $\square$

Combining corollaries 6.2 and 7.5 gives a complete endpoint statement.

Corollary 7.6 (Endpoint geometry of the rational spectrum). *Let $\mathcal{K}_{\text{rat}}$ be the set of exponents of all nonzero rational rays. Then*

$$\min \mathcal{K}_{\text{rat}} = \kappa_{\min}, \quad \sup \mathcal{K}_{\text{rat}} = +\infty. \quad (144)$$

The minimum is attained only by denominator 3.

8 A direct half-base q -binomial bridge

The Mersenne factorial (91) is the denominator-clearing normalization behind the half-base Gaussian coefficients. The preceding product asymptotics show that the same arithmetic factors normalize special cyclotomic ray multipliers.

Set

$$\mathcal{C}_* := (1/2; 1/2)_\infty h = 0.1599225518776068325779014528557986690 \dots, \quad (145)$$

where

$$(1/2; 1/2)_\infty = 0.2887880950866024212788997219292307801 \dots \quad (146)$$

Theorem 8.1 (Mersenne–cyclotomic–sinc bridge). *As $d \rightarrow \infty$,*

$$\left[\frac{\mathfrak{M}_d A_d^-}{(2\pi)^d} \rightarrow \mathcal{C}_*, \quad \left[\frac{\mathfrak{M}_d \sqrt{A_d^+}}{(2\pi)^d} \rightarrow \mathcal{C}_*. \right. \quad (147)$$

More precisely,

$$\frac{\mathfrak{M}_d A_d^-}{(2\pi)^d} = \mathcal{C}_* \exp\left((d + C_{\cot} + 1)2^{-d} + \mathcal{O}(d4^{-d})\right), \quad (148)$$

$$\frac{\mathfrak{M}_d \sqrt{A_d^+}}{(2\pi)^d} = \mathcal{C}_* \exp\left((1 - d - C_{\cot})2^{-d} + \mathcal{O}(d4^{-d})\right). \quad (149)$$

NEW RESULT PROVED HERE.

Proof. Using (92) and (130),

$$\frac{\mathfrak{M}_d B_d}{(2\pi)^d} = (1/2; 1/2)_d H_d(1).$$

Now multiply by A_d^-/B_d or by $\sqrt{A_d^+}/B_d$ and use theorem 7.4. The limits follow because $(1/2; 1/2)_d \rightarrow (1/2; 1/2)_\infty$ and $H_d(1) \rightarrow h$.

For the refined forms, note

$$\log \frac{(1/2; 1/2)_d}{(1/2; 1/2)_\infty} = - \sum_{k>d} \log(1 - 2^{-k}) = 2^{-d} + \mathcal{O}(4^{-d}),$$

while (134) contributes only $\mathcal{O}(4^{-d})$. Adding the exponent corrections from (136) and (137) proves (148)–(149). $\square$

Interpretation. The limit (147) is a literal bridge among the three recurrent structures of the repository:

$$\underbrace{\mathfrak{M}_d}_{\text{half-base Gaussian normalization}} \times \underbrace{A_d^- \text{ or } \sqrt{A_d^+}}_{\text{cyclotomic Thue–Morse orbit norm}} \times \underbrace{(2\pi)^{-d}}_{\text{sinc scaling}} \longrightarrow \underbrace{(1/2; 1/2)_\infty (-\mathcal{U}'(1))}_{\text{global constant}}.$$

This is more than a shared appearance of powers of two: it is a quantitative asymptotic identity tying the q -binomial and Fourier sides together.

8.1 Numerical convergence of the twin exponents

d	$2^d - 1$	κ_d^-	$2^d + 1$	κ_d^+	K_d
2	3	2.35901487911	5	2.57101410575	2.42631893591
3	7	2.68360364246	9	2.88733571269	2.78421262394
4	15	3.15149612947	17	3.27378362740	3.21315946796
5	31	3.63677701126	33	3.70378409640	3.67052757436
6	63	4.12430649454	65	4.15974429455	4.14210631197
7	127	4.61257768880	129	4.63098586366	4.62180541026
8	255	5.10184195614	257	5.11130441973	5.10657973398
9	511	5.59231992724	513	5.59715163452	5.59473754131
10	1023	6.08403530020	1025	6.08649134438	6.08526378718
11	2047	6.57689018061	2049	6.57813464441	6.57751253380
12	4095	7.07073858328	4097	7.07136766593	7.07105315599
13	8191	7.56542880068	8193	7.56574624039	7.56558752860
14	16383	8.06082272205	16385	8.06098268408	8.06090270513

Table 7: Mersenne and plus-denominator exponents bracketing the common main term K_d . The gap has the predicted size $2^{1-d}/\log 2$ to first order.

9 Exact computations and orbit-polynomial data

The theory is computationally concrete. For a given odd q one needs only the multiplicative order of 2, a finite product at roots of unity, and either a resultant or Ramanujan power sums. No numerical integration of up is involved.

9.1 Two exact algorithms

Algorithm 9.1 (Resultant route). Given odd $q > 1$:

1. compute $d = \text{ord}_q(2)$;
2. form $F_d(z) = \prod_{j=0}^{d-1} (1 - z^{2^j})$;
3. compute $R(X) = \text{Res}_z(\Phi_q(z), X - F_d(z))$ over $\mathbb{Z}[X]$;
4. factor R and divide every factor exponent by d ;
5. the resulting exact polynomial is $\mathcal{P}_{q,2}(X)$;
6. take the absolute values of its complex roots and insert them into (104).

The perfect- d th-power assertion in (114) is a built-in check.

Algorithm 9.2 (Ramanujan–Newton route). Let $c_q^* = \varphi(q)/d$. Compute the power sums $p_r = \sum_C \Gamma_C^r$ for $1 \leq r \leq c_q^*$ by (116), then apply Newton’s identities to recover the elementary symmetric functions and hence $\mathcal{P}_{q,2}(X)$. This route uses integer arithmetic and can exploit fast convolution of the finite Thue–Morse block.

The resultant route is simplest for modest d . The Ramanujan route becomes attractive when 2^d is too large to expand naively but q has a simple divisor structure, because $c_q(m)$ depends only on (q, m) .

9.2 Exact orbit polynomials

q	$d = \text{ord}_q(2)$	$\mathcal{P}_{q,2}(X)$
3	2	$X - 3$
5	4	$X - 5$
7	3	$X^2 + 7$
9	6	$X - 3$
11	10	$X - 11$
15	4	$(X + 1)^2$
17	8	$X^2 - 34X + 17$
21	6	$(X + 1)^2$
23	11	$X^2 + 23$
31	5	$X^6 + 93X^4 + 899X^2 + 31$

Table 8: Exact orbit polynomials obtained by the resultant method. Repeated roots at $q = 15$ and 21 reflect a larger stabilizer than $\langle 2 \rangle$.

Several structural features are visible immediately.

- If 2 generates all of G_q , the orbit polynomial is linear and the multiplier is $\Phi_q(1)$; this explains $q = 3, 5, 9, 11$.
- For $q = 7$ and 23 , the two multipliers are purely imaginary conjugates and both amplitudes equal $\sqrt{q}$.
- At $q = 17$, the two multipliers are real and split as $17 \pm 4\sqrt{17}$.
- For $q = 31$, the odd order $d = 5$ makes the multiplier roots occur in imaginary pairs; the squared amplitudes are the positive roots of $Y^3 - 93Y^2 + 899Y - 31$.

q	multiplicity	amplitude $A_{a,q}$	exponent $\kappa_{a,q}$
3	1	3	2.3590148791117
5	1	5	2.5710141057505
7	2	$\sqrt{7} = 2.6457513110646\dots$	2.6836036424627
9	1	3	2.8873357126855
11	1	11	2.8055529676086
15	2	1	3.1514961294723
17	1	$17 - 4\sqrt{17} = 0.50757749752936\dots$	3.2737836273969
17	1	$17 + 4\sqrt{17} = 33.492422502471\dots$	2.5182757763915
21	2	1	3.1514961294723
23	2	$\sqrt{23} = 4.7958315233127\dots$	2.9458796769243
31	2	0.18602850302165...	3.6367770112610
31	2	3.3042170276289...	2.8066344396753
31	2	9.0580098934805...	2.5156573064420

Table 9: Amplitude classes and rational-ray exponents. Multiplicity is the number of H_q -cosets with that amplitude.

9.3 Exact checks performed

The calculations behind tables 8 and 9 were checked in three independent ways:

1. the symbolic resultant was factored and verified to be a perfect d th power;
2. high-precision direct products $\prod_{j < d} (1 - \zeta_q^{a2^j})$ were evaluated for every coset and substituted into the exact integer polynomial;
3. the product of all orbit amplitudes was checked against $\Phi_q(1)$, and the numerical mean of the exponents was checked against (121).

The fixed-mantissa factorization (95) was also evaluated at $y = 4/3, 6/5, 8/7$ and $N = 0, 1, 3, 10$ with eighty decimal digits. The largest relative discrepancy was below 3×10^{-78} .

9.4 Reproducible symbolic code

The following compact SymPy program implements the resultant route. It is included as mathematical pseudocode rather than as a software dependency of the document.

Listing 3: Orbit-polynomial computation.

```
import sympy as sp
from math import gcd

def order_mod_2(q: int) -> int:
    x = 1
    for d in range(1, 2*q + 1):
        x = (2*x) % q
        if x == 1:
            return d
    raise ArithmeticError("order not found")

def orbit_polynomial(q: int):
    if q <= 1 or q % 2 == 0:
        raise ValueError("q must be odd and > 1")
    z, X = sp.symbols("z X")
    d = order_mod_2(q)
    F = sp.prod(1 - z**(2**j) for j in range(d))
    Phi = sp.cyclotomic_poly(q, z)
    R = sp.Poly(sp.resultant(Phi, X - F, z), X)
    factors = sp.factor_list(R.as_expr(), X)[1]
```

```

P = sp.Integer(1)
for factor, exponent in factors:
    if exponent % d:
        raise ArithmeticError("resultant is not a d-th power")
    P *= factor**(exponent // d)
return d, sp.Poly(sp.expand(P), X)

for q in [3, 5, 7, 9, 11, 15, 17, 21, 23, 31]:
    d, P = orbit_polynomial(q)
    print(q, d, sp.factor(P.as_expr()))

```

10 A general-base extension

The arithmetic mechanism does not depend on base two until the sharp endpoint argument. For an integer $b \geq 2$, define

$$\mathcal{U}_b(x) := \prod_{n=0}^{\infty} \operatorname{sinc}\left(\frac{\pi x}{b^n}\right). \quad (150)$$

For $1 < y < 2$ and $t = y - 1$, put

$$H_b(y) := \prod_{r=1}^{\infty} \operatorname{sinc}\left(\frac{\pi y}{b^r}\right), \quad Q_{b,n}(t) := \prod_{j=0}^{n-1} |2 \sin(\pi b^j t)|. \quad (151)$$

(The interval $1 < y < 2$ is enough for a canonical family of rays; one may use $1 < y < b$ with only notational changes.)

Theorem 10.1 (Exact rational rays in base b). *For every $N \geq 0$,*

$$|\mathcal{U}_b(b^N y)| = b^{-N(N+1)/2} (\pi y)^{-(N+1)} H_b(y) 2^{-(N+1)} Q_{b,N+1}(t). \quad (152)$$

Let $q > 1$, $(a, q) = 1$, $(b, q) = 1$, $d = \operatorname{ord}_q(b)$, and

$$A_{a,q}^{(b)} := \left| \prod_{j=0}^{d-1} (1 - \zeta_q^{ab^j}) \right|. \quad (153)$$

On the ray $x_N = b^N(1 + a/q)$ there is an exact periodic law

$$\log |\mathcal{U}_b(x_N)| = -\frac{(\log x_N)^2}{2 \log b} - \kappa_{a,q}^{(b)} \log x_N + \Omega_{a,q}^{(b)}(N), \quad (154)$$

where $\Omega_{a,q}^{(b)}$ has period dividing d and

$$\kappa_{a,q}^{(b)} = \frac{1}{2} + \frac{\log(2\pi)}{\log b} - \frac{\log A_{a,q}^{(b)}}{d \log b}. \quad (155)$$

If $H_{q,b} = \langle b \rangle \subseteq (\mathbb{Z}/q\mathbb{Z})^\times$, then the mean over its cosets is

$$\frac{1}{|G_q/H_{q,b}|} \sum_C \kappa_C^{(b)} = \frac{1}{2} + \frac{\log(2\pi)}{\log b} - \frac{\log \Phi_q(1)}{\varphi(q) \log b}. \quad (156)$$

NEW RESULT PROVED HERE.

Proof. Split the product at $n = N$ exactly as in [theorem 3.1](#); the denominator now contributes $b^{N(N+1)/2}$, while the factor $2^{-(N+1)}$ still comes from replacing $|\sin|$ by $|2\sin|$. If $t = a/q$, multiplication by b has period d modulo q , so $Q_{b,n+d} = A_{a,q}^{(b)} Q_{b,n}$. Completing the square in $\log x_N = N \log b + \log y$ gives (155). The mean law follows from the same norm identity because the $H_{q,b}$ -cosets partition G_q . $\square$

Remark 10.2 (What is genuinely binary). The resultant, relative norm, exact periodic remainder, and mean-exponent law are base-independent. The sharp $\sqrt{3}$ endpoint uses the special identity $\cos(2x) = 2\cos^2 x - 1$ and the fact that squaring permutes the cycle. Finding the corresponding sharp constant for $b \geq 3$ is a distinct ergodic-optimization problem.

Lean crosswalk: finite base arithmetic only. The five public theorems of `BaseDigitMultiplicity.lean` are `sum_range_weightedScaleMultiplicity_of_log_lt`, `sum_range_weightedScaleMultiplicity_log`, `sum_range_div_pow_log_eq_self_add_tail`, `sub_one_mul_sum_padicValNat_succ_add_digitSum`, and `sum_range_padicValNat_succ_eq_sub_digitSum_div`. Together they prove, for every natural base $b > 1$, including composite bases, and every $N \geq 0$, including $N = 0$, the sharp finite logarithmic truncation and the two exact forms

$$(b-1) \sum_{n=1}^N (1 + \nu_b(n)) + s_b(N) = bN, \quad \sum_{n=1}^N (1 + \nu_b(n)) = \frac{bN - s_b(N)}{b-1}.$$

This is an exhaustive five-theorem *finite arithmetic* API. It neither constructs $\mathcal{U}_b$ nor proves convergence of (150), an analytic zero order or zero set, any sign law, or the rational-ray factorization and asymptotics of [theorem 10.1](#); all of those remain at the manuscript boundary.

11 A Lean formalization blueprint

The new results are deliberately organized so that most proof obligations are finite algebra, elementary real analysis, or existing cyclotomic infrastructure. This section gives a concrete path from the report to the repository’s formalized core.

11.1 Suggested module boundary

`Analysis/FabiusFunction/ThueMorse/CyclotomicOrbit.lean`

Define T_n , prove its product formula, evaluate it on a root of unity, and prove the block recurrence $T_{n+d} = \Gamma T_n$.

`Analysis/FabiusFunction/Fourier/RationalRay.lean`

Split the infinite sinc product at N , prove (95), and derive the exact periodic log law.

`Analysis/FabiusFunction/NumberTheory/OrbitPolynomial.lean`

Define $H_q = \langle 2 \rangle$, the relative norm, and the orbit polynomial. Prove integrality, the norm product, and the mean exponent identity.

`Analysis/FabiusFunction/Analysis/MersenneRay.lean`

Establish the exact products (127)–(128), the convergence to h , and then the asymptotic estimates.

`Analysis/FabiusFunction/QCalculus/RayBridge.lean`

Combine the Mersenne factorial identity with the sinc-tail limit to formalize (147).

11.2 Core theorem signatures

The exact identities can be formalized before any logarithms, avoiding extended real values at zeros. Indicative signatures are:

Listing 4: Indicative Lean theorem interfaces.

```
-- Finite Thue-Morse polynomial at a root of unity.
theorem tmPoly_eval_cycle
  (hq : Odd q) (ha : Nat.Coprime a q) :
  tmPoly (n + orderOfTwo q) (zeta q ^ a) =
    cycleMultiplier a q * tmPoly n (zeta q ^ a)

-- Exact absolute factorization of the Rvachev Fourier product.
theorem upFourier_abs_dyadic_mantissa
  (hy : 1 < y) (hy2 : y < 2) :
  abs (upFourier ((2 : R) ^ N * y)) =
    2 ^ (-(N * (N + 1) / 2 : Z)) *
    (Real.pi * y) ^ (-(N + 1 : Z)) *
    sincTail y * 2 ^ (-(N + 1 : Z)) *
    orbitSineProduct (N + 1) (y - 1)

-- Product over G/H and the cyclotomic value at one.
theorem prod_cycleMultiplier_cosets :
  Finset.prod (allCosets q) (fun C =>
    cycleMultiplier C.rep q) = cyclotomic q 1

-- Sharp periodic-orbit bound.
theorem cycleAmplitude_geomMean_le_sqrt_three :
  Real.rpow (cycleAmplitude a q) (1 / orderOfTwo q) <= Real.sqrt 3
```

The displayed syntax is schematic. The main implementation decision is whether to represent roots of unity in $\mathbb{C}$ or in an abstract cyclotomic extension. The resultant theorem is easiest abstractly; the absolute-value ray law is easiest in $\mathbb{C}$ with the standard embedding.

11.3 Proof dependency order

A low-risk formalization sequence is:

1. finite product and block recurrence;
2. exact fixed-mantissa factorization without logarithms;
3. rational preperiod/period dichotomy;
4. norm product and mean exponent;
5. sharp $\sqrt{3}$ inequality, which uses only finite real sums and AM–GM;
6. analytic Mersenne estimates;
7. resultant and Ramanujan trace formulas, which can be added independently.

The first five steps already formalize the central qualitative discoveries and do not depend on delicate asymptotic libraries.

12 Further frontier problems

The exact arithmetic-ray theory raises a set of focused questions that are sharper than the generic request to “understand Fourier decay.”

12.1 The closure of the ray-exponent spectrum

Conjecture 12.1 (Interval closure). *Let*

$$\mathcal{K}_{\text{rat}} = \{\kappa_{a,q} : q > 1 \text{ odd}, (a, q) = 1\}.$$

Its closure may be the whole interval $[\kappa_{\min}, \infty)$.

OPEN OR CONJECTURAL.

Periodic orbit measures are dense among invariant measures of the doubling map for continuous observables, but $\log |2 \sin(\pi t)|$ has logarithmic singularities. The conjecture therefore requires a controlled specification argument that keeps periodic words away from zero while prescribing their cycle average. The Mersenne family gives the unbounded direction, and the denominator-three orbit gives the lower endpoint; the missing issue is interpolation.

12.2 Extremizers at fixed period

For each d , determine

$$A_d^{\max} := \max_{\text{ord}_q(2)=d} A_{a,q}, \quad A_d^{\min} := \min_{\text{ord}_q(2)=d} A_{a,q}.$$

The sharp global maximum of $A^{1/d}$ is known from [theorem 6.1](#), but the finite-period problem may have a rich symbolic structure. A successful solution would give the sharpest and slowest rational rays at every scale period.

12.3 Stabilizers and repeated orbit-polynomial roots

Characterize the subgroup

$$\text{Stab}(\Gamma_{1,q}) = \{u \in G_q : \Gamma_{u,q} = \Gamma_{1,q}\}.$$

It contains H_q but can be larger, as $q = 15$ and 21 show. Equivalently, determine when $\mathcal{P}_{q,2}$ is squarefree and when it is a proper power. This is a concrete problem in cyclotomic units and automatic polynomial values.

12.4 Regulator statistics

When q has at least two prime divisors, [\(120\)](#) says $\sum_C \log A_C = 0$. The vector $(\log A_C)_C$ lies in the logarithmic embedding of a cyclotomic-unit group. Its covariance, extreme coordinates, and limiting distribution as $q \rightarrow \infty$ should connect ray decay with regulators and unit lattices. Prime-power denominators add the deterministic shift $\log p/\varphi(p^s)$ from [\(121\)](#).

12.5 Beyond the first Mersenne correction

The proof of [theorem 7.4](#) can be iterated. Taylor expansion in $\delta = 2^{-d}$ produces an all-orders transseries whose coefficients are convergent cotangent/cosecant sums over dyadic angles, together with the tail expansion of $H_d(1)$. Determining a closed algebra for these coefficients would parallel the periodic saddle-jet machinery already developed for the small-argument Fabius asymptotic.

12.6 A wider q -binomial bridge

The bridge theorem uses the terminal Mersenne factor $2^d - 1$ and the full product $\mathfrak{M}_d = \prod_{j \leq d} (2^j - 1)$. A broader goal is to connect general cyclotomic orbit norms $A_{a,q}$ to coefficient extraction in the half-base q -binomial formulas. Possible entry points are:

- denominators dividing products of Mersenne integers;
- discrete Fourier transforms of the Gaussian interpolation blocks at roots of unity;
- resultant identities in which $F_d(z)$ is paired with the half-base Pochhammer polynomial.

A direct finite identity here would join the exact dyadic values of F to the arithmetic rays of $\widehat{\mathfrak{up}}$, rather than linking them only asymptotically.

12.7 Sharp constants in other bases

For $b \geq 3$, determine

$$\sup_{q,a} \left(A_{a,q}^{(b)} \right)^{1/\text{ord}_q(b)}$$

and classify the maximizing periodic orbits. The binary proof is exceptional because the real part of $z \mapsto z^2$ closes quadratically. The general problem looks like a finite-dimensional ergodic-optimization problem for $\log |1 - z|$ under $z \mapsto z^b$.

13 Conclusion

The reviewed corpus already contains two powerful but largely parallel bodies of work: exact $q = 1/2$ arithmetic for Fabius values and dynamical Fourier decay for the Rvachev product. The arithmetic-ray viewpoint inserts the finite Thue–Morse polynomial between them. On rational mantissas, the doubling cocycle closes; its cycle multiplier is a relative cyclotomic norm, and the Fourier decay becomes an exact log-quadratic formula with a finite periodic remainder.

This perspective yields several rigid structures at once:

- an integral orbit polynomial and a perfect-power resultant;
- Ramanujan-sum formulas for every multiplier trace;
- a one-integer mean-exponent law governed by $\Phi_q(1)$;
- a sharp rational-ray minimum from the orbit $\{1/3, 2/3\}$;
- unbounded exponents from Mersenne denominators;
- a twin Mersenne/plus-denominator expansion with opposite first corrections;
- a direct limit identity joining cyclotomic norms, the half-base q -Pochhammer constant, and $-\mathcal{U}'(1)$.

The results are elementary enough to be formalized in stages, yet arithmetic enough to open new questions about cyclotomic units, regulators, periodic-orbit spectra, and Gaussian q -binomial transforms. They therefore fit the repository's intended frontier role: mathematically testable statements, clearly separated from imported results, and organized for eventual migration into Lean.

A Corpus inventory

The audit counted fifty-seven TeX files: twenty-five in the living documentation tree and thirty-two frozen historical versions. The list below records the path relative to `Analysis/FabiusFunction/docs`. Historical items are grouped by the archive directory; filenames follow the base name shown.

A.1 Living documentation: 25 TeX files

No.	Relative path
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	semi-formalized-research-frontiers/Non_Formalized_Research_Frontiers.tex
6	semi-formalized-research-frontiers/drafts/Fabius_Thue_Morse_Formula_Atlas/Fabius_Thue_Morse_Formula_Atlas.tex
7	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay.tex
8	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Fourier_decay_of_Rvachev_up.tex
9	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay.tex
14	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay.tex
15	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/Rvachev_Up_Fourier_Decay_Final.tex
16	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/Rvachev_Up_Fourier_Decay_Final_Synthesis.tex
17	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay_final.tex
18	semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay_final.tex
19	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
20	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
21	papers/arXiv-1502.06738v2/Lacunary_products.tex
22	papers/arXiv-1702.05442v1/09-Function.tex
23	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24	papers/arXiv-2211.13570v1/document.tex

No.	Relative path
25	<code>papers/ubiq15/ubiq15-corrected.tex</code>

A.2 Frozen archive: 32 TeX files

No.	Relative path under archive/
1	<code>early-notes/FabiusAsymptotic/FabiusAsymptotic.tex</code>
2	<code>early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex</code>
3	<code>standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex</code>
4	<code>standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex</code>
5	<code>standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex</code>
6	<code>primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex</code>
7	<code>primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex</code>
8	<code>primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex</code>
9	<code>primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex</code>
10	<code>primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex</code>
11	<code>primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex</code>
12	<code>primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex</code>
13	<code>primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex</code>
14	<code>q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex</code>
15	<code>q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex</code>
16	<code>lean-walkthrough/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex</code>
17	<code>lean-walkthrough/fabius_lean_walkthrough/fabius_lean_walkthrough.tex</code>
18	<code>glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex</code>
19	<code>glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex</code>
20	<code>research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex</code>
21	<code>research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex</code>

No.	Relative path under archive/
22	research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
23	research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
24	research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
25	research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
26	research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
27	research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
28	research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex
29	research-frontiers/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
30	research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
31	research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
32	research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

B Constants and numerical reference

For convenient independent checking:

$$\begin{aligned}
\kappa_0 &= 3.1514961294723187980432792951080073350\dots, \\
\kappa_{\min} &= 2.3590148791117407073164098231340990806\dots, \\
h &= 0.5537712758888100953442174163445488813\dots, \\
-\log_2 h &= 0.8526378718208337174366143795551934080\dots, \\
C_{\cot} &= -1.2837124730461260445945438959726042474\dots, \\
(1/2; 1/2)_{\infty} &= 0.2887880950866024212788997219292307801\dots, \\
\mathcal{C}_* &= 0.1599225518776068325779014528557986690\dots
\end{aligned}$$

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Part III

Spectral Arithmetic and the Pascal–Rvachev Hierarchy

Abstract

The existing Fabius-function corpus gives a remarkably broad account of the Rvachev up-function, exact dyadic values, q -binomial formulae, signed Thue–Morse summation, endpoint asymptotics, and the pointwise and metric decay of the Fourier product [8]

$$\Phi_1(z) = \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right), \quad \operatorname{sinc} z = \frac{\sin(\pi z)}{\pi z}.$$

The present report begins at a boundary not developed in that corpus: the arithmetic of the entire zero divisor and its interaction with repeated dyadic summation. For the classical product, the cumulative multiplicity of positive zeros up to N is exactly $2N - s_2(N)$; hence the Thue–Morse sign is the parity of a spectral counting function. This observation extends to a new Pascal–Rvachev hierarchy

$$\Phi_r(z) = \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{\binom{h}{r-1}}, \quad r \geq 1.$$

Each member is the Fourier transform of a C_c^∞ probability density on $[-1, 1]$ and satisfies the refinement ladder $\Phi_r(2z) = \Phi_r(z)\Phi_{r-1}(z)$, with $\Phi_0(z) = \operatorname{sinc}(2z)$. Its zero at an integer $n \neq 0$ has multiplicity $\binom{\nu_2(n)+1}{r}$, and the corresponding zero-divisor Dirichlet series is

$$\mathcal{Z}_r(s) = \zeta(s) \frac{2^{-(r-1)s}}{(1 - 2^{-s})^r}.$$

The cumulative count is the exact digital quantity $2N - S_r(N)$, where

$$S_r(N) = \sum_{j \geq 0} b_j(N) \sum_{k=0}^{r-1} \binom{j+1}{k}.$$

Modulo two, the inner weight reduces to $\binom{j}{r-1}$. Consequently, the signs between consecutive integer zeros form a 2-automatic generalized Thue–Morse sequence with an explicit Mahler equation and exact Prouhet cancellation order.

The same hierarchy has cumulants

$$\kappa_{2k}(X_r) = \frac{B_{2k}}{2k} \frac{2^{2k}}{(2^{2k} - 1)^r}, \quad \text{Var}(X_r) = 3^{-(r+1)}.$$

After variance normalization it converges to the standard Gaussian, with an exponential Berry–Esseen bound. Its Fourier envelope obeys the sharp law

$$\limsup_{|x| \rightarrow \infty} \frac{\log |\Phi_r(x)|}{(\log |x|)^{r+1}} = -\frac{1}{(r+1)!(\log 2)^r}.$$

Finally, the left-endpoint Laplace exponent admits a Mellin representation whose poles are exactly the same dyadic lattice that appears in $\mathcal{Z}_r$. The pole at zero supplies a degree- $(r+1)$ logarithmic polynomial, the nonzero lattice poles supply degree- $(r-1)$ periodic polynomials, and the leading saddle is solved by the lower branch W_{-1} of Lambert’s function. Complete proofs are given for the exact algebraic, digital, probabilistic, Fourier, and Mellin statements. The final endpoint transseries beyond leading Tauberian order is isolated as a conjectural program rather than asserted.

Frontier and proof status

Known inputs. The classical up-function sinc product, its probability interpretation, its canonical integer zero set with multiplicities $1 + \nu_2(n)$, its Thue–Morse sign law, and its rank-one endpoint/Fourier asymptotics are treated in the surveyed corpus and literature.

Proved in this report. The cumulative spectral counting identities; the Pascal–Rvachev hierarchy; its refinement, support, zero divisor, zeta function, exact digital count, automatic signs, Prouhet partitions, q -scale identities, cumulants, Gaussian rank limit, sharp Fourier envelope, and Mellin residue expansion.

External theorem used. The leading small-ball statement invokes the classical exponential Tauberian theorem of de Bruijn. All hypotheses needed here are displayed explicitly.

Not yet formalized. None of the new statements is claimed to have a Lean counterpart. A proposed formalization order appears in [appendix 11](#).

Novelty qualification. “New” means not found in the complete repository corpus or in targeted exact-phrase and formula searches carried out for this report. Infinite sums of independent uniform variables are a known general class. No claim of universal bibliographic priority is made for every specialization below.

1 Scope, corpus boundary, and normalization

1.1 What was reviewed

The source base is larger than a single exposition [1]. The living tree contains the primary article on the Fabius and Rvachev functions, a mathematical glossary, a Lean walkthrough, a non-elementarity paper, a consolidated non-formalized frontier volume, a large Thue–Morse formula atlas, twelve successive Fourier-decay dossiers and audits, and six embedded literature folders containing seven LaTeX sources. The archive adds first versions, parallel drafts, supplements, q -calculus articles, repeated-integration studies, dyadic-asymptotic bridges, and earlier Thue–Morse atlases. [Appendix B](#) gives a grouped inventory.

The review was used in three ways. First, it fixed normalization: several superficially different formulae in the drafts differ only by a Fourier convention, a translation, or a factor of two. Second, it set a novelty boundary: the extensive rank-one Fourier analysis and the full Fabius endpoint machinery were treated as prior input, not rediscovered. Third, archived documents were used as a negative control. A proposed result was not counted as new merely because it had disappeared from the living exposition.

The repository itself imposes a useful epistemic distinction. Formalization-backed claims belong in the primary exposition; non-formalized mathematics is quarantined in the frontier volume until an exact Lean theorem can be identified. This report follows that convention. A conventional paper proof is called a proof, but it is never described as formalized.

1.2 Fourier and probabilistic conventions

Throughout,

$$\operatorname{sinc} z := \begin{cases} \frac{\sin(\pi z)}{\pi z}, & z \neq 0, \\ 1, & z = 0, \end{cases} \quad \widehat{f}(\xi) := \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} \, dx.$$

Thus the Fourier transform of the uniform probability density on $[-a/2, a/2]$ is $\operatorname{sinc}(a\xi)$. The 2-adic valuation of a positive integer is $\nu_2(n)$, and

$$n = \sum_{j \geq 0} b_j(n) 2^j, \quad b_j(n) \in \{0, 1\}, \quad s_2(n) := \sum_{j \geq 0} b_j(n).$$

We use the generalized polynomial binomial coefficient $\binom{y}{k} = y(y-1) \cdots (y-k+1)/k!$, and set ordinary binomial coefficients to zero outside their natural range.

The classical Rvachev Fourier product is

$$\Phi_1(z) := \prod_{h=0}^{\infty} \operatorname{sinc}\left(\frac{z}{2^h}\right). \quad (157)$$

It is the characteristic function, in the above Fourier convention, of

$$X_1 = \sum_{h=0}^{\infty} U_h, \quad U_h \sim \operatorname{Unif}\left[-2^{-h-1}, 2^{-h-1}\right] \quad (158)$$

with independent summands. Its density is the normalized Rvachev up-function, supported on $[-1, 1]$. On the left half of that support, the standard Fabius function is

$$F(x) = f_1(x-1), \quad 0 \leq x \leq 1. \quad (159)$$

1.3 The established rank-one architecture

The surveyed sources [2, 3, 6, 7] already establish, discuss, or audit all of the following rank-one facts:

- (a) the probability and infinite-convolution realization of (157);
- (b) the two-scale identities for the up-function and the Fabius function;
- (c) exact dyadic values involving q -Pochhammer symbols and Gaussian q -binomial coefficients at $q = 1/2$;
- (d) moment formulae and arithmetic denominator questions;
- (e) the canonical product

$$\Phi_1(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{1+\nu_2(n)};$$

- (f) the sign law $\text{sgn } \Phi_1(x) = (-1)^{s_2(N)}$ for $N < x < N + 1$;
- (g) pointwise, metric, mean, extremal, annular, and near-zero analyses of the Fourier decay;
- (h) the lower-Lambert endpoint coordinate and the small periodic correction in logarithmic scale;
- (i) the digital, Mahler, Prouhet, rarefaction, and arithmetic-subsequence theory of the classical Thue–Morse sequence [9, 10, 11].

The new work below does not compete with those results. It asks what becomes visible when the zero divisor itself is counted, and then promotes the dyadic product to an entire Pascal-weighted ladder. That change of viewpoint creates simultaneous extensions in spectral arithmetic, automatic sequences, probability, Fourier analysis, and endpoint asymptotics.

2 The classical product as a spectral Thue–Morse object

2.1 Cumulative multiplicity and parity

For a positive integer n , the zero multiplicity of (157) is

$$\mu_1(n) = 1 + \nu_2(n).$$

Define the cumulative positive-zero multiplicity

$$\mathcal{N}_1(N) := \sum_{n=1}^N \mu_1(n).$$

Theorem 2.1 (Exact classical zero count). *For every integer $N \geq 0$,*

$$\mathcal{N}_1(N) = 2N - s_2(N). \tag{160}$$

Consequently,

$$(-1)^{\mathcal{N}_1(N)} = (-1)^{s_2(N)}. \tag{161}$$

Thus the Thue–Morse sign between N and $N + 1$ is exactly the parity of the number of positive zeros counted with multiplicity up to N .

Proof. Every factor indexed by h contributes a simple zero at each positive multiple of 2^h . Hence

$$\mathcal{N}_1(N) = \sum_{h \geq 0} \left\lfloor \frac{N}{2^h} \right\rfloor = N + \sum_{h \geq 1} \left\lfloor \frac{N}{2^h} \right\rfloor.$$

Legendre’s digit formula gives $\sum_{h \geq 1} \lfloor N/2^h \rfloor = N - s_2(N)$, proving (160). Since $2N$ is even, (161) follows. Finally, $\Phi_1(0) > 0$, and crossing the zero at n changes the sign exactly when $\mu_1(n)$ is odd. Therefore the sign on $(N, N+1)$ is $(-1)^{\mathcal{N}_1(N)}$. $\square$

Remark 2.2. The familiar sign law is therefore not merely an incidental binary identity. It is a mod-2 shadow of an exact Weyl-type counting formula whose discrepancy from the linear main term is the binary digit sum.

2.2 Finite products and a two-scale sign correction

Let

$$\Phi_{1,M}(z) := \prod_{h=0}^{M-1} \operatorname{sinc}\left(\frac{z}{2^h}\right), \quad M \geq 0.$$

At $n \geq 1$ its zero multiplicity is

$$\mu_{1,M}(n) = \min\{M, 1 + \nu_2(n)\}.$$

Proposition 2.3 (Finite-scale count). *Put $Q = \lfloor N/2^M \rfloor$. Then*

$$\mathcal{N}_{1,M}(N) := \sum_{n=1}^N \mu_{1,M}(n) = \sum_{h=0}^{M-1} \left\lfloor \frac{N}{2^h} \right\rfloor \quad (162)$$

$$= 2N - s_2(N) - 2Q + s_2(Q). \quad (163)$$

In particular,

$$(-1)^{\mathcal{N}_{1,M}(N)} = (-1)^{s_2(N)} (-1)^{s_2(Q)}. \quad (164)$$

Proof. The first equality counts the zeros supplied by each of the first M factors. The omitted tail is

$$\sum_{h \geq M} \left\lfloor \frac{N}{2^h} \right\rfloor = \sum_{k \geq 0} \left\lfloor \frac{Q}{2^k} \right\rfloor = 2Q - s_2(Q)$$

by [theorem 2.1](#). Subtracting it from $2N - s_2(N)$ proves the result. $\square$

The factor $(-1)^{s_2(Q)}$ in (164) is a coarse-scale correction: a truncation remembers both the Thue–Morse parity at N and the parity of the block index at scale 2^M .

2.3 A zero-divisor Dirichlet series

The canonical product suggests treating the zero multiplicities as an arithmetic function.

Definition 2.4. For $\Re s > 1$, define the zero-divisor Dirichlet series

$$\mathcal{Z}_1(s) := \sum_{n \geq 1} \frac{1 + \nu_2(n)}{n^s}, \quad \mathcal{Z}_{1,M}(s) := \sum_{n \geq 1} \frac{\mu_{1,M}(n)}{n^s}.$$

The word “spectral” below refers to the zero divisor of an entire Fourier transform, not to the spectrum of a specified operator.

Theorem 2.5 (Classical zero-divisor zeta function). *For $\Re s > 1$,*

$$\mathcal{Z}_1(s) = \frac{\zeta(s)}{1 - 2^{-s}}, \quad \mathcal{Z}_{1,M}(s) = \zeta(s) \frac{1 - 2^{-Ms}}{1 - 2^{-s}}. \quad (165)$$

Both identities extend meromorphically to $\mathbb{C}$. The infinite series has a simple pole at $s = 1$ with residue 2, and simple poles at

$$s_k := \frac{2\pi i k}{\log 2}, \quad k \in \mathbb{Z}. \quad (166)$$

At $s = s_k$ its residue is $\zeta(s_k)/\log 2$.

Proof. Write $1 + \nu_2(n) = \sum_{h \geq 0} \mathbf{1}_{2^h | n}$. Absolute convergence permits interchange:

$$\mathcal{Z}_1(s) = \sum_{h \geq 0} \sum_{m \geq 1} (2^h m)^{-s} = \zeta(s) \sum_{h \geq 0} 2^{-hs} = \frac{\zeta(s)}{1 - 2^{-s}}.$$

Truncating the geometric sum gives the second formula. The pole statements follow from the meromorphic continuation of ζ and from $1 - 2^{-s} = (s - s_k) \log 2 + O((s - s_k)^2)$ near s_k . On the imaginary axis ζ has no zeros: this follows from the functional equation and the classical zero-free line $\Re s = 1$. Thus the displayed poles are genuine. $\square$

First new bridge

The same dyadic denominator $1 - 2^{-s}$ has now appeared in three guises: repeated divisibility by 2, the zero divisor of the sinc product, and the q -integer $[M]_{2^{-s}} = (1 - 2^{-Ms})/(1 - 2^{-s})$. The next sections show that replacing the geometric weight 1 by Pascal weights $\binom{h}{r-1}$ raises that denominator to the r th power and simultaneously raises the Fourier and endpoint logarithmic orders.

3 The Pascal–Rvachev hierarchy

3.1 Definition and convergence

The exponent 1 in the classical product is the first column of Pascal’s triangle. Replacing it by successive columns gives the central construction of this report.

Definition 3.1 (Pascal–Rvachev products). Set

$$\Phi_0(z) := \text{sinc}(2z). \quad (167)$$

For every integer $r \geq 1$, define

$$a_{r,h} := \binom{h}{r-1}, \quad \Phi_r(z) := \prod_{h=0}^{\infty} \text{sinc}\left(\frac{z}{2^h}\right)^{a_{r,h}}. \quad (168)$$

The convention $\binom{h}{k} = 0$ for $h < k$ makes the product start effectively at $h = r - 1$.

Theorem 3.2 (Entire product and random-series model). *For each $r \geq 1$, the product (168) converges locally uniformly on $\mathbb{C}$ and defines an even entire function with $\Phi_r(0) = 1$. Let*

$$X_r := \sum_{h=0}^{\infty} \sum_{\ell=1}^{a_{r,h}} U_{h,\ell}, \quad U_{h,\ell} \sim \text{Unif}\left[-2^{-h-1}, 2^{-h-1}\right] \quad (169)$$

with all summands independent. Then the series converges absolutely almost surely and

$$\mathbb{E} e^{-2\pi i z X_r} = \Phi_r(z). \quad (170)$$

The law of X_r has a bounded density f_r , and

$$\text{supp } f_r = [-1, 1]. \quad (171)$$

Proof. For z in a fixed compact set and h sufficiently large,

$$\log \operatorname{sinc}(z/2^h) = \mathcal{O}_K(4^{-h}).$$

Since $a_{r,h} = \mathcal{O}_r(h^{r-1})$, the series $\sum_h a_{r,h} \log \operatorname{sinc}(z/2^h)$ converges normally away from the finitely many zeros of the initial factors; equivalently, the associated Weierstrass product converges locally uniformly. Evenness and normalization are immediate.

For the random series, the total sum of the absolute bounds is deterministic:

$$\begin{aligned} \sum_{h \geq 0} a_{r,h} 2^{-h-1} &= \frac{1}{2} \sum_{h \geq 0} \binom{h}{r-1} 2^{-h} \\ &= \frac{1}{2} \frac{2^{-(r-1)}}{(1 - 1/2)^r} = 1. \end{aligned} \quad (172)$$

Thus (169) converges absolutely and takes values in $[-1, 1]$. Multiplication of the characteristic functions gives (170). At least one summand is uniformly distributed on a nondegenerate interval, so convolution with the law of the remaining series produces a bounded density. The support of a convolution of interval-supported nondegenerate measures is the Minkowski sum of the supports. Equality in (172) therefore gives the full interval $[-1, 1]$. $\square$

Remark 3.3. General infinite sums of independent uniform variables occur elsewhere in probability and in the theory of limit laws for q -hook formulae [5, 12]. The distinctive feature here is the Pascal multiplicity $\binom{h}{r-1}$, which forces all of the exact arithmetic and refinement identities below.

3.2 The refinement ladder

Theorem 3.4 (Fourier refinement ladder). *For every $r \geq 1$,*

$$\Phi_r(2z) = \Phi_r(z) \Phi_{r-1}(z). \quad (173)$$

Equivalently, if X'_{r-1} is an independent copy of X_{r-1} independent of X_r , then

$$2X_r \stackrel{d}{=} X_r + X'_{r-1}. \quad (174)$$

For the densities,

$$f_r(x) = 2(f_r * f_{r-1})(2x), \quad f_0(x) = \frac{1}{2} \mathbf{1}_{[-1,1]}(x). \quad (175)$$

Proof. Shift the index in (168). For $r \geq 2$ the initial exponent $a_{r,0}$ vanishes, and Pascal's identity gives

$$\begin{aligned} \Phi_r(2z) &= \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{\binom{h+1}{r-1}} \\ &= \prod_{h \geq 0} \operatorname{sinc}\left(\frac{z}{2^h}\right)^{\binom{h}{r-1} + \binom{h}{r-2}} = \Phi_r(z) \Phi_{r-1}(z). \end{aligned}$$

For $r = 1$, the factor removed by the shift is $\operatorname{sinc}(2z) = \Phi_0(z)$, so the same formula holds. Fourier uniqueness proves (174). The density of $2X_r$ is $x \mapsto \frac{1}{2} f_r(x/2)$, whereas the right side of (174) has density $f_r * f_{r-1}$. Replacing x by $2x$ yields (175). $\square$

Corollary 3.5 (A moment ladder). *Let $m_{r,n} := \mathbb{E}[X_r^n]$. Odd moments vanish. For $n \geq 1$,*

$$(2^{2n} - 1)m_{r,2n} = m_{r-1,2n} + \sum_{j=1}^{n-1} \binom{2n}{2j} m_{r,2j} m_{r-1,2n-2j}, \quad (176)$$

with $m_{0,2n} = 1/(2n + 1)$. In particular, every moment of every X_r is rational.

Proof. Raise (174) to the $2n$ th power and take expectations. Symmetry removes odd terms. The term with $j = n$ is $m_{r,2n}$ and is moved to the left. The base law $X_0 \sim \operatorname{Unif}[-1, 1]$ has the stated moments. $\square$

3.3 The shifted endpoint variables

For endpoint analysis it is useful to translate the support to $[0, 2]$:

$$D_r := X_r + 1. \quad (177)$$

Writing $U_{h,\ell} = 2^{-h}(V_{h,\ell} - 1/2)$ with $V_{h,\ell} \sim \text{Unif}[0, 1]$ and using $\sum_h a_{r,h} 2^{-h} = 2$, we obtain the positive random series

$$D_r = \sum_{h \geq 0} \sum_{\ell=1}^{a_{r,h}} 2^{-h} V_{h,\ell}. \quad (178)$$

The refinement law becomes

$$2D_r \stackrel{d}{=} D_r + D'_{r-1}. \quad (179)$$

This is the endpoint counterpart of the Fourier ladder and will drive the Mellin and Lambert analysis in [appendix 8](#).

4 Zero divisors, digital counts, and generalized Thue–Morse signs

4.1 Integer zero multiplicities

Theorem 4.1 (Pascal zero multiplicity). *Let $r \geq 1$ and $n \in \mathbb{Z} \setminus \{0\}$. The multiplicity of the zero of Φ_r at n is*

$$\mu_r(n) = \sum_{h=0}^{\nu_2(|n|)} \binom{h}{r-1} = \binom{\nu_2(|n|) + 1}{r}. \quad (180)$$

Consequently,

$$\Phi_r(z) = \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right)^{\mu_r(n)}. \quad (181)$$

The product converges normally on compact sets.

Proof. The factor at scale h vanishes at n precisely when $2^h \mid n$, and then contributes multiplicity $\binom{h}{r-1}$. Summing over $0 \leq h \leq \nu_2(n)$ and applying the hockey-stick identity gives (180). Expanding each sinc by Euler’s product and collecting equal zeros yields (181). Normal convergence follows from $\sum_n \mu_r(n) n^{-2} < \infty$, a consequence of [theorem 4.3](#) below. $\square$

Formalization boundary. The finite arithmetic equality in (180) is now machine-checked: we `ightedScaleMultiplicity_choose` in `WeightedScaleMultiplicity.lean`, specialized to base 2 and Lean’s Pascal index $r - 1$, proves $\sum_{h=0}^{\nu_2(n)} \binom{h}{r-1} = \binom{\nu_2(n)+1}{r}$. It does *not* by itself prove that this number is the analytic zero order of Φ_r , nor does it establish (181) or normal convergence. Those assertions remain in the analytic frontier.

Thus Φ_r has a zero at n exactly when $2^{r-1} \mid n$. Higher rank does not spread the zeros; it thins their locations while increasing their multiplicities according to the r th binomial coefficient in $\nu_2(n) + 1$.

4.2 Finite scale and a Pascal tail formula

Define

$$\Phi_{r,M}(z) := \prod_{h=0}^{M-1} \text{sinc}\left(\frac{z}{2^h}\right)^{\binom{h}{r-1}}. \quad (182)$$

Its zero multiplicity is

$$\mu_{r,M}(n) = \binom{\min\{M, \nu_2(n) + 1\}}{r}. \quad (183)$$

Let

$$\mathcal{N}_{r,M}(N) := \sum_{n=1}^N \mu_{r,M}(n), \quad \mathcal{N}_r(N) := \sum_{n=1}^N \mu_r(n).$$

Proposition 4.2 (Exact finite-scale tail). *For $Q = \lfloor N/2^M \rfloor$,*

$$\mathcal{N}_{r,M}(N) = \mathcal{N}_r(N) - \sum_{a=0}^{r-1} \binom{M}{a} \mathcal{N}_{r-a}(Q). \quad (184)$$

For $r = 1$ this reduces to (163).

Proof. The omitted tail is

$$\begin{aligned} \mathcal{N}_r(N) - \mathcal{N}_{r,M}(N) &= \sum_{k \geq 0} \binom{M+k}{r-1} \left\lfloor \frac{Q}{2^k} \right\rfloor \\ &= \sum_{a=0}^{r-1} \binom{M}{a} \sum_{k \geq 0} \binom{k}{r-1-a} \left\lfloor \frac{Q}{2^k} \right\rfloor, \end{aligned}$$

where the second line is Vandermonde's identity. The inner sum is $\mathcal{N}_{r-a}(Q)$ by direct zero counting. $\square$

4.3 The zero-divisor zeta hierarchy

Theorem 4.3 (Zero-divisor zeta functions). *For $\Re s > 1$ and $r \geq 1$,*

$$\mathcal{Z}_r(s) := \sum_{n \geq 1} \frac{\mu_r(n)}{n^s} = \zeta(s) \frac{2^{-(r-1)s}}{(1 - 2^{-s})^r}. \quad (185)$$

For the finite product,

$$\mathcal{Z}_{r,M}(s) = \zeta(s) \sum_{h=0}^{M-1} \binom{h}{r-1} 2^{-hs}. \quad (186)$$

The meromorphic continuation of $\mathcal{Z}_r$ has:

- (i) a simple pole at $s = 1$ with residue 2, independent of r ;
- (ii) a pole of exact order r at every $s_k = 2\pi i k / \log 2$;
- (iii) leading Laurent coefficient

$$\mathcal{Z}_r(s) = \frac{\zeta(s_k)}{(\log 2)^r (s - s_k)^r} + O((s - s_k)^{1-r}) \quad (187)$$

near s_k .

Proof. From (180), or directly by counting scale contributions,

$$\begin{aligned} \mathcal{Z}_r(s) &= \sum_{h \geq 0} \binom{h}{r-1} \sum_{m \geq 1} (2^h m)^{-s} \\ &= \zeta(s) \sum_{h \geq 0} \binom{h}{r-1} (2^{-s})^h = \zeta(s) \frac{2^{-(r-1)s}}{(1 - 2^{-s})^r}. \end{aligned}$$

The finite identity follows by stopping at $h = M - 1$. At $s = 1$ the non-zeta factor equals $2^{-(r-1)}/(1 - 1/2)^r = 2$, so the residue is 2. Near s_k , the numerator $2^{-(r-1)s_k} = 1$ and $1 - 2^{-s} = (s - s_k) \log 2 + O((s - s_k)^2)$. As in theorem 2.5, $\zeta(s_k) \neq 0$, proving exact order and (187). $\square$

The factor

$$\mathcal{A}_r(s) := \frac{2^{-(r-1)s}}{(1 - 2^{-s})^r} \quad (188)$$

will reappear, unchanged, in the endpoint Mellin transform. It is the scale zeta function of the hierarchy; multiplication by $\zeta(s)$ converts scale multiplicity into integer-zero multiplicity.

Corollary 4.4 (Rank resolvent). *For $|u2^{-s}| < |1 - 2^{-s}|$,*

$$\sum_{r \geq 1} \mathcal{Z}_r(s) u^{r-1} = \frac{\zeta(s)}{1 - (1+u)2^{-s}}. \quad (189)$$

At each fixed integer $n \geq 1$,

$$\sum_{r \geq 1} \mu_r(n) u^{r-1} = \frac{(1+u)^{\nu_2(n)+1} - 1}{u}. \quad (190)$$

Proof. Sum the geometric series in r using (185). The second identity is the binomial theorem applied to (180). $\square$

4.4 An exact digital zero-counting law

The following finite combinatorial identity is the key.

Lemma 4.5 (Pascal tail identity). *For integers $r \geq 1$ and $j \geq 0$, put*

$$Q_r(j) := \sum_{k=0}^{r-1} \binom{j+1}{k}. \quad (191)$$

Then

$$\sum_{h=0}^j \binom{h}{r-1} 2^{j-h} = 2^{j+1} - Q_r(j). \quad (192)$$

Equivalently,

$$2^j \sum_{h=j+1}^{\infty} \binom{h}{r-1} 2^{-h} = Q_r(j).$$

Proof. The right side of (192) counts the subsets of a $(j+1)$ -element set having cardinality at least r . Classify such a subset by its r th smallest element h . Choose the $r-1$ smaller elements in $\binom{h}{r-1}$ ways and choose an arbitrary subset of the $j-h$ larger elements in 2^{j-h} ways. Summing over h gives the left side. $\square$

Theorem 4.6 (Exact digital spectral count). *For every $r \geq 1$ and $N \geq 0$,*

$$\mathcal{N}_r(N) = \sum_{h \geq 0} \binom{h}{r-1} \left\lfloor \frac{N}{2^h} \right\rfloor, \quad (193)$$

while the binary expansion $N = \sum_j b_j(N) 2^j$ gives

$$\boxed{\mathcal{N}_r(N) = 2N - S_r(N)}, \quad S_r(N) := \sum_{j \geq 0} b_j(N) Q_r(j). \quad (194)$$

For $r = 1$, $Q_1(j) = 1$ and $S_1(N) = s_2(N)$.

Proof. Equation (193) counts the zeros supplied by each scale. Expanding each floor in binary,

$$\begin{aligned}\mathcal{N}_r(N) &= \sum_{h \geq 0} \binom{h}{r-1} \sum_{j \geq h} b_j(N) 2^{j-h} \\ &= \sum_{j \geq 0} b_j(N) \sum_{h=0}^j \binom{h}{r-1} 2^{j-h}.\end{aligned}$$

Apply lemma 4.5. The contribution of 2^{j+1} is $2N$, and the remaining term is $S_r(N)$. $\square$

The floor-sum half (193) is now formally covered by `sum_range_padicValNat_choose` in `WeightedScaleMultiplicity.lean`; its finite sum over $h < N$ equals the displayed infinite sum because all later natural quotients vanish. The binary digit formula (194), the sharp discrepancy statements, and their parity consequences are not consequences of that generic layer-cake theorem and remain to be formalized.

Corollary 4.7 (Sharp discrepancy range). *For $0 \leq N < 2^m$,*

$$0 \leq 2N - \mathcal{N}_r(N) \leq \sum_{\ell=1}^r \binom{m+1}{\ell} - 1. \quad (195)$$

The upper bound is attained at $N = 2^m - 1$. In particular, for fixed r ,

$$\mathcal{N}_r(N) = 2N + \mathcal{O}_r((\log N)^r),$$

and the exponent r in the error term is optimal.

Proof. All weights $Q_r(j)$ are nonnegative, so among m -bit integers $S_r(N)$ is maximized when every bit is 1. The hockey-stick identity gives

$$\sum_{j=0}^{m-1} Q_r(j) = \sum_{k=0}^{r-1} \sum_{j=0}^{m-1} \binom{j+1}{k} = \sum_{\ell=1}^r \binom{m+1}{\ell} - 1.$$

Its leading term is $m^r/r!$. $\square$

4.5 Pascal parity and automatic sign sequences

The apparently complicated weight $Q_r(j)$ has a simple parity.

Lemma 4.8 (Parity collapse). *For all $r \geq 1$ and $j \geq 0$,*

$$Q_r(j) \equiv \binom{j}{r-1} \pmod{2}. \quad (196)$$

Proof. The exact alternating partial-sum identity

$$\sum_{k=0}^{r-1} (-1)^{r-1-k} \binom{j+1}{k} = \binom{j}{r-1}$$

reduces modulo 2 to (196). $\square$

Definition 4.9 (Spectral sign sequence). Define

$$\varepsilon_r(N) := (-1)^{\mathcal{N}_r(N)} = (-1)^{S_r(N)} = (-1)^{\sum_{j \geq 0} b_j(N) \binom{j}{r-1}}. \quad (197)$$

Theorem 4.10 (Generalized Thue–Morse sign law). *For $N < x < N + 1$,*

$$\operatorname{sgn} \Phi_r(x) = \varepsilon_r(N). \quad (198)$$

Let

$$m_r := \lceil \log_2 r \rceil, \quad P_r := 2^{m_r}, \quad d_r := 2^{m_r - s_2(r-1)}. \quad (199)$$

Then the digit-position weights

$$\sigma_{r,j} := (-1)^{\binom{j}{r-1}} \quad (200)$$

are periodic in j with period P_r , and exactly d_r positions in one period have value -1 . Consequently, ε_r is 2-automatic and strongly 2^{P_r} -multiplicative.

Proof. The sign formula follows as in [theorem 2.1](#): start positive at zero and count integer zeros with multiplicity. By [lemma 4.8](#), $\varepsilon_r(N) = \prod_{j: b_j(N)=1} \sigma_{r,j}$.

Since $r - 1 < P_r$, Lucas's theorem shows that the parity of $\binom{j}{r-1}$ depends only on the lowest m_r binary digits of j . Hence P_r is a period. Lucas's theorem also says that $\binom{j}{r-1}$ is odd precisely when every 1-bit of $r - 1$ is also a 1-bit of j . Among the $P_r = 2^{m_r}$ residue classes, the remaining $m_r - s_2(r - 1)$ bits are free, giving d_r negative weights.

A finite automaton need only remember the current digit position modulo P_r and the accumulated sign, proving 2-automaticity. Grouping binary digits into blocks of length P_r shows strong 2^{P_r} -multiplicativity. $\square$

Corollary 4.11 (Mahler equation). *For $|z| < 1$, let*

$$G_r(z) := \sum_{N \geq 0} \varepsilon_r(N) z^N, \quad A_r(z) := \prod_{j=0}^{P_r-1} (1 + \sigma_{r,j} z^{2^j}). \quad (201)$$

Then

$$G_r(z) = \prod_{j \geq 0} (1 + \sigma_{r,j} z^{2^j}) \quad (202)$$

and

$$G_r(z) = A_r(z) G_r(z^{2^{P_r}}). \quad (203)$$

Proof. Expanding the product in (202) selects a binary digit set, and the coefficient is exactly the product of the corresponding $\sigma_{r,j}$. Split the product into the first P_r positions and the periodic tail. $\square$

Corollary 4.12 (Powers of two). *If $r = 2^m$, then*

$$\binom{j}{r-1} \equiv 1 \pmod{2} \iff j \equiv -1 \pmod{2^m}. \quad (204)$$

Hence $\varepsilon_{2^m}(N)$ is the parity of the binary digits of N in positions congruent to $2^m - 1$ modulo 2^m .

Proof. The binary expansion of $r - 1 = 2^m - 1$ consists of m ones. Lucas's criterion therefore requires all lowest m bits of j to be one. $\square$

Table 12: The first spectral sign masks. A minus sign means that a 1-bit in that position flips ε_r .

r	P_r	d_r	$\text{Var}(X_r)$	$(\sigma_{r,0}, \dots, \sigma_{r,P_r-1})$
1	1	1	1/9	—
2	2	1	1/27	+—
3	4	2	1/81	++--
4	4	1	1/243	+++—
5	8	4	1/729	++++--
6	8	2	1/2187	+++++—
7	8	2	1/6561	+++++—
8	8	1	1/19683	+++++—

4.6 Spectral Prouhet partitions

The classical Thue–Morse signs produce Prouhet partitions [9, 10]. The spectral signs do the same, with an order dictated by the number d_r of negative positions in a period.

Theorem 4.13 (Exact Prouhet cancellation order). *Let $r \geq 1$ and $B \geq 1$. Put $L = 2^{P_r B}$. Then*

$$\sum_{N=0}^{L-1} \varepsilon_r(N) N^q = 0, \quad 0 \leq q < Bd_r, \quad (205)$$

and the sum is nonzero for $q = Bd_r$. Thus the two sets

$$\{0 \leq N < L : \varepsilon_r(N) = 1\}, \quad \{0 \leq N < L : \varepsilon_r(N) = -1\}$$

have equal sums of like powers through the exact degree $Bd_r - 1$.

Proof. The finite generating polynomial factors as

$$\begin{aligned} T_{r,B}(z) &:= \sum_{N=0}^{L-1} \varepsilon_r(N) z^N = \prod_{j=0}^{P_r B-1} (1 + \sigma_{r,j} z^{2^j}) \\ &= \prod_{b=0}^{B-1} A_r(z^{2^{P_r b}}). \end{aligned} \quad (206)$$

In each period exactly d_r factors have the form $1 - z^{2^j}$ and vanish simply at $z = 1$; the remaining factors tend to 2. Hence $T_{r,B}$ has a zero of exact order Bd_r at 1. Its q th derivative at 1 is the signed sum of falling factorials $N(N-1) \cdots (N-q+1)$. Vanishing for all $q < Bd_r$ is therefore equivalent, by triangular change of basis, to (205) for ordinary powers. Exact order gives nonvanishing at $q = Bd_r$. $\square$

Example 4.14. For $r = 1$, $P_1 = d_1 = 1$ and [theorem 4.13](#) is the classical length- 2^B Prouhet–Thue–Morse cancellation through degree $B - 1$. For $r = 3$, $P_3 = 4$ and $d_3 = 2$: on the interval $0 \leq N < 16^B$, the spectral signs cancel powers through degree $2B - 1$.

4.7 Statistics of the spectral discrepancy

The exact discrepancy $S_r(N) = 2N - \mathcal{N}_r(N)$ is a polynomially weighted digit sum. It therefore has its own central limit theorem.

Theorem 4.15 (Digital central limit theorem). *Let U_m be uniform on $\{0, 1, \dots, 2^m - 1\}$. Then*

$$M_{r,m} := \mathbb{E}S_r(U_m) = \frac{1}{2} \left(\sum_{\ell=1}^r \binom{m+1}{\ell} - 1 \right), \quad (207)$$

$$V_{r,m} := \text{Var}(S_r(U_m)) = \frac{1}{4} \sum_{j=0}^{m-1} Q_r(j)^2. \quad (208)$$

For fixed r ,

$$M_{r,m} \sim \frac{m^r}{2r!}, \quad V_{r,m} \sim \frac{m^{2r-1}}{4(2r-1)((r-1)!)^2}. \quad (209)$$

Moreover,

$$\frac{S_r(U_m) - M_{r,m}}{\sqrt{V_{r,m}}} \implies \mathcal{N}(0, 1), \quad (210)$$

with Kolmogorov error $\mathcal{O}_r(m^{-1/2})$.

Proof. The bits of U_m are independent Bernoulli variables B_j of parameter $1/2$, and $S_r(U_m) = \sum_{j < m} Q_r(j) B_j$. The mean and variance follow immediately; the closed mean formula was computed in [corollary 4.7](#). Since $Q_r(j) \sim j^{r-1}/(r-1)!$, integral comparison gives (209).

The largest centered summand divided by the standard deviation is $\mathcal{O}(m^{-1/2})$, so Lindeberg's condition holds. The Berry–Esseen theorem [17] gives the stated error because

$$\frac{\sum_{j < m} Q_r(j)^3}{\left(\sum_{j < m} Q_r(j)^2 \right)^{3/2}} = \mathcal{O}_r(m^{-1/2}).$$

□

Spectral arithmetic summary

The integer zero divisor now carries three nested layers of binary information:

1. the full multiplicity $\binom{\nu_2(n)+1}{r}$;
2. the exact cumulative discrepancy $S_r(N)$, a polynomially weighted digit sum;
3. the mod-2 sign mask $\binom{j}{r-1} \bmod 2$, a periodic row of Pascal's triangle generating an automatic sequence and exact Prouhet partitions.

For $r = 1$ these collapse respectively to $1 + \nu_2(n)$, $s_2(N)$, and the ordinary Thue–Morse sign.

5 q -scale calculus: Pascal maxima versus Gaussian sums

The repository's dyadic-value formulae use Gaussian q -binomial coefficients because distinct scale selections are weighted by the *sum* of their indices. The Pascal–Rvachev hierarchy introduces a companion statistic: the *maximum* selected scale. This section makes that distinction exact.

5.1 The finite Pascal kernel

For $M \geq 0$, $r \geq 1$, and an indeterminate q , define

$$P_{r,M}(q) := \sum_{h=0}^{M-1} \binom{h}{r-1} q^h. \quad (211)$$

Proposition 5.1 (Three forms of the finite kernel). *One has*

$$P_{r,M}(q) = [u^{r-1}] \frac{1 - (q(1+u))^M}{1 - q(1+u)}, \quad (212)$$

$$= \frac{q^{r-1}}{(r-1)!} \left(\frac{d}{dq} \right)^{r-1} \frac{1 - q^M}{1 - q}, \quad (213)$$

$$= \sum_{0 \leq h_1 < \dots < h_r < M} q^{h_r}. \quad (214)$$

For $|q| < 1$,

$$P_r(q) := \sum_{h \geq 0} \binom{h}{r-1} q^h = \frac{q^{r-1}}{(1-q)^r}. \quad (215)$$

Proof. Expand $\sum_{h=0}^{M-1} q^h (1+u)^h$ and extract u^{r-1} to obtain (212). Differentiating the finite geometric series gives (213). For (214), fix the maximum $h_r = h$ and choose the remaining $r-1$ indices below it in $\binom{h}{r-1}$ ways. Letting $M \rightarrow \infty$ in the coefficient form gives (215). $\square$

With $q = 2^{-s}$, (186) becomes simply

$$\mathcal{Z}_{r,M}(s) = \zeta(s) P_{r,M}(2^{-s}), \quad \mathcal{Z}_r(s) = \zeta(s) P_r(2^{-s}). \quad (216)$$

The rank-one finite kernel is the q -integer $P_{1,M}(q) = [M]_q$.

5.2 The Gaussian companion

The finite q -binomial theorem gives

$$\prod_{h=0}^{M-1} (1 + uq^h) = \sum_{r=0}^M q^{\binom{r}{2}} \begin{bmatrix} M \\ r \end{bmatrix}_q u^r, \quad (217)$$

where

$$\begin{bmatrix} M \\ r \end{bmatrix}_q = \frac{(1 - q^M)(1 - q^{M-1}) \dots (1 - q^{M-r+1})}{(1 - q)(1 - q^2) \dots (1 - q^r)}.$$

Equating coefficients gives

$$\sum_{0 \leq h_1 < \dots < h_r < M} q^{h_1 + \dots + h_r} = q^{\binom{r}{2}} \begin{bmatrix} M \\ r \end{bmatrix}_q. \quad (218)$$

Compare (214) and (218):

selection statistic	enumerator	role in the Fabius–Rvachev theory
$h_1 + \dots + h_r$	$q^{\binom{r}{2}} \begin{bmatrix} M \\ r \end{bmatrix}_q$	Gaussian q -binomial dyadic formulae
$\max(h_1, \dots, h_r)$	$P_{r,M}(q)$	Pascal scale multiplicity and zero zeta

A max/sum bridge

The ordinary Pascal coefficient $\binom{h}{r-1}$ and the Gaussian coefficient $\begin{bmatrix} M \\ r \end{bmatrix}_q$ are not rival appearances of “binomial coefficients.” They enumerate the same increasing scale tuples with two different statistics. The Pascal–Rvachev hierarchy weights a tuple only by its terminal scale; the classical Gaussian identity weights it by the sum of all selected scales. This provides a concrete route

for transferring coefficient-extraction techniques from the dyadic Fabius formulae to the new hierarchy.

5.3 Local logarithms and the same q -kernel

Euler's product for the sine gives, for $|z| < 2^{r-1}$,

$$\log \Phi_r(z) = - \sum_{k=1}^{\infty} \frac{\zeta(2k)}{k} \frac{2^{-2k(r-1)}}{(1 - 2^{-2k})^r} z^{2k} = - \sum_{k=1}^{\infty} \frac{\zeta(2k)}{k} \frac{2^{2k}}{(2^{2k} - 1)^r} z^{2k}. \quad (219)$$

Thus the same infinite Pascal kernel $P_r(q) = q^{r-1}(1 - q)^{-r}$ controls both the integer-zero Dirichlet series at $q = 2^{-s}$ and the Taylor logarithm at $q = 2^{-2k}$.

6 Cumulants, concentration, and a Gaussian rank limit

6.1 Exact cumulants

Let B_{2k} denote the Bernoulli numbers with $B_2 = 1/6$ and $B_4 = -1/30$.

Theorem 6.1 (Cumulant formula). *All odd cumulants of X_r vanish, and for $k \geq 1$,*

$$\kappa_{2k}(X_r) = \frac{B_{2k}}{2k} \frac{2^{-2k(r-1)}}{(1 - 2^{-2k})^r} = \frac{B_{2k}}{2k} \frac{2^{2k}}{(2^{2k} - 1)^r}. \quad (220)$$

In particular,

$$\text{Var}(X_r) = \frac{1}{3^{r+1}}, \quad \kappa_4(X_r) = -\frac{2}{15^{r+1}}, \quad \kappa_6(X_r) = \frac{16}{63^{r+1}}. \quad (221)$$

Proof. A centered uniform variable of width a has even cumulants $B_{2k}a^{2k}/(2k)$. Cumulants add under independent summation, so (169) gives

$$\kappa_{2k}(X_r) = \frac{B_{2k}}{2k} \sum_{h \geq 0} \binom{h}{r-1} 2^{-2kh}.$$

Apply (215) with $q = 2^{-2k}$ and simplify. The displayed special cases follow directly. $\square$

Corollary 6.2 (Rational moments and exact standardized cumulants). *Let*

$$Y_r := 3^{(r+1)/2} X_r, \quad (222)$$

so that $\text{Var}(Y_r) = 1$. For $k \geq 2$,

$$\kappa_{2k}(Y_r) = \frac{B_{2k}}{2k} 2^{2k} 3^k \left(\frac{3^k}{2^{2k} - 1} \right)^r. \quad (223)$$

Since $3^k < 2^{2k} - 1$ for $k \geq 2$, every standardized cumulant of order greater than two tends exponentially to zero.

6.2 A quantitative central limit theorem in the rank

Although every unscaled density has the same compact support $[-1, 1]$, its variance shrinks by a factor $1/3$ at each rank. After the natural dilation, the shapes become Gaussian.

Theorem 6.3 (Gaussian rank limit). *As $r \rightarrow \infty$,*

$$Y_r = 3^{(r+1)/2} X_r \implies Z, \quad Z \sim \mathcal{N}(0, 1). \quad (224)$$

Equivalently, for every fixed $z \in \mathbb{R}$,

$$\Phi_r\left(3^{(r+1)/2} z\right) \longrightarrow e^{-2\pi^2 z^2}. \quad (225)$$

More quantitatively, there is an absolute Berry–Esseen constant C_{BE} such that

$$\sup_x |\mathbb{P}(Y_r \leq x) - \mathbb{P}(Z \leq x)| \leq C_{\text{BE}} \frac{3^{3/2}}{4} \left(\frac{3^{3/2}}{7}\right)^r. \quad (226)$$

Proof. Regard the independent summands in (169), after division by $\sigma_r = 3^{-(r+1)/2}$, as a triangular array. The largest absolute normalized summand is

$$\frac{2^{-r}}{\sigma_r} = \sqrt{3} \left(\frac{\sqrt{3}}{2}\right)^r \longrightarrow 0,$$

while the total variance is one. Lindeberg’s condition follows, proving (224); (225) is the characteristic-function form.

For a centered uniform variable of width a , $\mathbb{E}|U|^3 = a^3/32$. Therefore the sum of third absolute moments before normalization is

$$\sum_{h \geq 0} \binom{h}{r-1} \frac{2^{-3h}}{32} = \frac{1}{32} \frac{2^{-3(r-1)}}{(1-2^{-3})^r} = \frac{1}{4 \cdot 7^r}.$$

Dividing by $\sigma_r^3 = 3^{-3(r+1)/2}$ and applying Berry–Esseen yields (226). Infinite sums follow by truncation, since the omitted variances and third moments tend to zero. $\square$

Remark 6.4. The exponential ratio $3^{3/2}/7 \approx 0.7423$ is not merely qualitative. It shows that the central profile stabilizes rapidly in rank even though the endpoint and zero-divisor structures become algebraically more complicated.

6.3 Two complementary asymptotic regimes

The rank limit and the fixed-rank frequency limit describe different geometries:

- (a) For $r \rightarrow \infty$ at central scale $|x| \asymp 3^{-(r+1)/2}$, the density is asymptotically Gaussian after dilation.
- (b) For fixed r and $|\xi| \rightarrow \infty$, the Fourier transform has a logarithmic stretched-exponential envelope of order $r+1$, proved next.
- (c) Near the support endpoint, fixed-rank small deviations are governed by the same order $r+1$, but with a periodic polynomial correction and a Lambert- W saddle.

The hierarchy therefore separates central Gaussian universality from dyadic endpoint arithmetic.

7 Sharp Fourier decay for every rank

7.1 A global upper envelope

Lemma 7.1 (Binomial summation identity). *For integers $H \geq 0$ and $r \geq 1$,*

$$\sum_{h=0}^H \binom{h}{r-1} (H-h) = \binom{H+1}{r+1}. \quad (227)$$

Proof. Both sides count pairs (S, t) where S is an r -element subset of $\{0, 1, \dots, H\}$ and t is an integer strictly larger than the largest element of S . Alternatively, it follows by two applications of the hockey-stick identity. $\square$

Theorem 7.2 (Global logarithmic decay). *For fixed $r \geq 1$ and $|x| \geq 2$,*

$$\log |\Phi_r(x)| \leq -\frac{(\log |x|)^{r+1}}{(r+1)!(\log 2)^r} + \mathcal{O}_r((\log |x|)^r). \quad (228)$$

Equivalently, for some constants $C_r, c_r > 0$,

$$|\Phi_r(x)| \leq C_r \exp\left(-\frac{(\log |x|)^{r+1}}{(r+1)!(\log 2)^r} + c_r(\log |x|)^r\right). \quad (229)$$

Proof. Let $H = \lfloor \log_2 |x| \rfloor$. For $0 \leq h \leq H$, $|x|/2^h \geq 1$ and

$$\left| \operatorname{sinc}\left(\frac{x}{2^h}\right) \right| \leq \frac{2^h}{\pi |x|}.$$

Taking logarithms and retaining only these factors gives

$$\begin{aligned} \log |\Phi_r(x)| &\leq -\sum_{h=0}^H \binom{h}{r-1} (\log |x| - h \log 2 + \log \pi) \\ &= -\log 2 \binom{H+1}{r+1} + \mathcal{O}_r(H^r), \end{aligned}$$

where [lemma 7.1](#) absorbs the leading distance from h to H , and the difference between $\log |x|$ and $H \log 2$ contributes only $\mathcal{O}_r(H^r)$. Since $H = (\log |x|)/(\log 2) + \mathcal{O}(1)$, the result follows. $\square$

7.2 A sharp dyadic ray

A global upper bound is useful only if its leading constant is attainable away from the zeros. The rational ray $2/3$ supplies exactly constant sine amplitudes under doubling.

Theorem 7.3 (Sharp ray and exact limsup). *Let*

$$x_m := \frac{2}{3} 2^m. \quad (230)$$

Then, as $m \rightarrow \infty$,

$$\log |\Phi_r(x_m)| = -\log 2 \binom{m+1}{r+1} + \log\left(\frac{3\sqrt{3}}{4\pi}\right) \binom{m+1}{r} + \mathcal{O}_r(m^{r-1}). \quad (231)$$

In particular,

$$\log |\Phi_r(x_m)| = -\frac{\log 2}{(r+1)!} m^{r+1} + \mathcal{O}_r(m^r), \quad (232)$$

and

$$\boxed{\limsup_{|x| \rightarrow \infty} \frac{\log |\Phi_r(x)|}{(\log |x|)^{r+1}} = -\frac{1}{(r+1)!(\log 2)^r}}. \quad (233)$$

Proof. For $0 \leq h \leq m$, powers of two modulo 3 alternate between 1 and 2, so

$$\left| \sin\left(\pi \frac{x_m}{2^h}\right) \right| = \frac{\sqrt{3}}{2}.$$

Consequently,

$$\log \left| \operatorname{sinc}\left(\frac{x_m}{2^h}\right) \right| = -(m-h) \log 2 + \log \left(\frac{3\sqrt{3}}{4\pi} \right).$$

Summing with weights $\binom{h}{r-1}$ and using

$$\sum_{h=0}^m \binom{h}{r-1} = \binom{m+1}{r}, \quad \sum_{h=0}^m \binom{h}{r-1} (m-h) = \binom{m+1}{r+1}$$

gives the first two terms. For $h = m+j, j \geq 1$, the argument is $\frac{2}{3}2^{-j}$ and $\log \operatorname{sinc} y = O(y^2)$. Hence the tail is

$$\mathcal{O}_r \left(\sum_{j \geq 1} (m+j)^{r-1} 4^{-j} \right) = \mathcal{O}_r(m^{r-1}).$$

This proves (231). Combine the resulting lower sequence for the limsup with the global upper bound in theorem 7.2. $\square$

Corollary 7.4 (Smooth compact densities). *For every $r \geq 1$, the density f_r belongs to $C_c^\infty(\mathbb{R})$.*

Proof. Equation (228) implies $|x|^A \Phi_r(x) \rightarrow 0$ for every $A > 0$. Hence $(2\pi i \xi)^k \Phi_r(\xi)$ is integrable for every k , and Fourier inversion permits differentiation of all orders. Compact support was proved in theorem 3.2. $\square$

Remark 7.5. The leading logarithmic order rises from 2 for the classical up-function to $r+1$ at rank r . Nevertheless, along the sharp ray the decay is slower than $e^{-|x|^\alpha}$ for every $\alpha > 0$. Thus every member is superalgebraically smooth but retains the characteristic lacunary-product gap between polynomial and stretched-exponential Fourier decay.

8 Endpoint Laplace asymptotics and Lambert W

8.1 Laplace products and a discrete-antiderivative ladder

Let

$$L_r(t) := \mathbb{E} e^{-tD_r}, \quad \Psi_r(t) := -\log L_r(t), \quad t > 0. \quad (234)$$

The positive random-series representation (178) gives

$$L_r(t) = \prod_{h=0}^{\infty} \left(\frac{1 - e^{-t2^{-h}}}{t2^{-h}} \right)^{\binom{h}{r-1}}. \quad (235)$$

For $D_0 \sim \operatorname{Unif}[0, 2]$,

$$L_0(t) = \frac{1 - e^{-2t}}{2t}. \quad (236)$$

Theorem 8.1 (Laplace refinement). *For every $r \geq 1$,*

$$L_r(2t) = L_r(t) L_{r-1}(t), \quad \Psi_r(2t) = \Psi_r(t) + \Psi_{r-1}(t). \quad (237)$$

If

$$\psi_r(y) := \Psi_r(2^y), \quad (238)$$

then

$$\psi_r(y+1) - \psi_r(y) = \psi_{r-1}(y), \quad (239)$$

with

$$\psi_0(y) = (y+1) \log 2 - \log(1 - e^{-2^{y+1}}). \quad (240)$$

The polynomial sequence

$$\mathcal{P}_r(y) := \log 2 \binom{y+1}{r+1} \quad (241)$$

satisfies the same ladder $\mathcal{P}_r(y+1) - \mathcal{P}_r(y) = \mathcal{P}_{r-1}(y)$ for the polynomial part of (240).

Proof. Equation (237) is the Laplace-transform form of $2D_r \stackrel{d}{=} D_r + D'_{r-1}$, or follows directly from (235) and Pascal's identity. The remaining statements are substitutions, together with $\binom{y+2}{r+1} - \binom{y+1}{r+1} = \binom{y+1}{r}$. $\square$

The base residual

$$-\log(1 - e^{-2^{y+1}})$$

decays doubly exponentially as $y \rightarrow +\infty$. Repeated discrete antiderivation therefore turns the rank-one periodic constant into a periodic polynomial of degree $r-1$. The Mellin harmonic-sum analysis [13] below gives an explicit Fourier series for those periodic coefficients and links them to the zero-divisor zeta function.

8.2 The common scale zeta factor

Define

$$\rho(u) := \log \frac{u}{1 - e^{-u}}, \quad u > 0. \quad (242)$$

Then

$$\Psi_r(t) = \sum_{h \geq 0} \binom{h}{r-1} \rho(t2^{-h}). \quad (243)$$

Lemma 8.2 (Mellin transform of the uniform Laplace kernel). *For $-1 < \Re s < 0$,*

$$\int_0^\infty \rho(u) u^{s-1} du = \Gamma(s) \zeta(s+1). \quad (244)$$

Proof. At zero, $\rho(u) = u/2 + O(u^2)$; at infinity, $\rho(u) = \log u + O(e^{-u})$, so the displayed strip is the fundamental strip. Since

$$\rho'(u) = \frac{1}{u} - \frac{1}{e^u - 1},$$

integration by parts and the standard analytically continued Bose–Einstein Mellin integral give

$$\int_0^\infty \rho(u) u^{s-1} du = -\frac{1}{s} \int_0^\infty \left(\frac{1}{u} - \frac{1}{e^u - 1} \right) u^s du = \frac{\Gamma(s+1)}{s} \zeta(s+1),$$

which is (244). $\square$

Theorem 8.3 (Mellin bridge). *For $0 < c < 1$,*

$$\Psi_r(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(-z) \zeta(1-z) \mathcal{A}_r(z) t^z \, dz, \quad (245)$$

where the scale zeta factor is the same function as in (188):

$$\mathcal{A}_r(z) = \frac{2^{-(r-1)z}}{(1-2^{-z})^r}. \quad (246)$$

Thus

$$\mathcal{Z}_r(z) = \zeta(z) \mathcal{A}_r(z), \quad \mathcal{M}_{\text{endpoint},r}(z) = \Gamma(-z) \zeta(1-z) \mathcal{A}_r(z). \quad (247)$$

Proof. Mellin inversion in lemma 8.2, followed by $z = -s$, gives

$$\rho(u) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} \Gamma(-z) \zeta(1-z) u^z \, dz.$$

Insert this in (243). Since $c > 0$,

$$\sum_{h \geq 0} \binom{h}{r-1} 2^{-hz} = \frac{2^{-(r-1)z}}{(1-2^{-z})^r}.$$

Absolute convergence on the initial line justifies interchange. This proves (245); (247) compares with theorem 4.3. $\square$

Exact spectral–endpoint bridge

The arithmetic zero divisor and the left-endpoint Laplace exponent differ only in the universal analytic multiplier attached to the same scale zeta function:

$$\zeta(z) \longleftrightarrow \Gamma(-z) \zeta(1-z).$$

The dyadic pole lattice is therefore not an analogy imported after the fact. It is literally common to the two transforms.

8.3 Residues and periodic logarithmic polynomials

Let

$$\omega := \frac{2\pi}{\log 2}, \quad z_k := ik\omega, \quad k \in \mathbb{Z}. \quad (248)$$

These are exactly the zeros of $1 - 2^{-z}$.

Theorem 8.4 (Complete logarithmic residue expansion). *For every $A > 0$, as $t \rightarrow \infty$,*

$$\Psi_r(t) = \sum_{k \in \mathbb{Z}} \text{Res}_{z=z_k} [\Gamma(-z) \zeta(1-z) \mathcal{A}_r(z) t^z] + \mathcal{O}_{r,A}(t^{-A}). \quad (249)$$

The residue series converges absolutely after collecting coefficients of powers of $\log t$. Equivalently, with $L = \log t$ and $u = \log_2 t$, there exist real constants $c_{r,j}$ and real-analytic 1-periodic zero-mean functions $p_{r,j}$ such that

$$\begin{aligned} \Psi_r(t) &= \frac{L^{r+1}}{(r+1)!(\log 2)^r} + \frac{1 - \frac{1}{2}r(r-1)}{(r+1)!(\log 2)^{r-1}} L^r \\ &\quad + \sum_{j=0}^{r-1} (c_{r,j} + p_{r,j}(u)) L^j + \mathcal{O}_{r,A}(t^{-A}). \end{aligned} \quad (250)$$

The highest periodic coefficient has the explicit Fourier series

$$p_{r,r-1}(u) = \frac{1}{(r-1)!(\log 2)^r} \sum_{k \neq 0} \Gamma(-ik\omega) \zeta(1-ik\omega) e^{2\pi iku}. \quad (251)$$

In particular, the leading periodic coefficient at every rank is a universal lift of the rank-one oscillation:

$$p_{r,r-1}(u) = \frac{p_{1,0}(u)}{(r-1)!(\log 2)^{r-1}}. \quad (252)$$

Proof. Shift the contour in (245) from $\Re z = c$ to $\Re z = -A$. The poles crossed are precisely z_k , all on the imaginary axis. At $z = 0$, the factors $\Gamma(-z)$ and $\zeta(1-z)$ each have a simple pole, while $\mathcal{A}_r$ has a pole of order r ; hence the total order is $r+2$ and its residue is a polynomial of degree $r+1$ in L . At $z_k \neq 0$, the gamma and zeta factors are regular and $\mathcal{A}_r$ has exact order r , so each residue is $e^{2\pi iku}$ times a polynomial of degree $r-1$ in L .

Stirling's formula gives exponential decay $|\Gamma(-ik\omega)| = \exp(-\pi|k|\omega/2)$ up to a polynomial factor. Standard bounds for ζ on vertical lines then give absolute convergence of every coefficient Fourier series and justify the contour shift. The integral on $\Re z = -A$ is $\mathcal{O}_{r,A}(t^{-A})$.

Near zero,

$$\Gamma(-z)\zeta(1-z) = \frac{1}{z^2} + \mathcal{O}(1)$$

(the coefficient of z^{-1} cancels), while

$$\mathcal{A}_r(z) = \frac{1}{(\log 2)^r z^r} \left(1 + \left(1 - \frac{r}{2} \right) z \log 2 + \mathcal{O}_r(z^2) \right).$$

Taking the residue after multiplying by e^{zL} yields the first two coefficients in (250).

At $z = z_k \neq 0$,

$$\mathcal{A}_r(z) = \frac{1}{(\log 2)^r (z - z_k)^r} + \mathcal{O}((z - z_k)^{1-r}),$$

because $2^{-(r-1)z_k} = 1$. The coefficient of L^{r-1} in the residue is therefore exactly (251). The case $r = 1$ gives (252). $\square$

Remark 8.5 (Why the oscillation is tiny). The first harmonic contains the factor $\Gamma(-i\omega)$, whose magnitude is on the scale $\exp(-\pi^2/\log 2)$. This is roughly 6×10^{-7} before the remaining algebraic factors. The very small periodic correction seen in rank-one endpoint computations is therefore forced by the distance between consecutive dyadic complex dimensions.

Corollary 8.6 (Rank-one specialization). For $r = 1$,

$$\Psi_1(t) = \frac{(\log t)^2}{2 \log 2} + \frac{1}{2} \log t + c_{1,0} + p_{1,0}(\log_2 t) + \mathcal{O}_A(t^{-A}). \quad (253)$$

For general r , the pole at zero raises the polynomial degree to $r+1$, while the nonzero lattice poles raise the periodic degree only to $r-1$.

8.4 Small deviations

Let

$$H_r(x) := \mathbb{P}(D_r \leq x), \quad 0 \leq x \leq 2. \quad (254)$$

Theorem 8.7 (Leading endpoint small-ball law). For every fixed $r \geq 1$,

$$-\log H_r(x) \sim \frac{(\log(1/x))^{r+1}}{(r+1)!(\log 2)^r}, \quad x \downarrow 0. \quad (255)$$

Proof. By [theorem 8.4](#),

$$\Psi_r(t) \sim C_r (\log t)^{r+1}, \quad C_r = \frac{1}{(r+1)! (\log 2)^r}.$$

This is a slowly varying Laplace exponent of logarithmic power greater than one. De Bruijn's exponential Tauberian theorem for nonnegative random variables [\[14\]](#) then gives $-\log H_r(x) \sim C_r (\log(1/x))^{r+1}$. $\square$

Remark 8.8. The elementary Chernoff inequality already shows why the constant is unchanged: at the optimizing scale, tx is of order $(\log t)^r$, one logarithmic degree below $\Psi_r(t)$. The Tauberian theorem supplies the matching reverse inequality.

8.5 Recovery of the Fabius endpoint

Let g_1 be the density of D_1 , so $g_1(x) = f_1(x-1)$. On $[0, 1]$, g_1 is the standard Fabius function F . Since D_0 is uniform on $[0, 2]$, the density form of [\(179\)](#) gives, for $0 \leq x \leq 1$,

$$g_1(x) = 2(g_1 * g_0)(2x) = \int_0^{2x} g_1(u) \, du = H_1(2x).$$

Therefore

$$H_1(x) = F(x/2), \quad 0 \leq x \leq 2. \quad (256)$$

The rank-one case of [theorem 8.7](#) is exactly the leading Fabius small-argument law, with the harmless factor 2 affecting only lower logarithmic orders.

8.6 The lower-Lambert saddle in every rank

Differentiating the leading term of [\(250\)](#) gives the model saddle equation

$$xt \sim \frac{(\log t)^r}{r! (\log 2)^r}. \quad (257)$$

Proposition 8.9 (Canonical Lambert coordinate). *For sufficiently small $x > 0$, define*

$$\Lambda_r(x) := -r W_{-1} \left(-\frac{(r!)^{1/r} \log 2}{r} x^{1/r} \right). \quad (258)$$

Then $\Lambda_r(x) > 0$ is the large solution of

$$x e^{\Lambda_r(x)} = \frac{\Lambda_r(x)^r}{r! (\log 2)^r}. \quad (259)$$

Moreover,

$$\Lambda_r(x) = \log \frac{1}{x} + r \log \log \frac{1}{x} - \log(r! (\log 2)^r) + o(1). \quad (260)$$

Proof. Taking the r th root of [\(259\)](#) and rearranging gives

$$\left(-\frac{\Lambda}{r} \right) e^{-\Lambda/r} = -\frac{(r!)^{1/r} \log 2}{r} x^{1/r}.$$

The solution tending to $+\infty$ uses the lower real branch W_{-1} . The standard expansion of W_{-1} at zero [\[15, 16\]](#) gives [\(260\)](#). $\square$

At this model saddle, the leading Legendre expression is

$$\mathcal{J}_r^{(0)}(x) := \frac{\Lambda_r(x)^{r+1}}{(r+1)!(\log 2)^r} - \frac{\Lambda_r(x)^r}{r!(\log 2)^r}. \quad (261)$$

Its first term gives [theorem 8.7](#); the second term is the universal $-tx$ correction. To continue beyond this level one must solve the saddle equation with the degree- r polynomial term and the degree- $(r-1)$ periodic polynomial in [\(250\)](#).

Conjecture 8.10 (Full Pascal–Lambert endpoint expansion). *For each $r \geq 2$, the exact small-ball rate $-\log H_r(x)$ admits an all-orders asymptotic expansion in descending powers of $\Lambda_r(x)$ whose coefficients are real-analytic 1-periodic functions of the saddle phase*

$$\frac{\log t_*(x)}{\log 2} \pmod{1},$$

where $t_*(x)$ is the exact positive saddle for $\Psi_r(t) - tx$. The highest periodic coefficient is determined by [\(251\)](#); lower coefficients are obtained recursively by saddle inversion. The rank-one construction developed in the existing corpus is the base case.

What is and is not claimed

The Mellin expansion [\(250\)](#), the leading Tauberian law [\(255\)](#), and the Lambert coordinate [\(258\)](#) are proved. A complete density or distribution transseries after nonlinear saddle inversion is not asserted for $r \geq 2$. That is the principal analytic frontier opened by the hierarchy.

9 A unified dictionary and further exact consequences

The preceding sections were developed from several viewpoints, but they are not merely parallel analogies. This section records exact identities that pass directly between rank, zeros, cumulants, and endpoints.

9.1 A master rank germ

The complete refinement ladder can be compressed into one two-variable analytic germ.

Theorem 9.1 (Master rank germ). *For $|z| < 1$ and $|1+u| < 4$, define*

$$\mathfrak{P}(u, z) := \exp \left(\sum_{h \geq 0} (1+u)^h \log \operatorname{sinc} \left(\frac{z}{2^h} \right) \right), \quad (262)$$

where the logarithm is the analytic branch vanishing at $z = 0$. Then

$$\mathfrak{P}(u, 2z) = \operatorname{sinc}(2z) \mathfrak{P}(u, z)^{1+u}. \quad (263)$$

For $|u| < 3$,

$$\log \mathfrak{P}(u, z) = \sum_{r \geq 1} u^{r-1} \log \Phi_r(z). \quad (264)$$

Thus the coefficient of u^{r-1} in the logarithm of a single nonlinear two-scale germ is exactly the rank- r Pascal–Rvachev product.

Proof. The series in (262) converges normally because $\log \operatorname{sinc}(z/2^h) = \mathcal{O}_z(4^{-h})$. Shifting the scale index gives

$$\log \mathfrak{P}(u, 2z) = \log \operatorname{sinc}(2z) + (1+u) \sum_{h \geq 0} (1+u)^h \log \operatorname{sinc}\left(\frac{z}{2^h}\right),$$

which is (263). Finally,

$$(1+u)^h = \sum_{r \geq 1} \binom{h}{r-1} u^{r-1}$$

with only finitely many nonzero terms for each h . Normal convergence permits coefficient extraction and yields (264). $\square$

Remark 9.2. At $u = 0$, (263) is the classical identity $\Phi_1(2z) = \operatorname{sinc}(2z)\Phi_1(z)$. Differentiating with respect to u at zero successively recovers the logarithmic refinement laws for all higher ranks. The singular boundary $u = 3$ reflects the variance denominator 3^r and is the first rank-plane obstruction visible in the local Taylor expansion.

9.2 A bivariate polynomial for all finite zero counts

Proposition 9.3 (Rank-generating spectral count). *For an integer $N \geq 0$, the finite polynomial*

$$\mathcal{R}_N(u) := \sum_{r \geq 1} \mathcal{N}_r(N) u^{r-1} \quad (265)$$

has the two exact forms

$$\mathcal{R}_N(u) = \sum_{n=1}^N \frac{(1+u)^{\nu_2(n)+1} - 1}{u}, \quad (266)$$

$$= \frac{2N - \sum_{j \geq 0} b_j(N)(1+u)^{j+1}}{1-u}. \quad (267)$$

All apparent singularities are removable. Formula (266) records the valuations of every integer up to N , while (267) reconstructs the same polynomial from the binary digits of N alone.

Proof. Sum (190) over $n \leq N$ to obtain (266). On the other hand,

$$\sum_{r \geq 1} Q_r(j) u^{r-1} = \sum_{k=0}^{j+1} \binom{j+1}{k} \sum_{r \geq k+1} u^{r-1} = \frac{(1+u)^{j+1}}{1-u}$$

for $|u| < 1$. Summing (194) in r gives (267); both sides are polynomials after cancellation, so the identity holds algebraically. At $u = 1$, the numerator vanishes because $\sum_j b_j(N) 2^{j+1} = 2N$. $\square$

9.3 A zero-divisor trace formula for cumulants

Theorem 9.4 (Spectral zeta-cumulant identity). *For $|z| < 2^{r-1}$,*

$$\log \Phi_r(z) = - \sum_{k \geq 1} \frac{\mathcal{Z}_r(2k)}{k} z^{2k}. \quad (268)$$

Consequently, for every $k \geq 1$,

$$\kappa_{2k}(X_r) = \frac{B_{2k}}{2k \zeta(2k)} \mathcal{Z}_r(2k) \quad (269)$$

$$= (-1)^{k+1} \frac{2(2k-1)!}{(2\pi)^{2k}} \mathcal{Z}_r(2k). \quad (270)$$

Thus every even cumulant is an explicit positive-even-value trace of the entire zero divisor.

Proof. Take the logarithm of the canonical product (181) and expand $\log(1-w) = -\sum_{k \geq 1} w^k/k$. The nearest nonzero zero has modulus 2^{r-1} , giving (268). Comparing (185) with (220) proves (269). Euler's evaluation

$$\zeta(2k) = (-1)^{k+1} \frac{B_{2k}(2\pi)^{2k}}{2(2k)!}$$

then gives (270). □

Corollary 9.5 (Rank resolvent for cumulants). For $|u| < 2^{2k} - 1$,

$$\sum_{r \geq 1} \kappa_{2k}(X_r) u^{r-1} = \frac{B_{2k}}{2k} \frac{2^{2k}}{2^{2k} - 1 - u}. \quad (271)$$

The nearest rank-plane pole occurs at $u = 3$ when $k = 1$, explaining why variance controls the analytic radius of the master germ.

9.4 The common Fourier–endpoint constant

Put

$$C_r := \frac{1}{(r+1)!(\log 2)^r}. \quad (272)$$

Combining theorems 7.3 and 8.7 yields the exact duality

$$\limsup_{|\xi| \rightarrow \infty} \frac{\log |\Phi_r(\xi)|}{(\log |\xi|)^{r+1}} = -C_r, \quad \lim_{x \downarrow 0} \frac{-\log H_r(x)}{(\log(1/x))^{r+1}} = C_r. \quad (273)$$

The equality of constants is not a generic uncertainty-principle statement. Both sides arise from the same Pascal scale sum and ultimately from the order- r pole of $\mathcal{A}_r$ at zero. Frequency growth counts the number of active coarse sinc factors; endpoint decay counts the cost of forcing the same dyadic uniform summands simultaneously toward zero.

9.5 Result map

Table 13: New deductions and their principal mathematical bridges.

Object	Exact result proved here	Bridge opened
Classical zero count	$\mathcal{N}_1(N) = 2N - s_2(N)$	Thue–Morse parity becomes spectral counting parity.
Pascal–Rvachev product	$\Phi_r(z) = \prod_h \text{sinc}(z/2^h)^{\binom{h}{r-1}}$	A rank ladder extending the up-function without changing support.

Object	Exact result proved here	Bridge opened
Refinement	$\Phi_r(2z) = \Phi_r(z)\Phi_{r-1}(z)$	Pascal's identity becomes convolution and moment recurrences.
Integer zero divisor	$\mu_r(n) = \binom{\nu_2(n)+1}{r}$	Valuations, entire products, and rank-generating polynomials.
Zero zeta	$\mathcal{Z}_r(s) = \zeta(s)2^{-(r-1)s}(1-2^{-s})^{-r}$	A dyadic complex-pole lattice of order r .
Digital discrepancy	$\mathcal{N}_r(N) = 2N - \sum_j b_j(N)Q_r(j)$	Exact $O((\log N)^r)$ Weyl discrepancy and a digital CLT.
Spectral signs	$\varepsilon_r(N) = (-1)^{\sum_j b_j(N)\binom{j}{r-1}}$	Automatic sequences, Mahler equations, and Pascal parity.
Prouhet partitions	Exact cancellation degree $Bd_r - 1$ on $0 \leq N < 2^{P_r B}$	A family whose efficiency is governed by $s_2(r-1)$.
q -scale calculus	$P_{r,M}(q) = \sum_{h < M} \binom{h}{r-1} q^h$	Max-weighted scale tuples complement Gaussian sum-weighted tuples.
Cumulants	$\kappa_{2k} = \frac{B_{2k}}{2k\zeta(2k)} \mathcal{Z}_r(2k)$	A trace formula from the zero divisor to probability.
Rank asymptotics	$3^{(r+1)/2} X_r \Rightarrow \mathcal{N}(0,1)$ exponentially fast	Compactly supported smooth laws approach a Gaussian centrally.
Fourier envelope	Sharp coefficient $-C_r$ at logarithmic order $r+1$	Pascal multiplicity raises the classical quadratic log law.
Endpoint Laplace exponent	A degree- $(r+1)$ polynomial plus periodic degree- $(r-1)$ terms	The same scale zeta controls endpoint complex dimensions.
Lambert coordinate	$\Lambda_r = -rW_{-1}(-c_r x^{1/r})$	Canonical inversion of the higher-rank endpoint saddle.

Central synthesis

One meromorphic scale factor,

$$\mathcal{A}_r(s) = \frac{2^{-(r-1)s}}{(1-2^{-s})^r},$$

organizes the hierarchy. At positive even integers it gives cumulants and the local Fourier logarithm; after multiplication by $\zeta(s)$ it gives the entire zero divisor; at its imaginary pole lattice it gives logarithmically periodic endpoint terms; and its order at zero fixes the common Fourier/endpoint exponent $r+1$. The generalized Thue–Morse signs are the mod-2 boundary trace of the same divisor.

10 Frontier program

The hierarchy turns several isolated questions in the existing rank-one corpus into systematic programs. The following problems are ordered roughly from the most immediately approachable to the most analytically demanding.

10.1 Exact higher-rank endpoint transseries

Problem A: nonlinear inversion of the residue expansion

Prove or disprove [conjecture 8.10](#). A complete solution requires:

- (i) expansion of the exact saddle $t_*(x)$ in powers of $\Lambda_r(x)$;
- (ii) propagation of the periodic polynomials $p_{r,j}$ through that inversion;
- (iii) control of the saddle prefactor for the distribution and, separately, for the density;
- (iv) an error estimate uniform in the dyadic phase.

The Mellin coefficients are explicit and exponentially convergent, so the obstruction is no longer coefficient discovery but rigorous nonlinear asymptotic inversion.

10.2 Dyadic values and higher-rank q -arithmetic

For $r = 1$, exact dyadic Fabius values are governed by Gaussian q -binomial coefficients and q -Pochhammer products. The refinement law $f_r(x) = 2(f_r * f_{r-1})(2x)$ suggests a triangular exact evaluation scheme at dyadic points, but convolution couples ranks.

Problem B: a Pascal/Gaussian dyadic formula

Find a closed form for $f_r(m/2^n)$ or $H_r(m/2^n)$ that displays simultaneously:

- (i) the max-enumerator $P_{r,M}(q)$ from [appendix 5](#);
- (ii) Gaussian q -binomial coefficients arising from sums of selected scales;
- (iii) the rank refinement $r \mapsto r - 1$.

A likely algebraic object is a two-statistic enumerator of increasing scale tuples by both their sum and their maximum. Its specializations recover the two q -structures already isolated in [\(214\)](#) and [\(218\)](#).

10.3 Arithmetic of moments and denominators

The moment ladder proves rationality but not optimal denominator bounds. The cumulant trace formula expresses moments through Bell polynomials in $\mathcal{Z}_r(2), \mathcal{Z}_r(4), \dots$; each is a rational multiple of a power of π , and the powers cancel exactly in cumulants.

Problem C: denominator geometry

Determine the exact prime valuations of the reduced denominators of $m_{r,2n}$. Natural targets are:

- (i) a von Staudt–Clausen-type description for cumulant denominators;
- (ii) a rank-recursive denominator bound from [\(176\)](#);
- (iii) eventual behavior in r for fixed n ;
- (iv) an integrality normalization extending the known rank-one dyadic arithmetic.

10.4 Automatic signs, rarefaction, and spectral measures

The sequence ε_r is strongly 2^{P_r} -multiplicative. Its one-period mask is explicit, so the standard matrix-product machinery for automatic sequences is available.

Problem D: harmonic analysis of the spectral signs

For each rank, determine:

- (i) the exact rarefied sums $\sum_{n < N, n \equiv a \pmod{m}} \varepsilon_r(n)$;
- (ii) the maximal Lyapunov exponents of the associated matrix cocycles;
- (iii) the diffraction or spectral measure and its pure-point/singular-continuous decomposition;
- (iv) the natural boundary behavior of $G_r(z)$ on $|z| = 1$.

The dependence of the mask density $d_r/P_r = 2^{-s_2(r-1)}$ on the binary weight of the rank predicts sharp arithmetic phase changes as r varies.

10.5 Smoothed zero counts and complex dimensions

The exact unsmoothed count has only a polylogarithmic discrepancy, whereas the Dirichlet series has an order- r imaginary pole lattice. A Perron or Mellin inversion with smoothing should turn those poles into explicit periodic polynomials in $\log_2 x$.

Problem E: a fractal explicit formula

For a smooth compactly supported test function w , obtain an explicit formula for

$$\sum_{n \geq 1} \mu_r(n) w(n/x)$$

whose secondary terms are indexed by $s_k = 2\pi i k / \log 2$. Quantify how the order- r poles produce degree- $(r-1)$ logarithmic amplitudes, and compare the coefficient functions directly with those in the endpoint expansion (250).

10.6 Local Gaussian limits and moderate deviations in rank

The Berry–Esseen estimate controls distribution functions, not densities or derivatives. Since every f_r is smooth and the normalized support grows like $3^{r/2}$, stronger central asymptotics should hold.

Problem F: a rank Edgeworth theory

Prove a local limit theorem, with derivatives, for

$$3^{-(r+1)/2} f_r \left(3^{-(r+1)/2} x \right),$$

and derive an Edgeworth expansion from (223). Determine the largest moderate-deviation window in which the Gaussian approximation remains uniform before the dyadic endpoint geometry becomes visible.

10.7 Beyond the binary Pascal ladder

The finite weighted arithmetic already works over every natural base, even a composite one: `WeightedScaleMultiplicity.lean` proves the scale-shift, divisibility-filter, layer-count, and Pascal floor

identities without a primality hypothesis. The separate five-theorem module `BaseDigitMultiplicity.lean` now also closes the unweighted digit recovery at rank one: for every $b > 1$ and $N \geq 0$, it proves both the additive and quotient forms of $\sum_{n=1}^N (1 + \nu_b(n)) = (bN - s_b(N))/(b-1)$, with no primality hypothesis and a total $N = 0$ boundary. It does not supply the weighted higher-rank digit formula (194). What remains speculative is the *analytic* base- b hierarchy—its entire products, zero divisors, Fourier asymptotics, and endpoint theory. Replacing integral rank by a fractional exponent would additionally require a branch-consistent entire-product theory.

11 Lean formalization roadmap

The new results have very different formalization costs. A useful strategy is to separate finite arithmetic from analytic limits instead of beginning with the infinite product.

11.1 Tier I: finite combinatorics and digital arithmetic

This tier is the fastest route to certified novelty and needs little measure theory.

I.1 Completed generically. The definition

`weightedScaleMultiplicity b w n`

works in every additive commutative monoid and specializes at $b = 2$, $w(h) = \binom{h}{r-1}$ to the proposed Pascal multiplicity, with a documented totalized convention at zero.

I.2 Completed generically. The hockey-stick identity behind (180) is `weightedScaleMultiplicity_choose`; the scale-count identity (193) is `sum_range_padicValNat_choose`.

I.3 Completed at rank one in every base. `BaseDigitMultiplicity.lean` proves the finite logarithmic truncation and the additive and quotient digit-sum identities for all $b > 1$, including composite bases, and all $N \geq 0$. This does not prove the analytic zero-order/sign interpretation or the weighted formula for $r \geq 2$.

I.4 Formalize lemma 4.5 as a finite subset count; this avoids delicate algebraic normalization.

I.5 Derive (194), the discrepancy extremum, and optimal order.

I.6 Prove parity collapse using the alternating partial-binomial identity or Lucas’s theorem already available in `Mathlib`.

I.7 Implement the periodic mask, count its negative entries, and certify the finite Prouhet factorization (206).

Candidate theorem names include

```
weightedScaleMultiplicity_choose
sum_range_padicValNat_choose
pascalZeroMult_eq_choose
zeroCount_eq_twoMul_sub_digitWeight
digitWeight_modTwo
spectralSign_powerSum_eq_zero
```

11.2 Tier II: finite products, moments, and rationality

Define $\Phi_{r,M}$ as a finite product and model the finite random sum by a product probability space. Then formalize:

II.1 the finite refinement identity;

- II.2** the exact finite zero multiplicity and tail formula;
- II.3** the finite moment recurrence;
- II.4** convergence of every fixed moment as $M \rightarrow \infty$;
- II.5** rationality of the limiting moments by the closed recurrence.

This tier can establish much of the algebra without invoking complex entire functions.

11.3 Tier III: probability and Fourier limits

The random-series construction is likely easier to formalize before the infinite entire product. Since the deterministic sum of absolute radii is 1, almost-sure convergence is immediate from comparison rather than a probabilistic convergence theorem. The milestones are:

- III.1** existence of X_r as an L^1 and almost-sure limit;
- III.2** support in $[-1, 1]$ and characteristic-function convergence;
- III.3** the distributional refinement law;
- III.4** variance and cumulants, initially through moments if a cumulant library is unavailable;
- III.5** the Fourier upper bound and the explicit sharp ray;
- III.6** C_c^∞ regularity by Fourier inversion.

11.4 Tier IV: complex analysis and Tauberian theory

The canonical product, meromorphic continuation, Mellin contour shift, and de Bruijn theorem are the expensive layer. They should be modularized as reusable infrastructure:

- IV.1** an Euler-product theorem for paired zeros with summable multiplicities;
- IV.2** a library lemma for binomially weighted geometric Dirichlet series;
- IV.3** Mellin inversion for $\rho(u) = \log(u/(1 - e^{-u}))$;
- IV.4** residue extraction for an infinite vertical lattice with exponentially decaying coefficients;
- IV.5** an exponential Tauberian theorem specialized to logarithmic powers.

The endpoint transseries should not be formalized before its paper proof is complete. By contrast, Tiers I and II are self-contained and already expose the main Thue–Morse and q -arithmetic novelty.

Recommended first formalization target

The weighted-scale and Pascal floor foundations are complete, and the rank-one digit formula is complete in every base. The best next finite target is therefore the still-open higher-rank digital part of [theorem 4.6](#), together with [lemmas 4.5](#) and [4.8](#) and [theorem 4.13](#). It has no analytic dependencies and would certify the generalized Thue–Morse sign layer. The probability hierarchy can then be developed with the generic arithmetic API already fixed.

12 Conclusion

The original rank-one sinc product contains more arithmetic than its pointwise sign pattern suggests. Counting its zeros with multiplicity gives the exact formula $2N - s_2(N)$, so the Thue–Morse sequence is the parity boundary of a spectral counting law. Promoting the constant scale multiplicity to the

Pascal column $\binom{h}{r-1}$ reveals that this is the first case of a coherent hierarchy rather than an isolated coincidence.

The hierarchy is unusually rigid. Its support remains $[-1, 1]$ at every rank; its refinement is exactly Pascal's identity; its zero multiplicities are binomial polynomials in ν_2 ; its zero zeta function is elementary; its finite counts are digital; its signs are automatic; its blocks solve exact Prouhet problems; its moments are rational; and its normalized central laws become Gaussian. In the two most delicate asymptotic regimes, the same coefficient $C_r = ((r+1)!(\log 2)^r)^{-1}$ governs sharp Fourier decay and endpoint small deviations.

The most consequential structural result is the common scale zeta factor. It converts a scale enumeration into the entire zero divisor, the local Fourier logarithm, all even cumulants, and the endpoint Mellin transform. Its real pole at zero determines the main logarithmic degree; its imaginary dyadic lattice determines periodic corrections; and the resulting endpoint saddle is inverted by W_{-1} . This places the Thue–Morse, q -binomial, sinc-product, and Lambert- W phenomena within one exact analytic mechanism.

The principal unresolved frontier is now sharply delimited: carry the explicit Mellin residue polynomial through nonlinear saddle inversion to obtain a complete higher-rank endpoint transseries, and connect its coefficients to exact dyadic q -formulae. The generic weighted-scale and Pascal floor foundations and the unweighted rank-one digit count in every base are formalized; the higher-rank finite digital/sign layer and the analytic endpoint program remain, with the latter supplying a longer-term test bed for new formal Mellin and Tauberian infrastructure.

A Reproducibility checks

The following independent script checks the finite identities over substantial ranges, verifies the exact sign masks and Prouhet orders, compares both forms of the rank-generating count polynomial at rational points, and numerically checks refinement, the spectral zeta–cumulant trace formula, and the Lambert saddle equation. It was run with Python 3 and mpmath; the output was

All exact and numerical frontier checks passed.

These checks are not substitutes for proofs, but they are effective protection against index shifts, normalization errors, and branch mistakes.

Listing 5: Exact and high-precision verification script.

```
from fractions import Fraction
from math import comb, factorial
import mpmath as mp

mp.mp.dps = 60

def C(n: int, k: int) -> int:
    return comb(n, k) if 0 <= k <= n else 0

def v2(n: int) -> int:
    assert n > 0
    e = 0
    while n % 2 == 0:
        n //= 2
        e += 1
    return e

def bit_sum(n: int) -> int:
```

```

    return n.bit_count()

def Q(r: int, j: int) -> int:
    return sum(C(j + 1, k) for k in range(r))

def mu(r: int, n: int) -> int:
    return C(v2(n) + 1, r)

def zero_count_direct(r: int, N: int) -> int:
    return sum(mu(r, n) for n in range(1, N + 1))

def zero_count_floor(r: int, N: int) -> int:
    total = 0
    h = 0
    while (N >> h) > 0:
        total += C(h, r - 1) * (N >> h)
        h += 1
    return total

def zero_count_digital(r: int, N: int) -> int:
    discrepancy = sum(
        ((N >> j) & 1) * Q(r, j) for j in range(max(1, N.bit_length()))
    )
    return 2 * N - discrepancy

def zero_count_finite(r: int, M: int, N: int) -> int:
    return sum(C(h, r - 1) * (N >> h) for h in range(M))

def epsilon(r: int, N: int) -> int:
    exponent = sum(
        ((N >> j) & 1) * C(j, r - 1)
        for j in range(max(1, N.bit_length()))
    )
    return -1 if exponent % 2 else 1

# Exact zero counts, digital formula, and finite Pascal tail.
for r in range(1, 10):
    for N in range(1000):
        a = zero_count_direct(r, N)
        assert a == zero_count_floor(r, N) == zero_count_digital(r, N)
        for M in range(0, 9):
            q = N >> M
            tail = sum(C(M, a0) * zero_count_direct(r - a0, q)
                       for a0 in range(r))
            assert zero_count_finite(r, M, N) == a - tail

# Parity masks and their exact Lucas counts.
for r in range(1, 33):
    m = (r - 1).bit_length()
    period = 1 << m
    mask = [(-1 if C(j, r - 1) % 2 else 1) for j in range(period)]
    assert sum(x == -1 for x in mask) == 1 << (m - bit_sum(r - 1))
    for N in range(2000):
        assert epsilon(r, N) == (-1 if zero_count_direct(r, N) % 2 else 1)

```

```

# Exact Prouhet cancellation and first nonzero moment.
for r in range(1, 9):
    m = (r - 1).bit_length()
    period = 1 << m
    d = 1 << (m - bit_sum(r - 1))
    L = 1 << period
    for q in range(d):
        assert sum(epsilon(r, n) * n**q for n in range(L)) == 0
        assert sum(epsilon(r, n) * n**d for n in range(L)) != 0

# The rank-generating count polynomial, tested at rational points.
for N in range(1, 300):
    for u in (Fraction(-1, 3), Fraction(1, 5), Fraction(2, 3)):
        lhs = sum(Fraction(zero_count_direct(r, N)) * u**(r - 1)
                  for r in range(1, N.bit_length() + 2))
        digit_poly = sum(((N >> j) & 1) * (1 + u)**(j + 1)
                          for j in range(N.bit_length()))
        rhs = (2 * N - digit_poly) / (1 - u)
        assert lhs == rhs

def sinc(x):
    return mp.mpf(1) if x == 0 else mp.sin(mp.pi * x) / (mp.pi * x)

def phi(r: int, z, H: int = 120):
    if r == 0:
        return sinc(2 * z)
    ans = mp.mpf(1)
    for h in range(H):
        a = C(h, r - 1)
        if a:
            ans *= sinc(z / 2**h) ** a
    return ans

# Numerical refinement and the spectral-zeta/cumulant trace formula.
for r in range(1, 7):
    for z in (mp.mpf('0.137'), mp.mpf('0.371')):
        assert mp.almosteq(phi(r, 2*z), phi(r, z) * phi(r - 1, z),
                           rel_eps=mp.mpf('1e-45'), abs_eps=mp.mpf('1e-45'))
    for k in range(1, 7):
        A = mp.power(2, -2*k*(r - 1)) / mp.power(1 - mp.power(2, -2*k), r)
        Z = mp.zeta(2*k) * A
        kappa = mp.bernoulli(2*k) * A / (2*k)
        trace = ((-1)**(k + 1) * 2 * factorial(2*k - 1)
                  * Z / mp.power(2*mp.pi, 2*k))
        assert mp.almosteq(kappa, trace, rel_eps=mp.mpf('1e-50'))

# Lambert coordinate solves the model saddle equation.
for r in range(1, 7):
    x = mp.mpf('1e-20')
    arg = -(mp.power(factorial(r), mp.mpf(1)/r) * mp.log(2) / r) \
          * mp.power(x, mp.mpf(1)/r)
    lam = -r * mp.lambertw(arg, -1)
    lhs = x * mp.e**lam
    rhs = lam**r / (factorial(r) * mp.log(2)**r)
    assert mp.almosteq(lhs, rhs, rel_eps=mp.mpf('1e-50'))

print('All exact and numerical frontier checks passed.')

```


B Complete LaTeX corpus inventory

The review covered the 57 LaTeX sources visible under Analysis/FabiusFunction/docs on 27 August 2026. The living tree and embedded-paper folders were read as current sources. The archive was also read because its policy is to retain first-introduced historical versions while pruning redundant duplicates; archived claims therefore matter when testing whether a proposed deduction is genuinely absent from the repository. Paths below are repository-relative.

Table 14: All reviewed *.tex sources.

No.	group	repository-relative path
1	live core	Analysis/FabiusFunction/docs/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	live core	Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	live core	Analysis/FabiusFunction/docs/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	live core	Analysis/FabiusFunction/docs/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
5	live core	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
6	live draft	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/Thue_Morse_Formula_Atlas/Thue_Morse_Formula_Atlas.tex
7	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-2/Rvachev_Up_Fourier_Decay-2.tex
8	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay-3/Rvachev_Up_Fourier_Decay-3.tex
9	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Comparative_Audit/Rvachev_Up_Fourier_Decay_Comparative_Audit.tex
10	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Gentle_Guide/Rvachev_Up_Fourier_Decay_Gentle_Guide.tex
11	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/Rvachev_Up_Fourier_Decay_Second_Wave_Audit/Rvachev_Up_Fourier_Decay_Second_Wave_Audit.tex
12	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions/asymptotic-decay-rate-of-an-infinite-product-of-sinc-functions.tex
13	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-1/rvachev_up_fourier_decay-1.tex
14	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-4/rvachev_up_fourier_decay-4.tex
15	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-5/rvachev_up_fourier_decay-5.tex
16	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-6/rvachev_up_fourier_decay-6.tex
17	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-7/rvachev_up_fourier_decay-7.tex
18	live Fourier	Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/rvachev_up_fourier_decay/rvachev_up_fourier_decay-8/rvachev_up_fourier_decay-8.tex
19	embedded paper	Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
20	embedded paper	Analysis/FabiusFunction/docs/papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex

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Table 14 — continued

No.	group	repository-relative path
21	embedded paper	Analysis/FabiusFunction/docs/papers/arXiv-1502.06738v2/Lacunary_products.tex
22	embedded paper	Analysis/FabiusFunction/docs/papers/arXiv-1702.05442v1/09-Function.tex
23	embedded paper	Analysis/FabiusFunction/docs/papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
24	embedded paper	Analysis/FabiusFunction/docs/papers/arXiv-2211.13570v1/document.tex
25	embedded paper	Analysis/FabiusFunction/docs/papers/ubiq15/ubiq15-corrected.tex
26	early notes	Analysis/FabiusFunction/docs/archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
27	early notes	Analysis/FabiusFunction/docs/archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
28	standalone	Analysis/FabiusFunction/docs/archive/standalone-studies/Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
29	standalone	Analysis/FabiusFunction/docs/archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
30	standalone	Analysis/FabiusFunction/docs/archive/standalone-studies/fabius-inverse-dyadic-closed-form/fabius-inverse-dyadic-closed-form.tex
31	primary history	Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
32	primary history	Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-2/Fabius_Function_and_Rvachev_Up.tex
33	primary history	Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_Function_and_Rvachev_Up-3/Fabius_Function_and_Rvachev_Up.tex
34	primary history	Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_and_Rvachev_Up-2/Fabius_and_Rvachev_Up.tex
35	primary history	Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_and_Rvachev_up/Fabius_and_Rvachev_up.tex
36	primary history	Analysis/FabiusFunction/docs/archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
37	supplement	Analysis/FabiusFunction/docs/archive/primary-supplements/Fabius_Function_Missing_Parts/Fabius_Function_Missing_Parts.tex
38	supplement	Analysis/FabiusFunction/docs/archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
39	q-calculus	Analysis/FabiusFunction/docs/archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex
40	q-calculus	Analysis/FabiusFunction/docs/archive/q-calculus/fabius-q-special-functions/fabius-q-special-functions.tex
41	Lean history	Analysis/FabiusFunction/docs/archive/lean/Fabius_Function_Lean_Walkthrough/Fabius_Function_Lean_Walkthrough.tex
42	Lean history	Analysis/FabiusFunction/docs/archive/lean/fabius_lean_walkthrough/fabius_lean_walkthrough.tex
43	glossary history	Analysis/FabiusFunction/docs/archive/glossary/FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
44	glossary history	Analysis/FabiusFunction/docs/archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
45	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Asymptotic_Bridge/Fabius_Dyadic_Asymptotic_Bridge.tex
46	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
47	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
48	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
49	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rate/Fabius_Thue_Morse_Convergence_Rate.tex
50	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Fabius_Thue_Morse_Convergence_Rates/Fabius_Thue_Morse_Convergence_Rates.tex
51	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Primary_Exposition_Gap_Register/Primary_Exposition_Gap_Register.tex
52	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Repeated_Integration_and_Rvachev_Up/Repeated_Integration_and_Rvachev_Up.tex
53	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Rvachev_Up_from_Repeated_Integration/Rvachev_Up_from_Repeated_Integration.tex

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Table 14 — continued

No.	group	repository-relative path
54	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex
55	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-1/Thue_Morse_Formula_Atlas.tex
56	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-2/Consolidated_Formula_Atlas_for_the_Thue_Morse_Sequence.tex
57	frontier history	Analysis/FabiusFunction/docs/archive/research-frontiers/Thue_Morse_Formula_Atlas-3/Thue_Morse_Formula_Atlas.tex

C Notation crosswalk

symbol	meaning
$\text{sinc } z$	$\sin(\pi z)/(\pi z)$, normalized to equal 1 at zero
Φ_r	rank- r Pascal–Rvachev Fourier product
X_r	centered random series with characteristic function Φ_r
f_r	density of X_r , supported on $[-1, 1]$
$D_r = X_r + 1$	shifted nonnegative series, supported on $[0, 2]$
H_r	distribution function $\mathbb{P}(D_r \leq x)$
$\mu_r(n)$	integer-zero multiplicity $\binom{\nu_2(n)+1}{r}$
$\mathcal{N}_r(N)$	cumulative positive-zero multiplicity through N
$Q_r(j)$	Pascal tail weight $\sum_{k < r} \binom{j+1}{k}$
$S_r(N)$	digital discrepancy $\sum_j b_j(N) Q_r(j)$
$\varepsilon_r(N)$	sign of Φ_r on $(N, N+1)$
P_r	a digit-position period $2^{\lceil \log_2 r \rceil}$
d_r	number of negative sign-mask entries in one period
$\mathcal{Z}_r$	zero-divisor Dirichlet series
$\mathcal{A}_r$	common scale zeta factor
L_r, Ψ_r	endpoint Laplace transform and negative logarithm
Λ_r	lower-Lambert model saddle coordinate

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Part IV

Derivative Norm Spectra and Dual Moment Geometries of the Fabius–Rvachev System

Abstract

Let F be the bounded Fabius distribution function, $G = F^{-1}$ its inverse on $[0, 1]$, $\text{up}(x) = 1 - F(|x|)$ on $[-1, 1]$, and

$$\Phi(\xi) = \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi\xi}{2^j}\right)$$

under the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i x \xi} dx$. The recursive documentation tree of the ProveIt Fabius development was audited at commit 0341bbf91cb867f4510bc8431cf59878208de771. Existing reports already contain the dyadic derivative tiling, integer-zero arithmetic, sharp Fourier-decay regimes, endpoint Lambert expansions, q -binomial acceleration, inverse-quantile reversion, and several spectral determinant constructions. Those results are treated as baseline rather than renamed.

The first new package extracted here begins with an exact equimeasurability theorem. If $T_n = n(n+1)/2$, then $2^{-T_n} |\text{up}^{(n)}|$ has exactly the same value distribution on $[-1, 1]$ as up . The signs on the 2^n cells are the length- 2^n Thue–Morse block. Consequently every rearrangement-invariant norm, including every L^p quasi-norm $0 < p \leq \infty$, obeys

$$\left\| \text{up}^{(n)} \right\|_p = 2^{T_n} \|\text{up}\|_p.$$

The level sets are also exact:

$$\left| \{x : |\text{up}^{(n)}(x)| > 2^{T_n} t\} \right| = 2G(1-t), \quad 0 < t < 1.$$

Thus the complete inverse Fabius function is encoded in the value statistics of every nonzero derivative.

Writing $A_p = \frac{1}{2} \|\text{up}\|_p^p$, one obtains the exact beta transform

$$A_p = p \int_0^1 (1-y)^{p-1} G(y) dy.$$

For $\mathcal{D} = y \, d/dy$, the dilation variable $Z_p = pY_p$ with $Y_p \sim \text{Beta}(1, p)$ has Mellin symbol

$$\mathbb{E} Z_p^s = p^s \frac{\Gamma(p+1)\Gamma(1+s)}{\Gamma(p+1+s)}.$$

This yields a new Euler–Gamma differential calculus

$$A_p \sim G - \gamma \mathcal{D}G + \frac{\gamma^2 + \pi^2/6}{2} \mathcal{D}^2 G - \frac{\gamma^3 + \gamma\pi^2/2 + 2\zeta(3)}{6} \mathcal{D}^3 G + \dots \quad \text{at } y = p^{-1}.$$

The coefficients are complete Bell polynomials in log-exponential cumulants; finite- p corrections are generated by Bernoulli polynomials through the gamma-ratio expansion. Combining this operator with the periodic inverse-Fabius endpoint expansion gives

$$\|\text{up}\|_p^p = 2G(p^{-1}) \left(1 - \frac{\gamma}{\rho} + \mathcal{O}(\rho^{-2}) \right), \quad \rho = \sqrt{\frac{2 \log p}{\log 2}},$$

and, more sharply, shows that Euler’s constant subtracts from the first periodic inverse correction. In particular

$$1 - \|\text{up}\|_p \sim \frac{\sqrt{2 \log 2 \log p}}{p}.$$

A phase-locked sequence of p -norms recovers the nonconstant Gamma–zeta endpoint fluctuation itself.

Parseval turns the same norm law into a second, frequency-side geometry. The pushforward of $|\Phi(\xi)|^2 d\xi / \|\text{up}\|_2^2$ by $z = (2\pi\xi)^2$ has moments

$$m_n = 2^{n(n+1)} = q^{-n(n+1)/2}, \quad q = \frac{1}{4}.$$

It is therefore an explicit nonclassical representing measure for a shifted Stieltjes–Wigert moment problem. A lognormal measure has the same moments but a different density, proving indeterminacy directly. We calculate the Hankel determinants, monic orthogonal polynomials, recurrence coefficients, and Gaussian- q -binomial formulae, and derive the spectral q -Pearson law

$$w_{\text{sp}}(4z) = \frac{1}{2} \text{sinc}^2(\sqrt{z}) w_{\text{sp}}(z).$$

The report closes with numerical checks, explicit conjectures, a phase-locked confluent q -Richardson program, and a Lean formalization roadmap. Claims labelled “new” are new relative to the pinned repository corpus after a formula- and terminology-level collision audit; no universal publication-priority claim is made.

1 Scope, corpus audit, and novelty boundary

1.1 What was inspected

The recursive tree under <https://github.com/VladimirReshetnikov/ProveIt/tree/0341bbf91cb867f4510bc8431cf59878208de771/Analysis/FabiusFunction/docs> contained 68 *.tex artifacts at the pinned commit. Every path was included in the audit. The living expositions, active frontier reports, Fourier-decay chain, imported paper transcriptions, and mathematically distinctive archival deltas were read. The archive deliberately preserves first versions and provenance; its own ledger explains that byte-identical or superseded descendants should not be treated as independent contemporary claims. Accordingly, archived variants were checked for distinctive content but deduplicated when building the novelty map. The complete path ledger appears in [Appendix A](#).

The most important collision checks were against the following current reports:

- the primary Fabius–Rvachev exposition and mathematical glossary;
- the consolidated non-formalized frontier volume;
- the inverse-frontier report, including periodic Lambert–Bell quantile reversion;
- the complete Fourier-decay synthesis and its comparative audits;
- the finite-block/q-Richardson/deconvolution report;
- the half-integer, arithmetic-ray, spectral determinant, Pascal–Rvachev, and Gamma-duality frontier reports;
- all archived q-calculus, dyadic-asymptotic, Thue–Morse, and repeated-integration studies; and
- the seven imported paper sources.

Targeted searches were then run for the exact concepts and formula shapes used below: “equimeasurable derivative”, “all- L^p derivative norm”, “Beta(1, p) inverse Fabius”, “ $T(1+\mathcal{D})$ ”, “Stieltjes–Wigert”, “Denjoy–Carleman”, and the exact moment sequence $2^{n(n+1)}$. No collision was found for the theorem package developed here. The underlying derivative tiling itself is present in the corpus and is explicitly marked as baseline.

1.2 Status convention

Three proof-status labels *Corpus baseline* means a statement already present, perhaps in another normalization, in the inspected repository or one of its imported papers. *New deduction* means a statement and proof not located in that corpus after the collision audit. *Conjecture* means a precise, numerically or structurally motivated statement for which this report does not claim a proof. None of these labels asserts global publication priority.

The following are corpus baseline and are not claimed as new:

- (i) the identities relating F , up, and the signed global extension;
- (ii) the differential-refinement law for up and its finite Thue–Morse derivative tiling;
- (iii) the infinite sinc product and its integer zero multiplicities;
- (iv) the exact dyadic values of F , their q -Pochhammer and Gaussian-binomial forms;
- (v) the corrected endpoint expansion with a nonconstant periodic Gamma–zeta term;

- (vi) the inverse endpoint expansion in the lower-Lambert coordinate, including Bell reversion and quantile elasticity;
- (vii) the distinct pointwise, metric, L^1 , L^2 , and extremal Fourier-decay exponents;
- (viii) the existing finite-block, q-Richardson, convolution-tail, spectral determinant, and Pascal–Rvachev constructions.

New deductions proved in this report The main new results are: exact derivative equimeasurability; scaling in every rearrangement-invariant norm; exact inverse-Fabius level-set laws; a Hausdorff reconstruction of F from integer derivative norms; the inverse-Fabius beta transform; an exact Mellin symbol and asymptotic Euler–Gamma/Bell/Bernoulli differential calculus; a large- p Lambert law with an Euler-constant shift and phase tomography; a shifted Stieltjes–Wigert spectral-energy measure; explicit Hankel determinants and q-binomial monic orthogonal polynomials; a nonclassical sinc-square q-Pearson equation; and the exact Denjoy–Carleman weight and associated periodic Legendre envelope generated by the derivative norms.

Post-audit Lean status: a completed derivative-distribution core The current source file `Analysis / FabiusFunction / Lean / FabiusFunction / RvachevDerivativeDistribution.lean` now formalizes the exact dyadic cell coordinate and tiling, every signed cell-midpoint value, the all-order signed finite-mixture identity for continuous tests with values in an arbitrary real Banach space, the normalized absolute continuous-test identity, the equality of the two absolute Borel pushforward measures on Lebesgue measure restricted to $[-1, 1]$, and every nonnegative real absolute moment. This is a post-audit proof-status update; it does not alter the pinned novelty comparison above.

The report as a whole is not thereby formalized. In particular, the simplified signed Rademacher law as a Borel pushforward, the `eLpNorm` and general rearrangement-invariant norm ladder, the explicit inverse-Fabius level-set laws, the beta transform and its asymptotics, and all spectral-moment, Hankel, q-Pearson, and q-binomial layers below remain without matching declarations.

2 Normalization and established structural identities

Let $F : \mathbb{R} \rightarrow [0, 1]$ be the bounded Fabius function, with $F = 0$ on $(-\infty, 0]$, $F = 1$ on $[1, \infty)$, symmetry

$$F(1 - x) = 1 - F(x), \quad 0 \leq x \leq 1, \quad (274)$$

and differential law $F'(x) = 2F(2x)$ on $0 < x < 1/2$, continued by symmetry. Let

$$\text{up}(x) = \begin{cases} 1 - F(|x|), & |x| \leq 1, \\ 0, & |x| \geq 1. \end{cases} \quad (275)$$

Then up is even, nonnegative, C^∞ , supported on $[-1, 1]$, satisfies $\text{up}(0) = 1$ and $\int \text{up} = 1$, and obeys

$$\text{up}'(x) = 2(\text{up}(2x + 1) - \text{up}(2x - 1)). \quad (276)$$

Its Fourier transform is

$$\Phi(\xi) := \widehat{\text{up}}(\xi) = \prod_{j=0}^{\infty} \text{sinc}\left(\frac{\pi \xi}{2^j}\right), \quad \text{sinc } z = \frac{\sin z}{z}. \quad (277)$$

Write

$$\varepsilon_k := (-1)^{s_2(k)}, \quad T_n := \binom{n+1}{2} = \frac{n(n+1)}{2}, \quad (278)$$

where $s_2(k)$ is the binary digit sum. The identities $\varepsilon_{2k} = \varepsilon_k$ and $\varepsilon_{2k+1} = -\varepsilon_k$ will be used repeatedly.

Theorem 2.1 (Corpus baseline: exact derivative tiling). *For every $n \geq 0$ and $x \in \mathbb{R}$,*

$$\text{up}^{(n)}(x) = 2^{T_n} \sum_{k=0}^{2^n-1} \varepsilon_k \text{up}(2^n x + 2^n - 1 - 2k). \quad (279)$$

Equivalently, for $0 \leq k < 2^n$ and $-1 \leq y \leq 1$,

$$\text{up}^{(n)}\left(-1 + \frac{2k+1+y}{2^n}\right) = 2^{T_n} \varepsilon_k \text{up}(y). \quad (280)$$

Proof. The case $n = 0$ is immediate. Differentiate (279). Each derivative of an inner copy contributes 2^n , and (276) contributes another factor 2. The positive branch has index $2k$ and the negative branch index $2k+1$; the Thue–Morse identities above give exactly the signs ε_{2k} and ε_{2k+1} . The scalar grows from 2^{T_n} to $2^{T_n+n+1} = 2^{T_{n+1}}$.

For the cell form, the support condition $|2^n x + 2^n - 1 - 2j| \leq 1$ partitions $[-1, 1]$ into the 2^n closed cells

$$I_{n,k} = \left[-1 + \frac{2k}{2^n}, -1 + \frac{2k+2}{2^n}\right].$$

On the interior of $I_{n,k}$ only the k th summand survives. At common boundaries every potentially overlapping copy is evaluated at ± 1 and hence vanishes to infinite order. Solving for the local coordinate gives (280). $\square$

Remark 2.2. The corpus already exploits this tiling for derivative values, zeros, and discrete Thue–Morse structure. The new step is to view (280) as a measure-preserving value-distribution identity and push that observation through normed function spaces, inverse quantiles, Mellin operators, and spectral moments.

Exact Lean parity for the dyadic cells The current definition `rvachevDerivativeCell` is exactly

$$c_{n,k}(y) := -1 + \frac{2k+1+y}{2^n}.$$

It is total in n and k . For $k < 2^n$ it maps $[-1, 1]$ into $[-1, 1]$; the first left endpoint is -1 , adjacent endpoints agree, and the left endpoint after all 2^n cells is 1. These facts are `rvachevDerivativeCell_mem_Icc`, `rvachevDerivativeCell_zero_neg_one`, `rvachevDerivativeCell_one_eq_succ_neg_one`, and `rvachevDerivativeCell_two_pow_neg_one`; the affine dilation about the midpoint is `rvachevDerivativeCell_eq_div_add`.

The signed and absolute formulas (280) are `iteratedDeriv_rvachev_cell` and `abs_iteratedDeriv_rvachev_cell`. They include $n = 0$, every valid cell, and both cell endpoints. The midpoint specialization

$$\text{up}^{(n)}\left(-1 + \frac{2k+1}{2^n}\right) = 2^{T_n} \varepsilon_k$$

is the separate theorem `iteratedDeriv_rvachev_cell_zero`. All declarations in this box are in `RvachevDerivativeDistribution.lean`.

3 Exact equimeasurability and the complete derivative norm spectrum

3.1 Absolute and signed value distributions

Let U be uniform on $[-1, 1]$ and let R be an independent Rademacher variable, $\mathbb{P}(R = 1) = \mathbb{P}(R = -1) = 1/2$.

Theorem 3.1 (Derivative equimeasurability). *For every $n \geq 0$,*

$$\boxed{2^{-T_n} \left| \text{up}^{(n)}(U) \right| \stackrel{d}{=} \text{up}(U)}. \quad (281)$$

For every $n \geq 1$,

$$\boxed{2^{-T_n} \text{up}^{(n)}(U) \stackrel{d}{=} R \text{up}(U)}. \quad (282)$$

Equivalently, for every Borel function H for which the integrals exist,

$$\frac{1}{2} \int_{-1}^1 H\left(2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right) dx = \frac{1}{2} \int_{-1}^1 H(\text{up}(y)) dy, \quad (283)$$

$$\frac{1}{2} \int_{-1}^1 H\left(2^{-T_n} \text{up}^{(n)}(x)\right) dx = \frac{1}{4} \int_{-1}^1 (H(\text{up}(y)) + H(-\text{up}(y))) dy \quad (n \geq 1). \quad (284)$$

Proof. Integrate cell by cell and use (280). The affine map from each cell to $[-1, 1]$ has Jacobian 2^{-n} , hence

$$\int_{I_{n,k}} H\left(2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right) dx = 2^{-n} \int_{-1}^1 H(\text{up}(y)) dy.$$

Summing 2^n equal terms proves (283). For $n \geq 1$, exactly half of the length- 2^n Thue–Morse block has sign $+1$ and half sign -1 , proving (284). $\square$

Exact Lean parity for the distribution identities The signed theorem presently exposed by Lean is the following all-order finite mixture, not yet its Rademacher simplification. If E is any real Banach space and $H : \mathbb{R} \rightarrow E$ is continuous, then for every $n \geq 0$,

$$\int_{-1}^1 H\left(\text{up}^{(n)}(x)\right) dx = 2^{-n} \sum_{k=0}^{2^n-1} \int_{-1}^1 H(2^{T_n} \varepsilon_k \text{up}(y)) dy. \quad (285)$$

This is `intervalIntegral_comp_iteratedDeriv_rvachev`. In particular, it includes the order-zero boundary without introducing a fictitious independent sign.

The same Banach-valued continuous-test API proves

$$\int_{-1}^1 H\left(\left| \text{up}^{(n)}(x) \right| \right) dx = \int_{-1}^1 H(2^{T_n} \text{up}(y)) dy, \quad (286)$$

$$\int_{-1}^1 H\left(2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right) dx = \int_{-1}^1 H(\text{up}(y)) dy. \quad (287)$$

These are `intervalIntegral_comp_abs_iteratedDeriv_rvachev` and `intervalIntegral_comp_normalized_abs_iteratedDeriv_rvachev`.

More strongly on the absolute side, let λ_I be Lebesgue measure restricted to $I = [-1, 1]$, and write $f_{\#}\mu$ for pushforward. Lean proves the literal Borel-measure equality

$$\left(x \mapsto 2^{-T_n} \left| \text{up}^{(n)}(x) \right| \right)_{\#} \lambda_I = \text{up}_{\#} \lambda_I. \quad (288)$$

Its declaration is `map_normalized_abs_iteratedDeriv_rvachev_restrict_1cc`. What is not yet a named theorem is the signed Borel pushforward simplified, for $n \geq 1$, to $R \text{ up}(U)$. Thus (282) and the full Borel formulation of (284) remain on the frontier even though (285) is their exact continuous-test finite-cell precursor.

Corollary 3.2 (All absolute moments and signed moments). *For every real $p \geq 0$, interpreting the powers as real powers and using $0^0 = 1$ at $p = 0$,*

$$\int_{-1}^1 |\text{up}^{(n)}(x)|^p dx = (2^{T_n})^p \int_{-1}^1 \text{up}(x)^p dx. \quad (289)$$

For every integer $m \geq 0$ and $n \geq 1$,

$$\int_{-1}^1 \text{up}^{(n)}(x)^m dx = \begin{cases} 0, & m \text{ odd}, \\ 2^{mT_n} \int_{-1}^1 \text{up}(x)^m dx, & m \text{ even}. \end{cases} \quad (290)$$

The absolute identity (289), with the exact hypothesis $p \geq 0$, is `intervalIntegral_abs_iteratedDeriv_rvachev_rpow`. At $p = 0$ this uses Lean's `Real.rpow` convention $0^0 = 1$. The signed parity simplification (290) is not separately formalized.

3.2 Every rearrangement-invariant norm scales exactly

A rearrangement-invariant quasi-norm on a finite measure space depends only on the distribution of $|f|$. This includes Lebesgue, Lorentz, Lorentz–Zygmund, Orlicz, and many interpolation spaces.

Theorem 3.3 (Universal norm ladder). *Let $X[-1, 1]$ be any absolutely homogeneous rearrangement-invariant quasi-normed function space. Then*

$$\left\| \text{up}^{(n)} \right\|_X = 2^{T_n} \left\| \text{up} \right\|_X, \quad n \geq 0. \quad (291)$$

In particular, for every $0 < p \leq \infty$,

$$\left\| \text{up}^{(n)} \right\|_{L^p(\mathbb{R})} = 2^{n(n+1)/2} \left\| \text{up} \right\|_{L^p(\mathbb{R})}. \quad (292)$$

Proof. The functions vanish outside $[-1, 1]$. By Theorem 3.1, $2^{-T_n} |\text{up}^{(n)}|$ and up have the same decreasing rearrangement. Rearrangement invariance and absolute homogeneity give (291); the L^p case follows. $\square$

Function-space norm boundary The real-power interval identity (289) is formal, but it is not yet packaged as a Mathlib `eLpNorm` equality. The $p = \infty$ statement and the general rearrangement-invariant quasi-norm theorem (291) likewise have no matching Lean declarations. Consequently the universal norm ladder in this subsection remains a frontier layer; the Borel pushforward equality (288) is its completed measure-theoretic input.

Corollary 3.4 (Exact sup norms, total variations, and derivative ratios). *Since $\|\text{up}\|_1 = \|\text{up}\|_\infty = 1$,*

$$\left\| \text{up}^{(n)} \right\|_1 = 2^{T_n}, \quad (293)$$

$$\left\| \text{up}^{(n)} \right\|_\infty = 2^{T_n}, \quad (294)$$

$$\text{TV}(\text{up}^{(n)}) = \left\| \text{up}^{(n+1)} \right\|_1 = 2^{T_{n+1}}. \quad (295)$$

For every $0 < p \leq \infty$,

$$\frac{\|\text{up}^{(n+1)}\|_p}{\|\text{up}^{(n)}\|_p} = 2^{n+1}. \quad (296)$$

3.3 Thue–Morse extremizers and exact level sets

Corollary 3.5 (Midpoint extrema). *For $0 \leq k < 2^n$, put*

$$x_{n,k} := -1 + \frac{2k+1}{2^n}. \quad (297)$$

Then

$$\boxed{\text{up}^{(n)}(x_{n,k}) = 2^{T_n} \varepsilon_k.} \quad (298)$$

These are precisely the points at which $|\text{up}^{(n)}|$ attains its global maximum. Thus the signs of the complete extremal set of the n th derivative form the first 2^n terms of the signed Thue–Morse sequence.

Proof. Set $y = 0$ in (280) and use $\text{up}(0) = 1$. Strict decrease of up on $[0, 1]$ makes $y = 0$ the unique maximum inside each cell. $\square$

The signed midpoint equation (298) is formally iterated `Deriv_rvachev_cell_zero`. This does not by itself formalize the exact inverse-Fabius level sets below.

Let $G = F^{-1} : [0, 1] \rightarrow [0, 1]$. Since $\text{up}(y) = 1 - F(|y|)$,

$$\{y \in [-1, 1] : \text{up}(y) > t\} = (-G(1-t), G(1-t)), \quad 0 < t < 1. \quad (299)$$

Theorem 3.6 (Exact inverse-Fabius level-set law). *For every $n \geq 0$ and $0 < t < 1$,*

$$\boxed{\left| \left\{ x \in \mathbb{R} : |\text{up}^{(n)}(x)| > 2^{T_n} t \right\} \right| = 2G(1-t).} \quad (300)$$

Consequently, if $V_n = 2^{-T_n} |\text{up}^{(n)}(U)|$, then

$$\mathbb{P}(V_n > t) = G(1-t), \quad (301)$$

$$\mathbb{P}(V_n \leq t) = 1 - G(1-t), \quad (302)$$

$$f_{V_n}(t) = G'(1-t), \quad 0 < t < 1. \quad (303)$$

For $n \geq 1$, the signed variable $2^{-T_n} \text{up}^{(n)}(U)$ has density

$$f_n(v) = \frac{1}{2} G'(1-|v|), \quad -1 < v < 1. \quad (304)$$

Proof. Combine equimeasurability with (299). The density follows by differentiating the distribution function on the open interval. The signed density is the symmetric Rademacher mixture. $\square$

Closed-form level-set boundary Equation (288) proves exact absolute equimeasurability as a Borel measure identity. The further identifications (300)–(304), which express the sets and densities explicitly through the totalized inverse G , do not yet have matching Lean theorems. In particular, the exact level-set and density package remains unformalized.

Remark 3.7 (A quantile coupling). If U is uniform on $[-1, 1]$, then $|U|$ is uniform on $[0, 1]$ and

$$G(1 - \text{up}(U)) = |U|. \quad (305)$$

Thus the inverse Fabius function is not merely asymptotically related to derivative values; it is their exact probability integral transform.

4 The norm spectrum determines the Fabius function

Define the normalized norm spectrum

$$A_p := \frac{1}{2} \|\text{up}\|_p^p = 2^{-1-pT_n} \left\| \text{up}^{(n)} \right\|_p^p, \quad p > 0. \quad (306)$$

The second expression is independent of n by (292). Set $A_0 := 1$. If $W = \text{up}(U)$, then $A_p = \mathbb{E}W^p$ for every $p \geq 0$.

Proposition 4.1 (Complete monotonicity and log-convexity). *The function $p \mapsto A_p$ is strictly completely monotone on $(0, \infty)$:*

$$(-1)^r A_p^{(r)} = \mathbb{E}[W^p (-\log W)^r] > 0, \quad r \geq 0. \quad (307)$$

It is strictly log-convex. Its integer samples are a completely monotone Hausdorff moment sequence:

$$(-1)^r \Delta^r A_m = \mathbb{E}[W^m (1 - W)^r] > 0, \quad m, r \geq 0, \quad (308)$$

where $\Delta A_m = A_{m+1} - A_m$.

Proof. Write $W^p = \exp(-pX)$ with $X = -\log W \geq 0$ and differentiate under the integral. Strict log-convexity follows from Hölder's inequality because W is nonconstant. The finite-difference formula follows by expanding $(1 - W)^r$. $\square$

Theorem 4.2 (Hausdorff reconstruction from integer derivative norms). *The sequence $\{A_m\}_{m \geq 0}$ uniquely determines G , hence F and up . More explicitly, define*

$$b_{N,k} := \binom{N}{k} \sum_{j=0}^{N-k} (-1)^j \binom{N-k}{j} A_{k+j}, \quad 0 \leq k \leq N. \quad (309)$$

Then $b_{N,k} \geq 0$, $\sum_k b_{N,k} = 1$, and at every $0 < t < 1$,

$$G(1 - t) = 1 - \lim_{N \rightarrow \infty} \sum_{k=0}^{\lfloor Nt \rfloor} b_{N,k}. \quad (310)$$

The same reconstruction applies to the normalized norms of any fixed derivative $\text{up}^{(n)}$ through (306).

Proof. By (308),

$$b_{N,k} = \binom{N}{k} \mathbb{E}[W^k (1 - W)^{N-k}].$$

Conditioned on $W = w$, these are the probabilities of a binomial random variable $K_N \sim \text{Bin}(N, w)$. Therefore

$$\sum_{k=0}^{\lfloor Nt \rfloor} b_{N,k} = \mathbb{E}\mathbb{P}(K_N/N \leq t \mid W).$$

The conditional law of large numbers makes the inner probability converge to $\mathbf{1}_{\{W \leq t\}}$ away from $w = t$. The distribution of W is continuous, so the expectation converges to $\mathbb{P}(W \leq t) = 1 - G(1 - t)$ by (302). Uniqueness is also the classical determinacy of the Hausdorff moment problem on $[0, 1]$. $\square$

Remark 4.3. This theorem gives a perhaps unexpected completeness statement: the scalar sequence $\|\text{up}^{(n)}\|_m$, $m = 1, 2, \dots$, for any one derivative order n , contains the entire Fabius function. The Thue–Morse oscillation disappears under absolute moments, yet no information about the underlying profile is lost.

5 An exact inverse-Fabius beta transform

Theorem 5.1 (Beta transform of the inverse Fabius function). *For every real $p > 0$,*

$$A_p = p \int_0^1 (1-y)^{p-1} G(y) \, dy. \quad (311)$$

Equivalently, if $Y_p \sim \text{Beta}(1, p)$, then

$$A_p = \mathbb{E}G(Y_p). \quad (312)$$

Proof. Using evenness and (275),

$$A_p = \int_0^1 (1-F(x))^p \, dx.$$

The monotone substitution $y = F(x)$, interpreted as a Lebesgue–Stieltjes substitution at the flat endpoints, gives

$$A_p = \int_0^1 (1-y)^p \, dG(y).$$

Integration by parts has no boundary term because $G(0) = 0$ and $(1-y)^p = 0$ at $y = 1$. This proves (311); the beta density is $p(1-y)^{p-1}$. $\square$

Corollary 5.2 (Layer-cake form). *For every $p > 0$ and every $n \geq 0$,*

$$\left\| \text{up}^{(n)} \right\|_p^p = 2^{pT_n+1} p \int_0^1 (1-y)^{p-1} G(y) \, dy. \quad (313)$$

The beta kernel concentrates at $y \asymp p^{-1}$. Because G is slowly varying at zero, (311) already implies $A_p \sim G(1/p)$. The next section replaces that first-order Abelian statement by an all-orders differential operator.

6 The Euler–Gamma differential operator

6.1 Exact Mellin symbol

Let

$$\mathcal{D} := y \frac{d}{dy} \quad (314)$$

be the Euler dilation derivative. Put $Z_p = pY_p$.

Proposition 6.1 (Mellin symbol of the beta dilation). *For $\Re s > -1$,*

$$M_p(s) := \mathbb{E}Z_p^s = p^s \frac{\Gamma(p+1)\Gamma(1+s)}{\Gamma(p+1+s)}. \quad (315)$$

For any bounded extension of G to $(1, \infty)$,

$$A_p = [M_p(\mathcal{D})G](p^{-1}), \quad (316)$$

where the operator is rigorously interpreted by the Bochner average

$$[M_p(\mathcal{D})H](y) := \mathbb{E}[H(Z_p y)]. \quad (317)$$

The value in (316) is independent of how G is extended, because $0 < Z_p/p = Y_p < 1$.

Proof. The beta integral gives

$$\mathbb{E}Z_p^s = p^s p \int_0^1 y^s (1-y)^{p-1} dy = p^s \frac{\Gamma(p+1)\Gamma(1+s)}{\Gamma(p+1+s)}.$$

The dilation semigroup satisfies $\exp(t\mathcal{D})H(y) = H(e^t y)$. Averaging with $t = \log Z_p$ and evaluating at $y = p^{-1}$ gives $\mathbb{E}G(Z_p/p) = \mathbb{E}G(Y_p) = A_p$. $\square$

As $p \rightarrow \infty$, Z_p converges to an exponential random variable Z of mean one, and $M_p(s) \rightarrow \Gamma(1+s)$. Thus $\Gamma(1+\mathcal{D})$ is not decorative notation: it is the Mellin symbol of the limiting random dilation.

6.2 Cumulants, Bell polynomials, and finite-p corrections

Let

$$K_p(s) := \log M_p(s). \quad (318)$$

The cumulants of $\log Z_p$ are

$$\kappa_{1,p} = \log p - \psi(p+1) - \gamma, \quad (319)$$

$$\kappa_{m,p} = \psi^{(m-1)}(1) - \psi^{(m-1)}(p+1), \quad m \geq 2. \quad (320)$$

If B_m denotes the complete exponential Bell polynomial, then

$$\mu_{m,p} := \mathbb{E}(\log Z_p)^m = B_m(\kappa_{1,p}, \dots, \kappa_{m,p}). \quad (321)$$

As $p \rightarrow \infty$,

$$\kappa_1 = -\gamma, \quad (322)$$

$$\kappa_m = (-1)^m (m-1)! \zeta(m), \quad m \geq 2, \quad (323)$$

which are the cumulants of $\log Z$ for $Z \sim \text{Exp}(1)$.

The uniform gamma-ratio expansion on compact s -sets is

$$K_p(s) \sim \log \Gamma(1+s) + \sum_{r=1}^{\infty} \frac{(-1)^{r+1} (B_{r+1}(1) - B_{r+1}(1+s))}{r(r+1)p^r}. \quad (324)$$

Here $B_m(x)$ are Bernoulli polynomials. The first correction is $-s(s+1)/(2p)$.

Theorem 6.2 (Local Gamma–Bell expansion). *Fix $N \geq 1$. As $p \rightarrow \infty$,*

$$A_p = \sum_{m=0}^{N-1} \frac{\mu_m}{m!} \mathcal{D}^m G(p^{-1}) + \mathcal{O}_N \left(G(p^{-1}) \frac{(\log p)^{c_N}}{\rho^N} \right), \quad (325)$$

where

$$\rho = \sqrt{\frac{2 \log p}{\log 2}}, \quad \mu_m = B_m(-\gamma, \zeta(2), -2\zeta(3), 6\zeta(4), \dots), \quad (326)$$

and c_N is a finite integer depending only on N . A finite- p expansion with corrections to every power of p^{-1} is obtained by replacing μ_m with (321) or, equivalently, substituting (324) into $\exp K_p(\mathcal{D})$.

Proof. Write $y = p^{-1}$ and $t = \log Z_p$. On a logarithmic neighborhood of y ,

$$G(e^t y) = \sum_{m=0}^{N-1} \frac{t^m}{m!} \mathcal{D}^m G(y) + R_N(t, y).$$

The all-orders inverse endpoint expansion in the corpus implies, for each fixed m , $\mathcal{D}^m G(y)/G(y)$ is bounded by a power of $\log \rho$ times ρ^{-m} . Fix a constant $C > L + 1$ and split the expectation into $|t| \leq C\rho$ and its complement. On the central range, every point $e^u y$ between y and $e^t y$ has endpoint coordinate $\rho + \mathcal{O}_C(1)$. Integrating the elasticity shows $G(e^u y)/G(y) = \mathcal{O}_C(1)$ there, so Taylor's integral remainder is

$$|R_N(t, y)| \leq C_N G(y) \frac{(\log \rho)^{c_N}}{\rho^N} |t|^N.$$

The moments $\mathbb{E}|\log Z_p|^N$ are bounded uniformly for $p \geq 2$: near zero the density of Z_p is at most one, while for $z \geq 1$ it is bounded by $e^{-z/2}$. For the omitted negative tail, $\mathbb{P}(Z_p < e^{-C\rho}) \leq e^{-C\rho}$; for the positive tail, $\mathbb{P}(Z_p > e^{C\rho}) \leq \exp(-e^{C\rho})$ once p is large. Since $G(p^{-1}) \asymp \rho e^{-L\rho}$, both tail contributions are $o(G(p^{-1})\rho^{-N})$. Taking expectations gives coefficients $\mu_{m,p}$; replacing them by μ_m changes the result by $\mathcal{O}(p^{-1}G(p^{-1}))$, which is beyond every power of ρ^{-1} relative to $G(p^{-1})$. The Bell and Bernoulli descriptions follow from differentiation of (318) and the standard gamma-ratio expansion. $\square$

The first five log-exponential moments are

$$\begin{aligned} \mu_0 &= 1, \\ \mu_1 &= -\gamma, \\ \mu_2 &= \gamma^2 + \zeta(2), \\ \mu_3 &= -\gamma^3 - 3\gamma\zeta(2) - 2\zeta(3), \\ \mu_4 &= \gamma^4 + 6\gamma^2\zeta(2) + 8\gamma\zeta(3) + 3\zeta(2)^2 + 6\zeta(4). \end{aligned} \tag{327}$$

Thus

$$\begin{aligned} A_p &= G - \gamma \mathcal{D}G + \frac{\gamma^2 + \pi^2/6}{2} \mathcal{D}^2 G \\ &\quad - \frac{\gamma^3 + \gamma\pi^2/2 + 2\zeta(3)}{6} \mathcal{D}^3 G \\ &\quad + \frac{\gamma^4 + \gamma^2\pi^2 + 8\gamma\zeta(3) + 3\pi^4/20}{24} \mathcal{D}^4 G + \dots, \quad y = p^{-1}. \end{aligned} \tag{328}$$

Remark 6.3 (Why Bell and Bernoulli polynomials appear). Bell polynomials convert the cumulants of the random logarithmic dilation into its ordinary moments, hence into powers of $\mathcal{D}$. Bernoulli polynomials govern the finite- p deviation of the beta dilation from the limiting exponential dilation through the quotient $\Gamma(p+1)/\Gamma(p+1+s)$. The two families enter for different structural reasons and then compose automatically.

7 Large- p norms, Lambert coordinates, and phase tomography

7.1 Imported inverse endpoint expansion

Put

$$L := \log 2, \quad T := \log p, \quad \rho := \sqrt{\frac{2T}{L}}, \quad \beta := \frac{1}{L} + \frac{1}{2}, \quad \theta := \rho + \beta. \tag{329}$$

The inverse-frontier report in the corpus proves

$$G_0(p^{-1}) := \rho e^{-L\rho - L\beta} = \frac{\sqrt{T}}{e\sqrt{L}} e^{-\sqrt{2LT}}, \quad (330)$$

and

$$\frac{G(p^{-1})}{G_0(p^{-1})} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta)}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right), \quad (331)$$

where Ψ is a nonconstant real-analytic one-periodic Gamma-zeta function. It also proves the endpoint elasticity

$$\frac{yG'(y)}{G(y)} = \frac{1}{\rho} + \frac{\Psi'(\rho + \beta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right), \quad y \downarrow 0. \quad (332)$$

These are baseline inputs; the norm consequences below are new.

7.2 Euler's constant shifts the inverse correction

Theorem 7.1 (Two-term norm-to-quantile ratio). *As $p \rightarrow \infty$,*

$$\boxed{\frac{A_p}{G(p^{-1})} = 1 - \frac{\gamma}{\rho} + \frac{1}{\rho^2} \left[\frac{\gamma^2 + \pi^2/6}{2} - \frac{\gamma(\Psi'(\theta) - 1)}{L} \right] + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right)}. \quad (333)$$

In particular,

$$\boxed{A_p \sim G(p^{-1})}. \quad (334)$$

Proof. By (328),

$$\frac{A_p}{G} = 1 - \gamma \frac{\mathcal{D}G}{G} + \frac{\gamma^2 + \pi^2/6}{2} \frac{\mathcal{D}^2G}{G} + \mathcal{O}(\rho^{-3} \log \rho).$$

Equation (332) gives

$$\frac{\mathcal{D}G}{G} = \frac{1}{\rho} + \frac{\Psi'(\theta) - 1}{L\rho^2} + \mathcal{O}\left(\frac{\log \rho}{\rho^3}\right).$$

If $a = \mathcal{D} \log G$, then $\mathcal{D}^2G/G = a^2 + \mathcal{D}a = \rho^{-2} + \mathcal{O}((\log \rho)/\rho^3)$. Substitution proves the formula. $\square$

Corollary 7.2 (First periodic norm expansion and Euler shift). *As $p \rightarrow \infty$,*

$$\boxed{\frac{\|\text{up}\|_p^p}{2G_0(p^{-1})} = 1 + \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta) - \gamma}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right)}. \quad (335)$$

Thus the first correction for the global L^p norm is the first inverse-quantile correction with the universal subtraction γ .

Proof. Multiply (333) by (331) and use $\|\text{up}\|_p^p = 2A_p$. $\square$

Corollary 7.3 (Approach to the sup norm). *As $p \rightarrow \infty$,*

$$\boxed{1 - \|\text{up}\|_p \sim \frac{\sqrt{2 \log 2 \log p}}{p}}. \quad (336)$$

More precisely,

$$-p \log \|\text{up}\|_p = L\rho - \log \rho + 1 - \frac{L}{2} - \frac{\frac{1}{2} \log \rho + \kappa_0 - \Psi(\theta) - \gamma}{\rho} + \mathcal{O}\left(\frac{(\log \rho)^2}{\rho^2}\right). \quad (337)$$

Proof. Take logarithms in (335) and use $G_0 = \rho \exp(-L\rho - L\beta)$ with $L\beta = 1 + L/2$. The leading term of $-\log \|\text{up}\|_p$ is $L\rho/p$; since it tends to zero, $1 - \|\text{up}\|_p \sim -\log \|\text{up}\|_p$. $\square$

7.3 Recovering the endpoint periodic function from global norms

Theorem 7.4 (Phase tomography by L^p norms). *Fix a phase $\vartheta \in \mathbb{R}/\mathbb{Z}$. For all sufficiently large integers j , define*

$$\rho_j := j + \vartheta - \beta, \quad p_j := \exp\left(\frac{L}{2}\rho_j^2\right). \quad (338)$$

Then

$$\lim_{j \rightarrow \infty} \left[\rho_j \left(\frac{\|\text{up}\|_{p_j}^{p_j}}{2G_0(p_j^{-1})} - 1 \right) - \frac{1}{2} \log \rho_j \right] = \kappa_0 - \gamma - \Psi(\vartheta). \quad (339)$$

Hence the nonconstant endpoint fluctuation Ψ can be reconstructed from a family of global norm measurements; pointwise evaluation or inversion of F is not required.

Proof. For (338), the parameter in (329) is exactly ρ_j and $\theta_j = \rho_j + \beta = j + \vartheta$. Periodicity gives $\Psi(\theta_j) = \Psi(\vartheta)$. Multiply (335) by ρ_j and subtract $\frac{1}{2} \log \rho_j$. $\square$

Remark 7.5 (Lambert interpretation). The Gaussian-logarithmic sampling law $\log p_j = (L/2)\rho_j^2$ is the inverse-side analogue of the lower-Lambert phase coordinate used in the endpoint expansion. It is precisely the sampling geometry that freezes the periodic saddle phase while sending the asymptotic scale to zero.

8 Fourier energy and a shifted Stieltjes–Wigert moment problem

8.1 Exact spectral moments

Let

$$C_2 := \|\text{up}\|_2^2 = \int_{\mathbb{R}} |\Phi(\xi)|^2 d\xi. \quad (340)$$

Numerically, $C_2 = 0.8088735359679707\dots$, but no decimal evaluation is needed below.

Theorem 8.1 (Shifted Stieltjes–Wigert spectral moments). *Define the probability measure*

$$d\nu(\xi) := \frac{|\Phi(\xi)|^2}{C_2} d\xi \quad (341)$$

and let μ_{sp} be its pushforward by $z = (2\pi\xi)^2$. Then, for every $n \geq 0$,

$$\int_0^\infty z^n d\mu_{\text{sp}}(z) = 2^{n(n+1)}. \quad (342)$$

With $q = 1/4$, this is

$$m_n = q^{-n(n+1)/2}. \quad (343)$$

Proof. Parseval and the Fourier differentiation rule give

$$\int_{\mathbb{R}} (2\pi\xi)^{2n} |\Phi(\xi)|^2 d\xi = \left\| \text{up}^{(n)} \right\|_2^2.$$

By (292) with $p = 2$, the right side is $2^{2T_n} C_2 = 2^{n(n+1)} C_2$. Divide by C_2 and push forward. $\square$

The measure has the explicit density

$$w_{\text{sp}}(z) = \frac{|\Phi(\sqrt{z}/(2\pi))|^2}{2\pi C_2 \sqrt{z}}, \quad z > 0. \quad (344)$$

8.2 A direct proof of moment indeterminacy

Let $L = \log 2$ and define the lognormal density

$$w_{\ln}(z) := \frac{1}{z\sqrt{4\pi L}} \exp\left(-\frac{(\log z - L)^2}{4L}\right), \quad z > 0. \quad (345)$$

This is the law of $\exp N(L, 2L)$ and therefore

$$\int_0^\infty z^n w_{\ln}(z) dz = \exp(nL + n^2 L) = 2^{n(n+1)}. \quad (346)$$

Corollary 8.2 (Explicit indeterminacy). *The Stieltjes moment sequence (342) is indeterminate. Indeed, $w_{\text{sp}}(z) dz$ and $w_{\ln}(z) dz$ are distinct probability measures with all moments equal. More generally,*

$$w_t := (1-t)w_{\ln} + tw_{\text{sp}}, \quad 0 \leq t \leq 1, \quad (347)$$

forms an explicit continuum of representing densities.

Proof. The moment equality is (342)–(346). The densities are different because w_{sp} vanishes whenever $z = (2\pi m)^2$, $m \in \mathbb{N}$, whereas $w_{\ln}$ is strictly positive on $(0, \infty)$. Convex combinations preserve every moment. $\square$

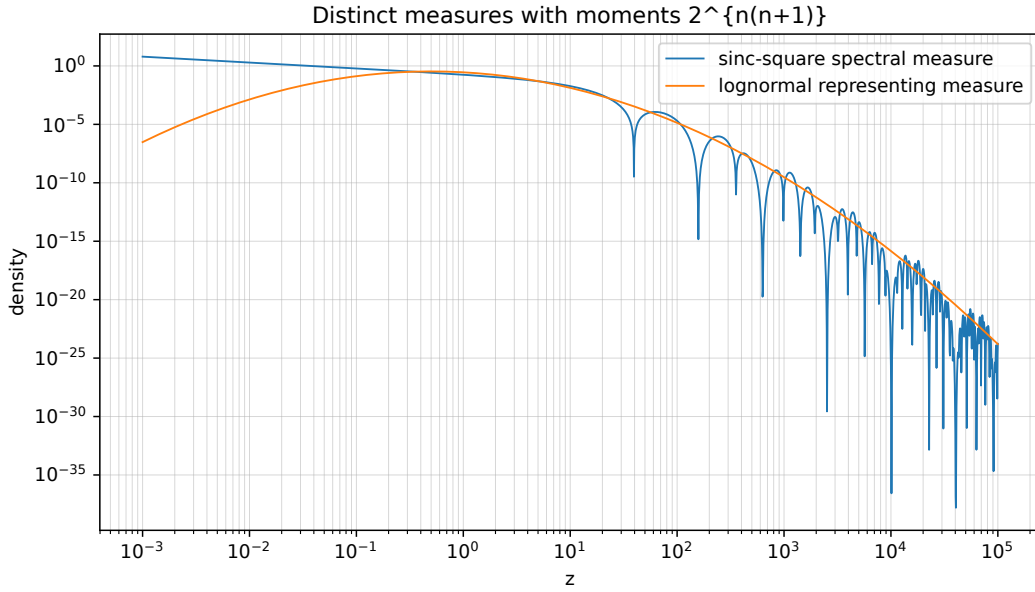


Figure 1: The sinc-square spectral density and a lognormal density have exactly the same nonnegative integer moments but visibly different local structures. The spectral zeros are inherited from the integer zeros of the infinite sinc product.

8.3 A nonclassical q-Pearson equation

The Fourier product satisfies

$$\Phi(2\xi) = \text{sinc}(2\pi\xi)\Phi(\xi). \quad (348)$$

Proposition 8.3 (Sinc-square q-Pearson law). *The spectral density (344) obeys*

$$w_{\text{sp}}(4z) = \frac{1}{2} \text{sinc}^2(\sqrt{z}) w_{\text{sp}}(z). \quad (349)$$

By contrast, the lognormal representative satisfies

$$w_{\ln}(4z) = \frac{1}{4z} w_{\ln}(z). \quad (350)$$

Proof. In (344), replacing z by $4z$ doubles the Fourier argument and multiplies the square-root denominator by 2. Apply (348) and note that $2\pi(\sqrt{z}/(2\pi)) = \sqrt{z}$. The lognormal identity follows by direct substitution in (345). $\square$

Remark 8.4. Classical Stieltjes–Wigert measures satisfy q-Pearson equations with monomial or rational multipliers. Equation (349) replaces that multiplier by an entire sinc-square factor carrying the complete integer-zero divisor. Determining its position in the Nevanlinna family is one of the principal open directions in [Appendix 12](#).

9 Hankel determinants, q-binomial polynomials, and Lagrange quadrature

Fix $0 < q < 1$ and the moment sequence

$$m_n = q^{-n(n+1)/2}. \quad (351)$$

The Fabius–Rvachev spectral case is $q = 1/4$.

Theorem 9.1 (Hankel determinants). *For $N \geq 0$,*

$$\Delta_N := \det[m_{i+j}]_{i,j=0}^N = q^{-N(N+1)(4N+5)/6} \prod_{k=1}^N (q; q)_k, \quad (352)$$

where $(q; q)_k = \prod_{r=1}^k (1 - q^r)$.

Proof. Factor

$$m_{i+j} = q^{-i(i+1)/2} q^{-j(j+1)/2} q^{-ij}.$$

The row and column factors contribute $q^{-\sum_{i=0}^N i(i+1)}$. The remaining determinant is the Vandermonde determinant in q^{-i} :

$$\det[(q^{-i})^j]_{i,j=0}^N = \prod_{0 \leq i < k \leq N} (q^{-k} - q^{-i}) = q^{-\sum_{k=1}^N k^2} \prod_{k=1}^N (q; q)_k.$$

Finally

$$\sum_{i=1}^N i(i+1) + \sum_{i=1}^N i^2 = \frac{N(N+1)(4N+5)}{6}.$$

$\square$

Theorem 9.2 (Monic shifted Stieltjes–Wigert polynomials). *The monic orthogonal polynomial of degree n for any representing measure of (351) is*

$$P_n(z) = \sum_{k=0}^n (-1)^{n-k} q^{k^2-n^2} \begin{bmatrix} n \\ k \end{bmatrix}_q z^k. \quad (353)$$

It satisfies

$$\int z^j P_n(z) d\mu(z) = 0, \quad 0 \leq j < n, \quad (354)$$

for every measure μ with moments (351), and

$$h_n := \int P_n(z)^2 d\mu(z) = q^{-(2n^2+n)}(q; q)_n. \quad (355)$$

The three-term recurrence is

$$P_{n+1}(z) = (z - a_n)P_n(z) - b_nP_{n-1}(z), \quad (356)$$

with

$$a_n = q^{-2n-1}(1 + q - q^{n+1}), \quad (357)$$

$$b_n = q^{-(4n-1)}(1 - q^n), \quad n \geq 1. \quad (358)$$

Proof. Let c_k be the coefficient of z^k in (353). For $j < n$,

$$\begin{aligned} \sum_{k=0}^n c_k m_{k+j} &= (-1)^n q^{-n^2-j(j+1)/2} \sum_{k=0}^n (-1)^k q^{k(k-1)/2} \begin{bmatrix} n \\ k \end{bmatrix}_q q^{-jk} \\ &= (-1)^n q^{-n^2-j(j+1)/2} (q^{-j}; q)_n = 0, \end{aligned}$$

by the finite q-binomial theorem; one factor is $1 - q^0$. The leading coefficient is one. The norm is the standard Gram-determinant quotient $h_n = \Delta_n / \Delta_{n-1}$, which gives (355). Hence $b_n = h_n / h_{n-1}$. Comparing the coefficients of z^n in the monic recurrence gives a_n ; the coefficient of z^{n-1} in P_n is $-q^{-2n+1}(1 - q^n)/(1 - q)$, yielding (357). $\square$

For $q = 1/4$, the first polynomials are

$$\begin{aligned} P_0(z) &= 1, \\ P_1(z) &= z - 4, \\ P_2(z) &= z^2 - 80z + 256, \\ P_3(z) &= (z - 64)(z^2 - 1280z + 4096). \end{aligned} \quad (359)$$

9.1 Gaussian quadrature as q-binomial Lagrange interpolation

Let $\lambda_{n,1}, \dots, \lambda_{n,n}$ be the zeros of P_n and let $\ell_{n,j}$ be the Lagrange cardinal polynomial at these nodes. The Gaussian weights are

$$\omega_{n,j} := \int \ell_{n,j}(z) d\mu(z) = \left(\sum_{r=0}^{n-1} \frac{P_r(\lambda_{n,j})^2}{h_r} \right)^{-1}. \quad (360)$$

Because the polynomials and h_r are determined by the moment sequence, the nodes and weights are identical for μ_{sp} , the lognormal measure, and every other representing measure.

Corollary 9.3 (Measure-independent Gaussian quadrature). *For every polynomial Q of degree at most $2n - 1$,*

$$\boxed{\int_0^\infty Q(z) d\mu_{\text{sp}}(z) = \sum_{j=1}^n \omega_{n,j} Q(\lambda_{n,j}) = \int_0^\infty Q(z) w_{\text{ln}}(z) dz.} \quad (361)$$

The q-binomial coefficients in (353) therefore generate exact finite Lagrange quadrature rules for polynomial spectral energies of the infinite sinc product.

Remark 9.4. Moment indeterminacy is invisible to every finite polynomial test: all representing measures produce the same Gaussian quadrature. The distinction lives in the orthogonal complement of the polynomial closure, exactly where the integer-zero geometry of Φ can enter.

10 The optimal derivative weight and a q-Legendre Fourier envelope

10.1 Exact Denjoy–Carleman weight

Define

$$M_n := 2^{T_n} = 2^{n(n+1)/2}. \quad (362)$$

By (294), up saturates this derivative weight exactly:

$$\left\| \text{up}^{(n)} \right\|_{\infty} = M_n. \quad (363)$$

The sequence is logarithmically convex and

$$\frac{M_n}{M_{n+1}} = 2^{-(n+1)}, \quad \sum_{n=0}^{\infty} \frac{M_n}{M_{n+1}} = 1 < \infty. \quad (364)$$

Thus the direct-weight Denjoy–Carleman class generated by M is nonquasianalytic, as the existence of the compactly supported nonzero function up requires.

Proposition 10.1 (Outside every finite Gevrey class). *For no finite $s > 0$ do constants $A, C > 0$ satisfy*

$$M_n \leq CA^n (n!)^s \quad (n \geq 0). \quad (365)$$

Hence up belongs to no finite-order Gevrey class on any interval containing its support.

Proof. The logarithm of M_n is $(\log 2)n^2/2 + \mathcal{O}(n)$, whereas $\log((n!)^s A^n) = sn \log n + \mathcal{O}(n)$. The quadratic term wins. $\square$

10.2 Exact associated function and periodic lattice correction

Define the associated function

$$\omega_M(t) := \sup_{n \geq 0} \log \frac{t^n}{M_n}, \quad t > 0. \quad (366)$$

Let $r = \log_2 t$.

Theorem 10.2 (Exact q-Legendre transform). *For $r \geq 1/2$,*

$$\omega_M(t) = \frac{\log 2}{2} \left[\left(r - \frac{1}{2} \right)^2 - \text{dist} \left(r - \frac{1}{2}, \mathbb{N}_0 \right)^2 \right]. \quad (367)$$

For $r \leq 1/2$, $\omega_M(t) = 0$. Thus the continuous log-square Legendre transform carries an exact bounded one-periodic lattice correction in $\log_2 t$.

Proof. For each integer $n \geq 0$,

$$\log \frac{t^n}{M_n} = \log 2 \left(nr - \frac{n(n+1)}{2} \right) = \frac{\log 2}{2} \left[\left(r - \frac{1}{2} \right)^2 - \left(n - r + \frac{1}{2} \right)^2 \right].$$

Maximizing means choosing the nearest nonnegative integer to $r - 1/2$. If $r \leq 1/2$, the maximum is attained at $n = 0$ and equals zero. $\square$

Corollary 10.3 (Optimized integration-by-parts envelope). *For every $\xi \neq 0$,*

$$|\Phi(\xi)| \leq \inf_{n \geq 0} \frac{M_n}{(2\pi|\xi|)^n} = \exp(-\omega_M(2\pi|\xi|)). \quad (368)$$

If $r = \log_2(2\pi|\xi|) \geq 1/2$, this is

$$|\Phi(\xi)| \leq 2^{-\frac{1}{2}(r-1/2)^2 + \frac{1}{2} \text{dist}(r-1/2, \mathbb{N}_0)^2}. \quad (369)$$

Proof. Integrating the Fourier transform by parts n times and using (293) gives $|\Phi(\xi)| \leq M_n/(2\pi|\xi|)^n$. Optimize with Theorem 10.2. $\square$

Warning 10.4. The envelope (369) is the best one obtainable from the exact L^1 derivative norms alone; it is not the sharp pointwise or annular Fourier envelope. The Fourier-decay reports in the corpus exploit the lacunary sine cocycle and obtain stronger linear-in-log $|\xi|$ corrections. The value of (369) is its exact functional-analytic duality with the derivative weight, not a replacement for those sharper results.

11 Numerical experiments

All code used here is in `numerical_experiments.py`. It is included with detailed comments in the accompanying archive. The script reconstructs `up` by inverse FFT on a periodic box of length 4 from 55 dyadic sinc factors, using 2^{19} samples. Since `up` is supported on $[-1, 1]$, the box supplies exact geometric zero padding. The run used

$$\Delta x = 2^{-17} = 7.62939453125 \times 10^{-6}, \quad \Delta \xi = \frac{1}{4}.$$

The reconstructed density had integral 1 and value `up(0) = 1` to displayed precision; the largest value outside the support was 3.33×10^{-16} .

11.1 Derivative tiling and norm checks

For $n = 0, \dots, 6$ and $p \in \{1, 2, 4, 8, \infty\}$, the script formed the derivative from the exact cell formula and compared its numerical norm with $2^{T_n} \|\text{up}\|_p$. The largest absolute deviation of the ratio from 1 was 2.22×10^{-16} .

11.2 Spectral moments

n	numerical $\int (2\pi\xi)^{2n} \Phi ^2$	predicted $2^{n(n+1)} C_2$	relative error
0	$8.088735359679706 \times 10^{-1}$	$8.088735359679707 \times 10^{-1}$	-1.11×10^{-16}
1	3.235494143871882	3.235494143871883	-2.22×10^{-16}
2	$5.176790630195011 \times 10^1$	$5.176790630195013 \times 10^1$	-2.22×10^{-16}
3	$3.313146003324810 \times 10^3$	$3.313146003324808 \times 10^3$	4.44×10^{-16}
4	$8.481653768511517 \times 10^5$	$8.481653768511509 \times 10^5$	8.88×10^{-16}
5	$8.685213458955796 \times 10^8$	$8.685213458955785 \times 10^8$	1.33×10^{-15}
6	$3.557463432788293 \times 10^{12}$	$3.557463432788290 \times 10^{12}$	8.88×10^{-16}
7	$5.828548088280333 \times 10^{16}$	$5.828548088280334 \times 10^{16}$	-1.11×10^{-16}

The near-machine equality is a stringent joint check of the FFT normalization, derivative scale, Parseval convention, and moment exponent.

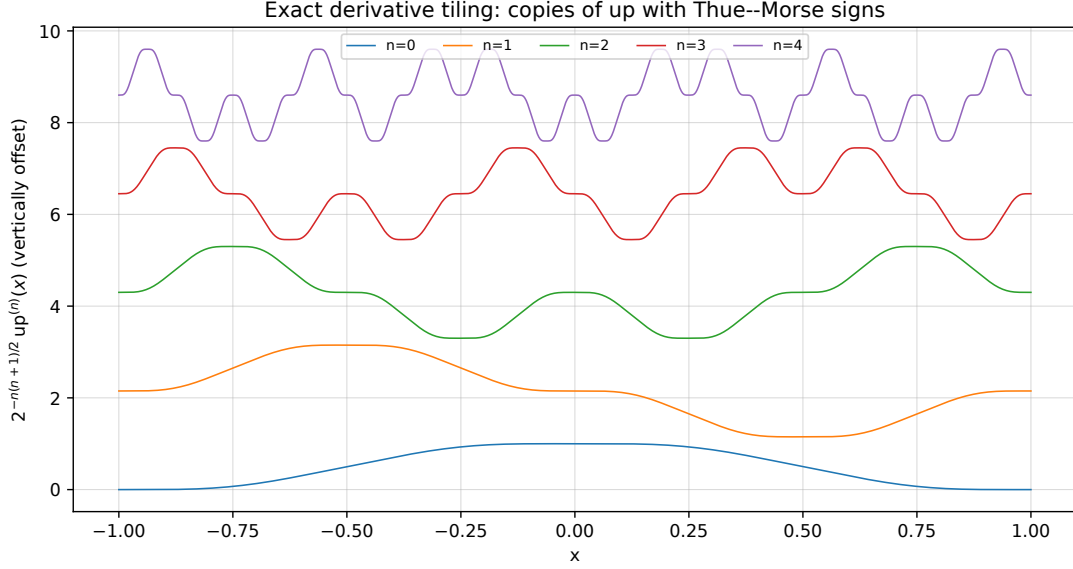


Figure 2: Normalized derivatives $2^{-T_n} \text{up}^{(n)}$, vertically offset. Each row is a concatenation of copies of the original bump; their signs are the length- 2^n Thue–Morse block.

11.3 Euler–Gamma operator convergence

The inverse interpolant was obtained directly from $G(1 - \text{up}(x)) = x$ on $0 \leq x \leq 1$. Logarithmic finite differences estimated $\mathcal{D}^k G(1/p)$ for $k \leq 3$. The following table lists relative errors of successive truncations of (328).

p	$A_p/G(1/p)$	G only	through $\mathcal{D}$	through $\mathcal{D}^2$	through $\mathcal{D}^3$
256	0.92618125	-7.38×10^{-2}	4.84×10^{-2}	-1.47×10^{-2}	6.46×10^{-3}
512	0.92793214	-7.21×10^{-2}	4.32×10^{-2}	-1.20×10^{-2}	6.37×10^{-3}
1024	0.92942377	-7.06×10^{-2}	3.91×10^{-2}	-9.78×10^{-3}	4.87×10^{-3}
2048	0.93089393	-6.91×10^{-2}	3.60×10^{-2}	-8.37×10^{-3}	3.35×10^{-3}
4096	0.93235311	-6.76×10^{-2}	3.33×10^{-2}	-7.55×10^{-3}	2.53×10^{-3}
8192	0.93370939	-6.63×10^{-2}	3.09×10^{-2}	-6.86×10^{-3}	2.39×10^{-3}
16384	0.93492704	-6.51×10^{-2}	2.87×10^{-2}	-6.10×10^{-3}	2.29×10^{-3}
32768	0.93604971	-6.40×10^{-2}	2.69×10^{-2}	-5.39×10^{-3}	1.97×10^{-3}
65536	0.93712995	-6.29×10^{-2}	2.53×10^{-2}	-4.83×10^{-3}	1.60×10^{-3}

11.4 Symbolic Hankel checks

SymPy evaluated the determinants in (352) exactly for $N = 0, \dots, 6$ at $q = 1/4$; every quotient of the direct determinant by the claimed closed form simplified to 1. The base-10 logarithms of the determinants for $N = 1, \dots, 6$ were

$$1.68124, 7.54887, 20.03233, 41.54498, 74.49635, 121.29499,$$

illustrating the rapid $q^{-\frac{2}{3}N^3 + \mathcal{O}(N^2)}$ growth.

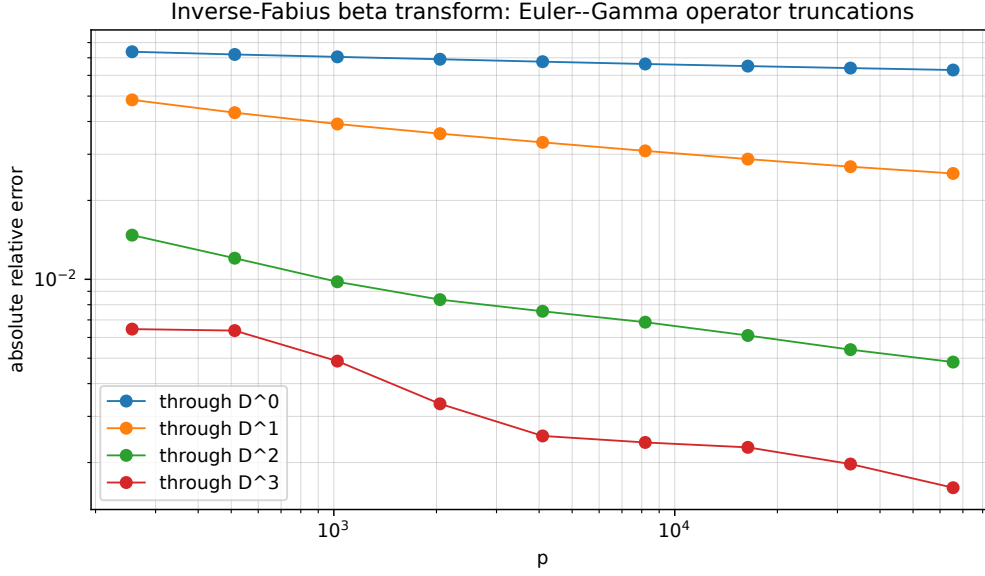


Figure 3: Absolute relative errors of the first four Euler–Gamma operator truncations. The slow scale $\rho^{-1} \asymp (\log p)^{-1/2}$ explains why very large p is needed before successive asymptotic orders separate cleanly.

12 Conjectures and promising research directions

12.1 A full norm transseries

The inverse report gives an all-orders expansion of G in the differential algebra generated by periodic saddle jets. The exact operator (316) should transport that algebra to the norm spectrum without introducing any new transcendental data beyond the log-exponential cumulants.

Conjecture 12.1 (All-orders Gamma–Lambert norm transseries). *There exist one-periodic real-analytic coefficient functions $C_{m,\ell}(\theta)$ such that*

$$\frac{\|\text{up}\|_p^p}{2G_0(p^{-1})} \sim 1 + \sum_{m=1}^{\infty} \rho^{-m} \sum_{\ell=0}^m C_{m,\ell}(\theta) (\log \rho)^\ell, \quad (370)$$

with $C_{1,1} = 1/2$ and $C_{1,0} = \kappa_0 - \gamma - \Psi(\theta)$. Every $C_{m,\ell}$ belongs to the differential algebra generated by the inverse saddle jets and the constants $\gamma, \zeta(2), \dots, \zeta(m)$; no additional periodic functions occur.

A proof should combine the formal inverse recursion already in the corpus with a tame remainder theorem for the operator $M_p(\mathcal{D})$. The key analytic task is uniform control of $\mathcal{D}^m G$ on logarithmic windows whose width grows slowly with ρ .

12.2 Nevanlinna parameter of the sinc-square measure

Problem 12.2 (Locate μ_{sp} in the Stieltjes–Wigert Nevanlinna family). *Determine the Nevanlinna parameter corresponding to the density (344). Equivalently, calculate the Cauchy transform*

$$S_{\text{sp}}(z) = \int_0^\infty \frac{d\mu_{\text{sp}}(t)}{z - t} \quad (371)$$

relative to the classical entire functions A, B, C, D of the shifted Stieltjes–Wigert moment problem.

The exact q -Pearson law (349) supplies unusually rigid extra data. One promising route is to Mellin-transform it, obtaining a functional equation coupling the spectral Mellin transform to the Mellin transform of $\text{sinc}^2(\sqrt{z})$. Another is to use the canonical product of Φ to express S_{sp} as a convergent sum over integer-zero lobes.

Conjecture 12.3 (Sinc-Pearson uniqueness). *Among absolutely continuous representing measures of $m_n = 2^{n(n+1)}$ whose density is locally bounded after multiplication by $\sqrt{z}$, the normalization, moment conditions, and q -Pearson equation (349) determine w_{sp} uniquely.*

12.3 Orthogonal complement and spectral zeros

For an indeterminate moment problem, the polynomials are not dense in every representing L^2 space. The spectral measure should possess an orthogonal complement reflecting the integer-zero structure of Φ .

Problem 12.4 (Explicit nonpolynomial orthogonal basis). *Construct functions $Q_j(z)$ such that*

$$\int_0^\infty z^n Q_j(z) w_{\text{sp}}(z) dz = 0 \quad (n \geq 0)$$

and such that the Q_j span the orthogonal complement of the shifted Stieltjes–Wigert polynomials in $L^2(\mu_{\text{sp}})$. Dyadic rescalings of sinc factors, integer-lobe indicator functions, and q -Bessel functions from the classical theory are the three natural candidate families.

12.4 Confluent q -Richardson phase tomography

The existing corpus develops Gaussian- q Richardson weights for pure power error germs. The norm expansion (370) contains powers multiplied by logarithms. Choose a fixed phase ϑ and a dyadic phase-locked sequence

$$\rho_j = M2^j + \delta, \quad \delta + \beta \equiv \vartheta \pmod{1}, \quad p_j = \exp(L\rho_j^2/2). \quad (372)$$

Then $h_j = 1/\rho_j$ is an analytic function of 2^{-j} , while the phase is constant. After subtracting the known logarithmic terms, the remaining errors are combinations of $j^\ell 2^{-mj}$.

Conjecture 12.5 (Confluent Gaussian- q filter). *For each truncation order r , there is a stable finite filter whose shift polynomial is a product of repeated factors*

$$\prod_{m=1}^r (E - 2^{-m})^{d_m}, \quad (373)$$

which annihilates every error mode $j^\ell 2^{-mj}$ with $0 \leq \ell < d_m$. Its coefficients can be written as derivatives or confluent limits of the Gaussian- q -binomial Lagrange weights

$$w_{r,j}(q) = (-1)^{r-j} \frac{q^{\binom{r-j+1}{2}}}{(q; q)_j (q; q)_{r-j}}. \quad (374)$$

Applied to (372), this filter recovers $\kappa_0 - \gamma - \Psi(\vartheta)$ with a rigorously controlled error of arbitrarily high order in 2^{-j} .

This would create a genuinely new Richardson pipeline: the data are global derivative norms, the target is a pointwise endpoint periodic function, and the logarithmic multiplicities are handled by confluent q -Lagrange interpolation.

12.5 Base- b and Pascal–Rvachev extensions

The Pascal–Rvachev hierarchy in the corpus replaces the single sinc exponent by binomial multiplicities. The present analysis suggests two extensions.

Problem 12.6 (Norm ladders for higher Rvachev ranks). *For each hierarchy level, identify a spatial refinement formula with disjoint or finitely overlapping cells, determine whether normalized derivatives are equimeasurable with a fixed profile, and calculate the resulting derivative weight. If exact equimeasurability fails, seek a finite-state transfer operator for value distributions.*

Conjecture 12.7 (Higher-rank lognormal moment geometry). *For the rank- r Pascal–Rvachev profile, the normalized spectral energy moments are q -hyperfactorial sequences whose Hankel determinants factor into products of q -Pochhammer symbols with exponents polynomial in the index. The associated orthogonal polynomials should be multilevel q -binomial deformations of Stieltjes–Wigert polynomials.*

12.6 Formalization roadmap

The roadmap is now partly discharged. The completion labels below refer only to the named declarations in `RvachevDerivativeDistribution.lean`; they do not mark the whole report as formalized. A practical order is:

1. **Completed.** The derivative tiling has a public exact cell coordinate, endpoint/range lemmas, signed and absolute cellwise formulas, and signed midpoint corollary.
2. **Absolute part completed.** The normalized absolute law is an exact Borel pushforward equality and has Banach-valued continuous-test forms. The signed finite-mixture identity is also formal for continuous Banach-valued tests and all n , but its simplified signed Rademacher Borel law remains open.
3. **Still open beyond completed prerequisites.** Use the all- $p \geq 0$ real-power moment theorem, midpoint signs, and existing pointwise bounds to derive `eLpNorm`, general rearrangement-invariant, total-variation, and exact inverse-Fabius level-set corollaries.
4. **Open.** Formalize the Stieltjes substitution and beta-transform identity for the totalized inverse.
5. **Open.** Prove the spectral moments by the existing Fourier differentiation API.
6. **Open.** Formalize the Vandermonde determinant and finite q -binomial orthogonality.
7. **Deferred.** Treat the all-orders asymptotic operator after the inverse derivative-jet bounds have a stable public interface.

13 Conclusion

The dyadic derivative formula contains more information than its usual pointwise reading suggests. Because the cells are disjoint and all have the same affine Jacobian, it is an exact renormalization fixed point for value distributions. The entire derivative tower has one absolute profile, amplified by $2^{n(n+1)/2}$ and signed by Thue–Morse. This immediately produces an exact norm spectrum in every rearrangement-invariant space and makes the inverse Fabius function the level-set law of every derivative.

Two dual transforms then emerge. In physical space, the large- p norms are a beta average of the inverse function. The beta dilation converges to an exponential dilation, whose Mellin symbol is $\Gamma(1+s)$; Bell polynomials, zeta values, and Bernoulli polynomials are therefore forced into the asymptotic operator. Combining it with the lower-Lambert endpoint coordinate reveals a universal Euler-constant shift and lets global norms perform phase tomography on the periodic saddle correction.

In frequency space, Parseval turns the same derivative scale into the moment sequence $2^{n(n+1)}$. The infinite sinc product thus supplies an explicit, highly nonclassical Stieltjes–Wigert representing measure. Its integer zeros distinguish it from the lognormal representative even though all polynomial moments agree; its q -Pearson law retains the full sinc-square factor. The resulting q -binomial orthogonal polynomials and Gaussian Lagrange rules give a concrete bridge between spectral energy, moment indeterminacy, and the q -calculus already pervasive in the Fabius theory.

The most promising analytic next step is to identify the sinc-square measure inside the Nevanlinna family. On the formal side, the dyadic-cell and normalized absolute equimeasurability core is now complete. The next compact targets are its $eLpNorm$ /rearrangement-invariant and inverse-Fabius level-set consequences, followed by the beta-transform layer. The signed Rademacher Borel simplification and every beta/spectral layer in this report remain frontier statements.

A Recursive TeX corpus ledger

The following 68 paths were present under Analysis/FabiusFunction/docs at commit 0341bbf91cb867f4510bc8431cf59878208de771. Archived versions were included in the collision audit but interpreted according to the archive’s provenance ledger.

#	Path
1	FabiusFunction_Mathematical_Glossary/FabiusFunction_Mathematical_Glossary.tex
2	Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex
3	Non_Elementarity_of_the_Fabius_Function/Non_Elementarity_of_the_Fabius_Function.tex
4	archive/early-notes/FabiusAsymptotic/FabiusAsymptotic.tex
5	archive/early-notes/K-foldsummationoverthesignedThue-Morsesequence/K-foldsummationoverthesignedThue-Morsesequence.tex
6	archive/glossary/FabiusFunction_Mathematical_Glossary-2/FabiusFunction_Mathematical_Glossary.tex
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15	archive/primary-exposition/Fabius_function_and_up_function/Fabius_function_and_up_function.tex
16	archive/primary-supplements/Fabius_Function_Missing_Parts-2/Fabius_Function_Missing_Parts.tex
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18	archive/q-calculus/Fabius_q_Special_Functions/Fabius_q_Special_Functions.tex

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21	archive/research-frontiers/Fabius_Dyadic_Formulae_and_Alternative_Representations/Fabius_Dyadic_Formulae_and_Alternative_Representations.tex
22	archive/research-frontiers/Fabius_Dyadic_Formulae_to_Asymptotics/Fabius_Dyadic_Formulae_to_Asymptotics.tex
23	archive/research-frontiers/Fabius_Dyadic_q_Connections/Fabius_Dyadic_q_Connections.tex
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32	archive/research-frontiers/non-formalized-research-frontiers/semi-formalized-research-frontiers.tex
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34	archive/standalone-studies/Small_Argument_Asymptotics/Small_Argument_Asymptotics.tex
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36	fabius_lean_walkthrough/fabius_lean_walkthrough.tex
37	semi-formalized-research-frontiers/semi-formalized-research-frontiers.tex
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62	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0-corrected.tex
63	papers/AIF_2017__67_2_637_0/AIF_2017__67_2_637_0.tex
64	papers/arXiv-1502.06738v2/Lacunary_products.tex
65	papers/arXiv-1702.05442v1/09-Function.tex
66	papers/arXiv-1702.06487v3/157-Arithmetic-v3.tex
67	papers/arXiv-2211.13570v1/document.tex
68	papers/ubiq15/ubiq15-corrected.tex

B Reproducibility files

The accompanying archive contains:

- Fabius_Derivative_Norm_Spectrum.tex – this source;
- Fabius_Derivative_Norm_Spectrum.pdf – the rendered report;
- numerical_experiments.py – commented reproduction code;

- figures/ – PDF and PNG versions of the three generated figures;
- data/derivative_norm_ratios.csv;
- data/spectral_moment_checks.csv;
- data/beta_operator_checks.csv;
- data/hankel_determinant_checks.csv; and
- data/run_summary.txt.

The default reproduction command is

```
python numerical_experiments.py
```

The grid size, number of sinc factors, derivative order, and output directory are configurable through command-line options documented by

```
python numerical_experiments.py --help
```

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